ATOMIC STRUCTURE

- Energy of orbit = $\frac{-13.6}{n^2}$ e·V [of Hydrogen]
- Angular momentum $(\vec{x} \times \vec{p}) = \vec{x} \cdot \vec{p} \cdot \sin 90°$. $\Rightarrow M_e V_e x = \frac{nh}{2\pi}$
- Velocity = $V_{1,H} \times \frac{Z}{n}$, $V_{1,H} = 2.18 \times 10^6$ m/s
- Radius = $x_{1,H} \times \frac{n^2}{Z}$, $R_{1,H} = 0.529$ Å
- $\mu = \frac{M_1 M_2}{M_1 + M_2}$ = reduced mass of the system.
- $T = \frac{2\pi x}{v} = \left(\frac{2\pi x_{1,H}}{V_{1,H}}\right) \times \frac{n^3}{Z^2}$ [of revolution]
- Energy = $E_{1,H} \times \frac{Z^2}{n^2}$, $E_{1,H} = -13.6$ e·v/atom $= -1312$ KJ/mole $= -313$ K cal/mole
  $= 2.18 \times 10^{-18} \frac{Z^2}{n^2}$ J/atom
- Energy of 1 quantum = $h\nu = \frac{hc}{\lambda}$
- Min^m value of energy supplied to remove e^- → threshold energy (ϕ) = work funcn
- $E = \phi + (K \cdot E)_{of\ e^-}$ [E = Energy of striking e^-]
  $\Rightarrow K \cdot E$ = stopping potential × charge on e^- = $(V \times e^-)$
- $\Rightarrow h\nu = h\nu_0 + (K \cdot E)_{e^-}$ [Photo electric effect $\Rightarrow V > V_0$]
- $\Rightarrow \frac{hc}{\lambda} = \frac{hc}{\lambda_0} + (KE)_{e^-}$ [Photo electric effect, $\lambda < \lambda_0$] λ_0 = threshold frequency.
- $E = \frac{hc}{\lambda}$ & $E = mc^2$ $\Rightarrow \lambda = \frac{h}{mc} = \frac{h}{p}$ [p = momentum of photon]
- no. of complete de broglie wave in single e^- specie = orbit no.
- $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(K \cdot E)}} = \frac{h}{\sqrt{2mqV}}$ [$M_e < M_p < M_\alpha$] (M_α (He^{2+} ion) = 1 amu (2p + 2n))
- De-broglie wave length of photo e^- = $\frac{h}{\sqrt{2m(E-\phi)}}$ ($\because (KE)_e = E - \phi$)
- $\frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] Z^2$, where R_H = Rydberg's const. = 1.1×10^7 m^{-1}
  n_1 → Lower orbit
  n_2 → higher orbit
  [$R_H = \frac{-E_{1,H}}{hc}$]
- $\lambda = \frac{910\ Å}{\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] Z^2}$
- Max^m no. of spectral line → in spectrum [single isolated atom = $(n-1)$; Largeness of atom = $\frac{n(n-1)}{2}$]
- $\Delta p \times \Delta x \geq \frac{h}{4\pi}$

Lyman, Balmer, Paschen, Brackett, Pfund, Humphrey

Δx = uncertainity in measurement of posi^n & Δp = uncertainity in measurement of momentum

+ Charge on e^- or $p^+ = 1.6 \times 10^{-19}$ C
* n = orbit no.
* Z = no. of proton + nt in nucleus of single e^- species.
* T·E = K·E + P·E
  (−x) (+x) (−2x)
- h → plank's const. → 6.7×10^{-34} J/sec
  [1 C/s = 1 Hz [frequency]
  1 K C/sec = 1000 C/sec = 1000 Hz.
  e·V × 1.6×10^{-19} ⇒ coulomb.

Ionisa^n energy → $E_\infty - E_1 = 0 - (-13.6 \times \frac{Z^2}{n^2})$ for 1st
→ $13.6\ Z^2$

Intensity = $\frac{\text{Power}}{\text{area}}$

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- Probability of finding electron at any pt = 0, ψ or wave func^n = 0 $\Rightarrow \psi = 0$ [Node]

- $L = \frac{\lambda}{2}$ node $\Rightarrow$ 0; $L = \lambda$ node = 1; $L = \frac{3\lambda}{2}$ node 2

$L = n \times \frac{\lambda}{2}$

no of node = n − 1

- $\lambda = \frac{h}{\sqrt{2 \times m \times KE}} \Rightarrow$ & $\lambda = \frac{2L}{n} \Rightarrow E = \frac{n^2 h^2}{8 m L^2}$ [$E_x + E_y + E_z = E$ (n_x, n_y, n_z)] [$\psi_1 = (2,1,1)$, $\psi_2 = (1,2,1)$, $\psi_3 = (1,1,2)$]
- Degeneracy of e^- = different wave func^n (ψ) but same energy value.
- Wave func^n of $e^- \rightarrow \psi = R(r) \times A(\theta, \phi)$ [Radial part → n, l; Angular part → m, l]
- Energy of e^- ↑ No of node ↑ — radial part; angular part.
- Both e^- + n^+ in an atomic orbital would have same wave function.
- Orbital angular momentum = $\frac{h}{2\pi}\sqrt{l(l+1)}$ l = angular node (l); l = 0 (s), l = 1 (p), L = 2 (d), L = 3 (f)
- magnetic quantum no. (direc^n) → $m = \sqrt{l(l+1)}\cos\theta$. [−l to l]
- n = (l+1), (l+2), ...
- Probability of finding e^- at pt. r dist from nucleus → prob. of finding e^- from nucleus = $(4\pi r^2 dr)\psi^2$
- no of radial node = (n − l − 1)

$\psi(1s)$, $w(2s)$, $w(4s)$, $w^2(2s)$, $w^2(4s)$

$4\pi r^2 dr\, w^2$ (3s) ⇔ r

[for given value of l, n↑ energy ↑. ⇒ 1s < 2s < 3s < 4s; 2p < 3p < 4p < 5p] (n+l)↑ energy more

- Degenerate orbitals: n, l, m − same but 'm' diff. [same radial part, diff. angular part] same energy for diff. θ, ψ.
- Hamiltonian operator: $H = \frac{-h}{8\pi^2 m}\left(\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2}\right)$ [same (n+l) more n more energy]
- Pauli's exclusion principle − no two e^- in an atom have same set of all 4 Q.N.
- Magnetic moment (μ) = $\sqrt{4s(s+1)} = \sqrt{4n_x(n_x+1)} = \sqrt{4n_x \frac{1}{2}\left(\frac{1}{2}+1\right)} = \sqrt{n(n+2)}$ Bohr magneton
  n → unpaired e^- (in no's), sum of spin Q.N = n_x
- 1 Bohr magnetism = $\frac{eh}{4\pi Mc}$ [fully & half filled orbitals are more stable]
- Ex → Cr (24) = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$ & (29), $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$
- magnetic mom. = Cr > Mn > Fe > Co > Ni [24 > 25 > 26 > 27 > 28]
- $Mn^{2+} = Fe^{3+} > Co^{2+} > Ni^{2+}$
- Spin multiplicity = 2S + 1 = n + 1 [n = no of unpaired e^-; S = sum of all spin Q.N's]

- L = 0, m = 0, spherical **S** ⊕
- l = 1, m = ±1, 0. dumbbell shape **P**: yz (nodal) (Px); xz (nodal) (Py); xy − nodal plane (Pz orbital)
- l = 2, m = ±2, ±1, 0. double dumbbell. **d**: (xz, yz nodal) dxy; (xy, xz) dyz; dxz; dx^2-y^2 (2 nodal planes); dz^2

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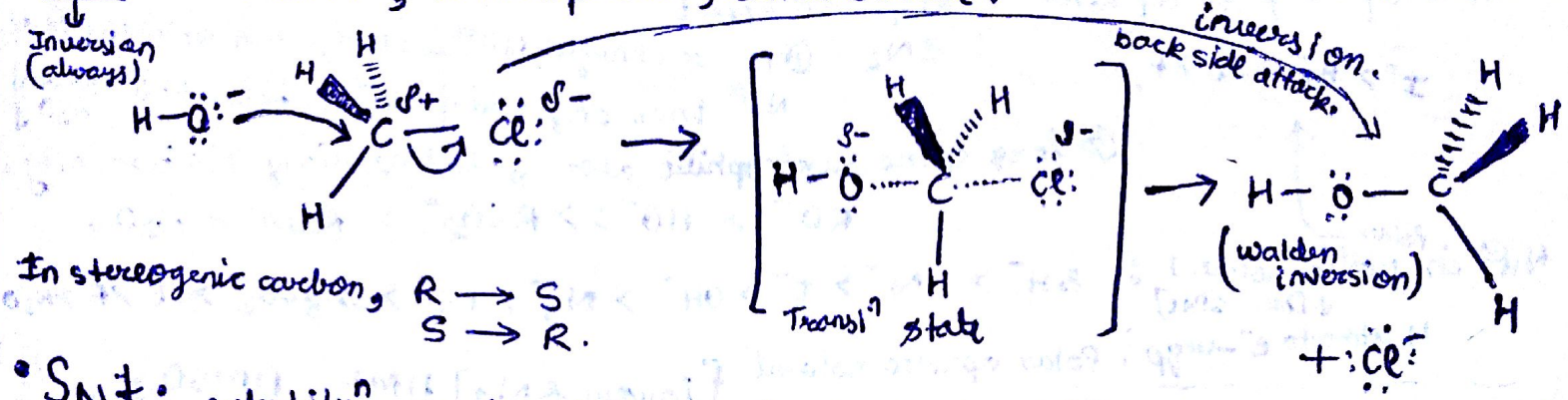
(16A)

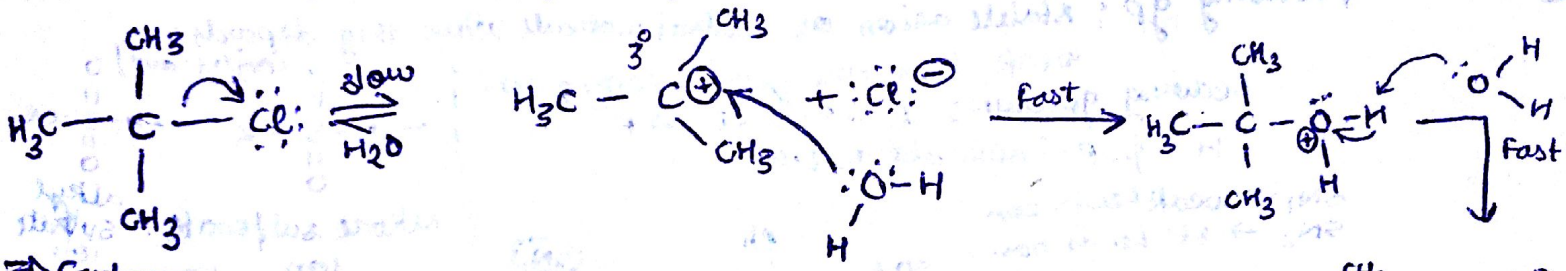
- On moving down in periodic table BL ↑ & BS ↓. [C–X]
- Vinylic or phenyl halides : halogen atom is bonded to sp^2 carbon. [$CH_2=CH-$ vinyl gp]
- Nucleophilic substitution Reac^n : $Nu:^{\ominus} + R-\ddot{X}: \longrightarrow R-Nu + :\ddot{X}:^-$
  ↳ seek +ve center.

- **S_N2** : substitu^n, nucleophilic, bimolecular.
  ↓ Inversion (always)



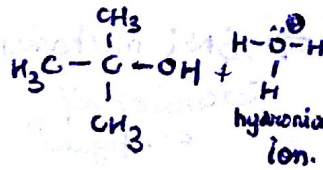
In stereogenic carbon, R → S
S → R.

- **S_N1** : substitu^n, nucleophilic, unimolecular.

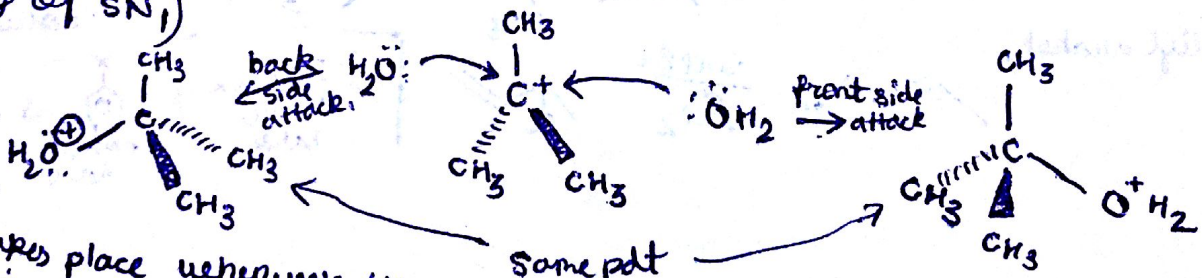


⇒ Carbocation : Ex : BF_3.
↓ trigonal planar (sp^2)

→ Stability → 3° > 2° > 1° > methyl.
(due to hyper conjugan^n)

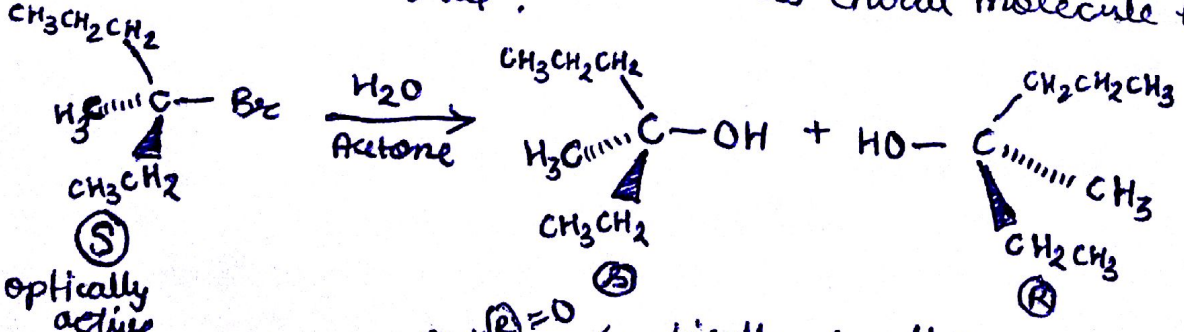


Stereochemistry of SN_1)



- Racemiza^n takes place whenever the rx^n causes chiral molecule to be converted in an achiral intermediate.

⇒



front & back side attack take place at equal rate.

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• Solvolysis (S_N1 rckn) : $(CH_3)_3C-Br + H_2O \rightarrow (CH_3)_3C-OH + HBr$.

: $(CH_3)_3-C-Cl + H-\overset{O}{\overset{\|}{C}}-OH \rightarrow (CH_3)_3-C-O-\overset{O}{\overset{\|}{C}}H + HCl$

⇒ Factors affecting rate of S_N1 & S_N2 →

① Structure of substrate : S_N2 : $0^\circ > 1^\circ > 2^\circ >>> 3^\circ$ (unreactive)
↳ steric hindrance

② Effect of CJ & st. of $Nu^\ominus$: S_N1 – no effect.

$I^- > Br^- > Cl^- > F^-$

: S_N2 – ① A –ve charged $Nu^\ominus$ is always a more reactive $Nu^\ominus$ than conj acid. [HO^- is better $Nu^\ominus$ than H_2O]

② for same nucleophilic atom, nucleophilicity ↑ basicity.

$RO^- > HO^- >> RCO_2^- > ROH > H_2O$,

$Nu^\ominus$ in polar protic solvent : ↓ Eff. S_N1 (H atom to e⁻-wdg)

$SH^- > CN^- > I^- > OH^- > N_3^- > Br^- > CH_3CO_2^- > Cl^- > F^- > H_2O$

Polar aprotic solvent (no H atom to e⁻-w gp.) [favour S_N2 & high CJ $Nu^\ominus$] DMF, DMSO, DMA, HMPA (pg 333 & 340)

③ Nature of leaving gp : stable anion or neutral molecule when they departs.

weak base are good Leaving gp.

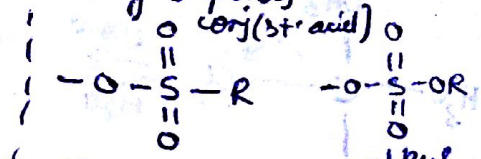
Leaving gp order : $I^- > Br^- > Cl^- >> F^-$

$H:^-$ & $R:^-$ never act as L.Gp.

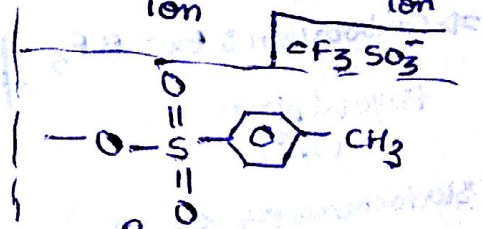
S_N1 → weak Lewis base

S_N2 → st. Lewis base.

conj(st. acid)



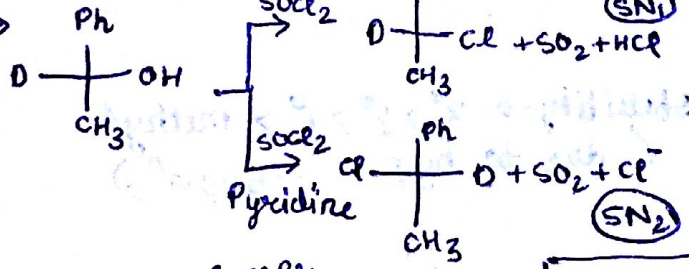
alkane sulfonate ion, alkyl sulfate ion



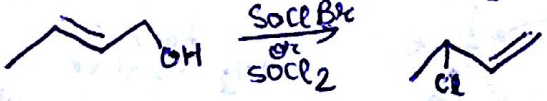
$CF_3SO_3^-$

P-toluene sulfonate ion

⇒ .S_Ni mechanism → Retension of configuration



for allyl alcohol,



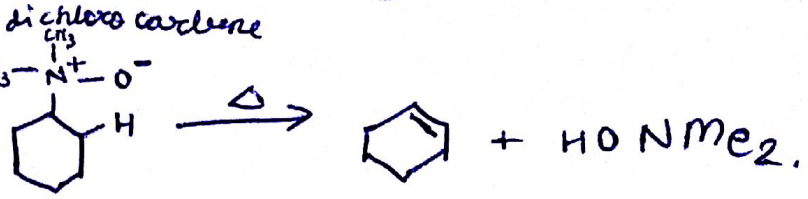
vinylic halide & phenyl halide ⇒ unreactive in S_N1 & S_N2

(2) (6B)

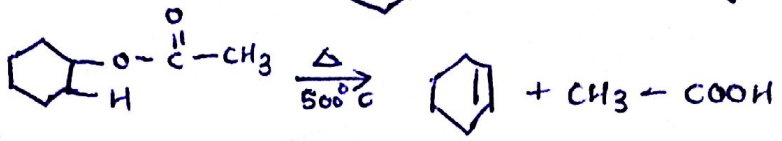
• Elimina^n rck^n → (A) 1,1-elimina^n or α-elimina^n – Removal of gp attached to same C.

$$CHCl_3 \xrightarrow[\Delta]{KOH} KCl + \underset{\text{dichloro carbene}}{Cl-\ddot{C}-Cl} + H_2O\uparrow$$

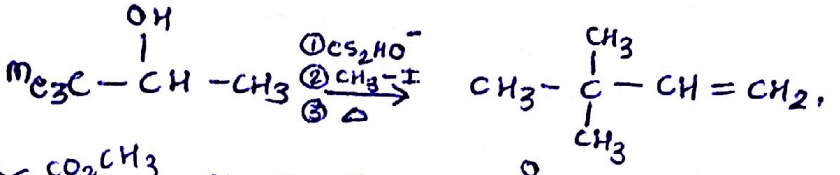
(B) β elimina^n → (i) Cope rck^n →



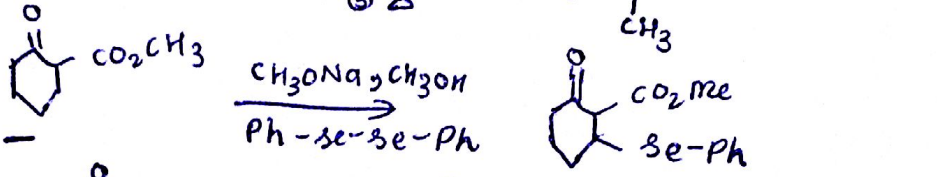
• Pyrolysis of ester. :→



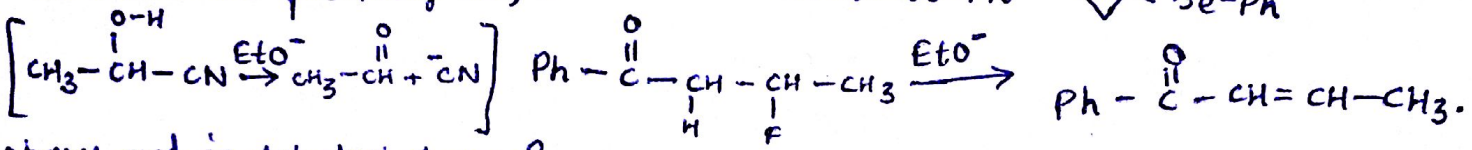
• Pyrolysis of Xanthates or chugaeu rck^n →



• Pyrolysis of selenoxide →



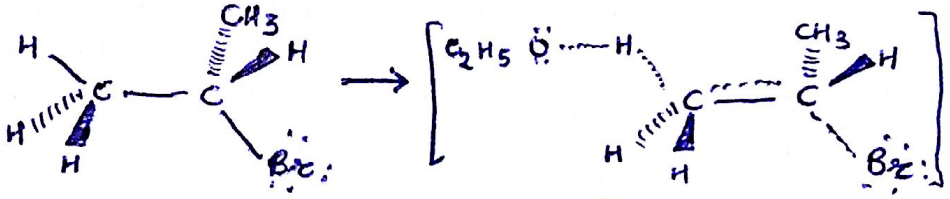
• E1cb (elimina^n from conj base) –



• bases used in dehydrohalogena^n – Na salt of alcohol, potassium tert-butoxide.

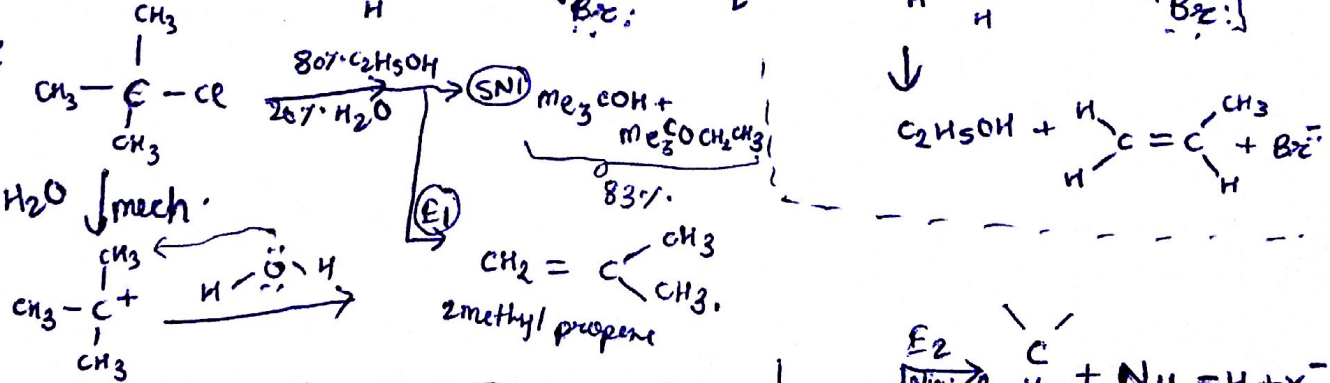
⇒ E_2 Rk^n :

(bimolecular) $C_2H_5O^-$



⇒ E_1 Rk^n :

Unimolecular.



• S_N2 Vs E_2 : Both favr by high [] $Nu^\ominus$

primary halide → unhindered base → S_N2 favr.

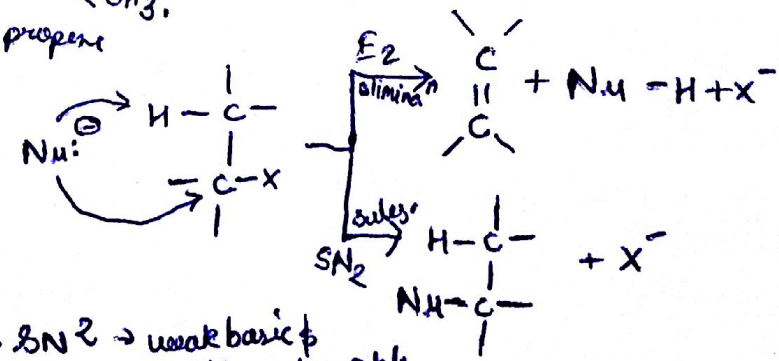
secon d. halide → S_N2 + E_2.

tertiary → E_2

↑temp favours elimina^n over substitu^n.

unhindered small base $Nu^\ominus$ → S_N2 → weak basic & highly polarable.

hindered base $Nu^\ominus$ → E_2. → strong base.



• S_N1 Vs E_1 → S_N1 –

E_1 – 3° ~~halide~~ carbocation., poor $Nu^\ominus$, weak base, polar solvent

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HYDROCARBONS

- Alkane → prep → ① from reduc^n : unsaturated + $H_2 \xrightarrow{Ni}$ saturated H·C | stability ↑ ∝ $\frac{1}{\text{Heat of hyd.}}$

② from decarboxylation :

$$R-\underset{\text{O}}{\overset{\|}{C}}-O-H \xrightarrow[CaO]{NaOH \rightleftharpoons Na^+ + \bar{O}H} RH + CO_2 + H_2O$$

[when gas is formed yield is good]

③ clamention reduc^n :- Aldehyde or ketone $\xrightarrow[\text{conc HCl}]{Zn/Hg}$ H_2O + Alkane. [Hg controls rxn] [Zn control oxidisn]

$$R-\underset{H}{\overset{H}{C}}=O \xrightarrow[\text{conc HCl}]{Zn/Hg} R-CH_3 + \underset{H}{\overset{H}{>}}O$$

($Zn/Hg \xrightarrow{mix}$ amulgum)

④ Wolf Krisnan reduc^n :-

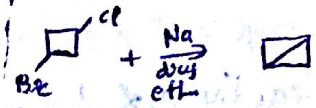
[$R-CH=O + H_2NNH_2$]

aldehyde / ketone + H_2NNH_2 (Hydrazine) $\xrightarrow{NaOH}$ $R-CH=N-NH_2 \xrightarrow[NaOH]{\Delta} N_2\uparrow +$ $R-CH_3$

⑤ Wurtz rxn → $R-X + X-R' \xrightarrow[\text{dry ether}]{2Na} R-R' + 2NaX$

free radicle mech.

symmetrical alkyl halide → good yield.

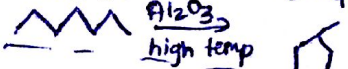


⑥ Kolbe's electrolysis – $2CH_3COOH \xrightarrow[-lysis]{electro} CH_3-CH_3 + 2CO_2 + H_2$

→ Chem prop – ① Halogenation

② Bromination (less reactive & more selective) ③ Nitration : $R-H + HO-NO_2 \xrightarrow[H_2SO_4]{Conc} H_2O + R-NO_2$

④ Isomerisation :



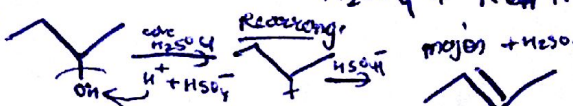
- Alkene → Prep – ① from dehydration : Alcohol $\xrightarrow[H^+ + HSO_4^-]{\text{conc } H_2SO_4}$ H_2O + $R-SO_4H \xrightarrow{\Delta} H_2SO_4 + R^{\oplus}H$

② from alkyl halide : $R-Cl \xrightarrow[\text{alco. KOH}]{}$ Alkene.

$\alpha \rightarrow X = Cl, Br...$ $\beta \rightarrow H$

$CH_3-\overset{\beta}{CH_2}-\overset{\alpha}{CH_2}-Cl \xrightarrow[KOH]{alco.} CH_3-\overset{\ominus}{C}H-CH_2-Cl \rightarrow CH_3CH_2=CH_2 + HCl + ROH$

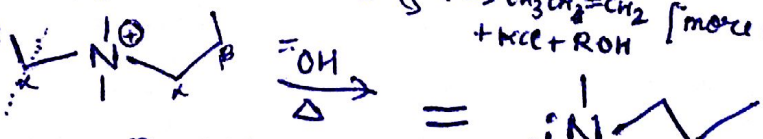
Ex:



major + H_2SO_4 (more HC)

[more HC about =]

③ from quadnery amine : [opp. of previous]



{ $\mu_{cis} > \mu_{trans}$; $(BP)_{cis} > (BP)_{trans}$; $(MP)_{cis} < (MP)_{trans}$

→ Chem. prop : ① Hydration of alkene –

② Addition of hydrogen halide –

[X → Br, Cl, I]

$CH_3 \rightarrow CH=CH_2 \xrightarrow[\text{Markco. Nico. Rule.}]{conc HCl}$ $CH_3-\underset{Cl}{CH}-CH_3$

[Anti Markco in presence of H_2O_2 / HBr

I-ue → more no of C

less no of C.

③ Ozonolysis (H_2O_2 oxidises – CH=O to –COOH)

Reductive (Zn dust) oxidative (Zn Ab)

$CH_3-CH=CH-CH_3 \xrightarrow[H_2O/CCl_4]{O_3/Zn}$ $CH_3-CHO + CH_2O$ or HCHO

[IF Zn Ab, CHO → COOH]

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⇒ ALKANE • Hydrogenation of alkene alkyne. (presence of Pt, Pd or Ni, solvent, press.)

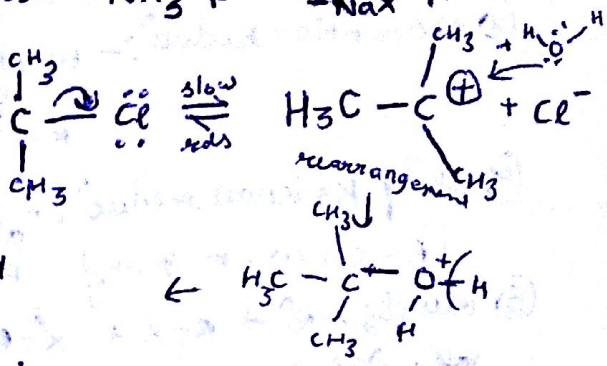
• Reduction of alkyl halide $R-X \xrightarrow[-ZnX_2]{Zn/HX} R-H$ [Zn → good reducing agent]

• Alkylation of terminal alkynes [R-X must be methyl or unbranched at β-carbon].

alkyne $\xrightarrow[-NH_3]{NaNH_2}$ $R-C\equiv C^-Na^+ \xrightarrow[-NaX]{R'-X}$ $R-C\equiv C-R'$

• $Nu:^- + R-\ddot{X}: \longrightarrow R-Nu + X^-$

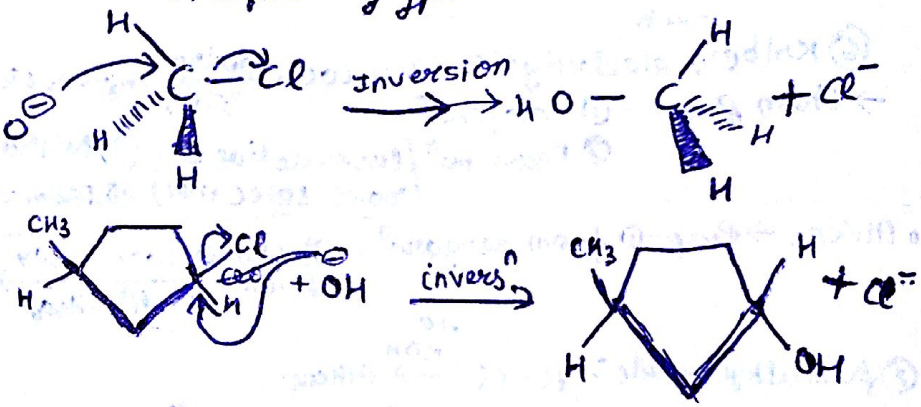
→ S_N1 [Substitution, nucleophilic, unimolecular] 3 > 2 > 1



→ S_N2 – $Nu^\ominus$ attacks from back side or opp of leaving gp.

↳ methyl > 1° > 2° > 3°

Reactivity ∝ $\frac{1}{\text{steric hindrance}}$



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s - Block → configuraⁿ = [Noble gas] ns¹

- In gp ① H & Fr are possessing diff. property than other alkali metals (Fr - radioactive)
- Gp ① is known as alkali metals → as aq. solⁿ of metal is alkaline in nature
- Highly explosive when brought in contact with H_2O. [Stored in → Li - wax & paraffin, other in Kerosene]

$t^{1/2}$ = 21 min

Properties –

- Atomic or ionic radius → ↑across gp & ↓across period [across period (Radius)max]
- Ionisaⁿ energy → energy req. to remove e⁻ from isolated gaseous state. of noble gas

Period → Li < Be < B < C < N < O < F

IE ∝ 1/R ⇒ Li < B < C < Be < O < F < N

[fully & half filled orbital are more stable]

- in period (IE)max = Noble gas, Ex.) IE of Na → $(IE)_2 > (IE)_3 > (IE)_1$
- Electronegativity ∝ 1/R ∴ gp → decrease; period → Increase
- electropositivity ∝ R • across gp. → increases Li < Na < K < Rb < Cs.
- Electron affinity or e⁻ gain enthalpy : it is the energy released or absorbed when an gaseous atom accepts one e⁻. ΔH < 0 exother, ΔH > 0 endoth.

EA ∝ 1/R ← Li Be B C N O F

* EA = F > O > C > B > N > Li > Be & Noble gas, IST EA = 0

≈ Cl > S > Si > Al > P > Na > Mg

↓ Lowest EA in period. ⇒ Li < Na > K < Rb < Cs

- Density = m/V as d ↑, M ↑ faster than V [Na > K as m/V ratio ↓]
- Metallic character – It is the tendency to loose its valence e⁻ & form cation ↑ down the gp. Li < Na < K < Rb < Cs
- Oxidaⁿ state → alkali metals oxidaⁿ state = +1 (always)
- Melting / Boiling pt. = as metallic strength ↓ down the gp. MP / BP ↓ [force of attracⁿ]
- Photo electric effect → Intensity ∝ no. of photon & frequency ∝ KE

Caesium is mostly used in photo electric cells but lithium is never used.

Photo electric effect ∝ tendency to loose e⁻ ↑ in gp. [Lustre also ↑ in gp]

- Electrical & thermal conductivity : ↑ down the gp.
- Malleability & ductility – All alkali metal are malleable & ducti – le as they are soft [they too have lattice structure [gap]]
- Flame colourisaⁿ – when e⁻ excited return to ground state & emit light of visible wavelength [balmer series]

Li - Crimson red; Na - yellow; K - violet; Rb - Red violet; Cs - Blue; Fr - yellow

- Hydraⁿ enthalpy ⇒ HE ∝ Polarising Power [P.P = charge/size or ability to pull] & HE ↓ in gp. [or solvaⁿ energy]

* Li > Na > K > Rb > Cs Group → PP ↓ HE ↓ in gp.

↳ more H_2O surrounded due to high P.P.

In gaseous state, size ↑ down gp but in Aq. solⁿ size ↓ down gp. [Aq → Li > Na > K > Rb > Cs]

however conductivity remains same, [small size faster movement]

- Reducing Prop. – oxidises itself & reduces other [Reducing agent] ⇒ * Li > Cs > Rb > K > Na

* Li is the most powerful reducing agent in periodic table due to more hydraⁿ energy

- Solubility – Alkali metals & its salt (except Li) are ionic hence soluble in H_2O, or polar solvent.

Polar solvent → high dielectric constant (ε) (ability to separate charges).

→ more ε, more soluble

→ ε_0 = H_2O (80) > NH_3 (33) > Diethanol > ethanol (26)

Organic comp.'s solubility ⇒ LE ∝ 1/solubility; HE ∝ solubility; LE > HE not soluble, LE < HE soluble

- Fazan's rule – give informaⁿ of covalent compounds. [Polarisaⁿ ↑ covalent character ↑ Solubility ↓]

factors : Size of cation ∝ 1/cc & Size of anion ∝ C·C (C·C)

: charge of anion/cation ∝ C·C & Pseudo e⁻ configuraⁿ [like Noble gas] → $[Cu^+ = [Ar] 4s^0 3d^{10}$

BCC - body centered cubic [1 in centre also]
no of atom = 1 + 8 × 1/8 = 2

simple cubic
atoms = 1/8 × 8.

Face centered cubic (FCC)
atom = 1/8 × 8 + 6 × 1/2 = 4
(vertex, face)

HCP/HCC
hexagonal close Packing or cubic.
atom = 2 × 1/2 + 3 + 12 × 1/6 = 6

* All alkali metal = BCC struc but at Low temp → Lithium have HCC struc.

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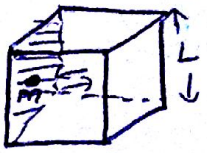
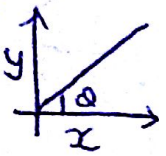
GASEOUS STATE

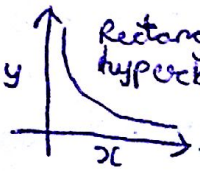


- Force $= \frac{mv^2}{L}$ & Pressure $= \frac{F}{A} = \frac{mv^2}{L^3}$
- Pressure $= \lambda \times h \times g$ (where λ = density of liquid) [1 J = 10^7 ergs]
- $PV = nRT$ → P in atm, V in lts, then $R = 0.0821$ lts atm/mol-K
  R ≈ 2 cal K^{-1} mol^{-1} (SI) → P in pascal, V in m^3 then $R = 8.314$ J/mol-K
- $PM_{gas} = d_{gas}RT$ where d = density & M is molar mass.
- $(\text{Molar mass})_{\text{mixture}} = x_1 \times M_1 + x_2 \times M_2$ [x → mole frac^n, M → molar mass]
- 1 atm = 1.015×10^5 Pa (pascal) $\overset{760 \text{ mm of Hg}}{\Rightarrow}$ 10 m of H_2O • $1 m^3 = 10^3$ L.
- Press. due to 'h' m H_2O = $S h g = 10^3 \times 10 \times h$ pa $= \left(\frac{h}{10}\right)$ atm.
- Boyle's law ⇒ $P \propto \frac{1}{V}$ • Avagadro's law = $V \propto n$
- Charles law ⇒ $V \propto T$ • Gay Lussac's law = $P \propto T$

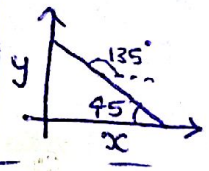
- $y \propto x$



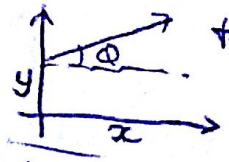
- $y \propto \frac{1}{x}$



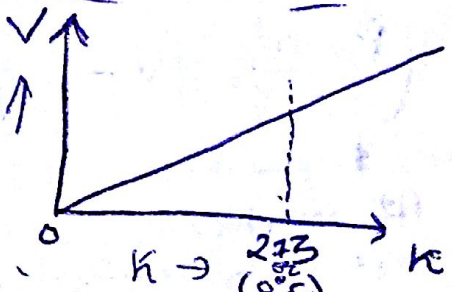
- $x + y = \text{const.}$



- $y = mx + c$



- $\frac{V_1}{V_2} = \frac{t_1 + 273}{t_2 + 273}$; $V_t = V_0 + \frac{V_0}{273} \times t$
  (y) (x) (m) (x)



- DALTON'S LAW OF PARTIAL PRESSURE → $P = P_1 + P_2 + \dots + P_n$
- $P_1 = x_1 \times P_T$ & $P_2 = x_2 \times P_T$
- $\frac{P_1}{P_T} = x_1$ & $\frac{P_1}{P_2} = \frac{n_1}{n_2}$

$PV = \frac{2}{3} KE$

- Press. exerted by liq. at eqb → Vapour press $\xrightarrow{\text{It is independent of}}$ amount of liq. Vol. of liq. Shape or SA of vessel
  ($T \uparrow \Rightarrow VP \uparrow$; ∈ Temp. $\xleftarrow{\text{depends on}}$
- V.P of H_2O is called aqueous tension.
- GRAHAM'S LAW ⇒ Rate of migra^n ∝ press diff. ∝ $\frac{1}{\sqrt{m}}$ [m = molar mass of gas]

$$\Rightarrow \frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{d_2}{d_1}}$$

$r \propto \frac{1}{\sqrt{d}}$; d → density

⇒ Rate of migra^n depends upon — * Press. diff.
* Temp. condi^n
* area & shape of orifice
* Nature of gas

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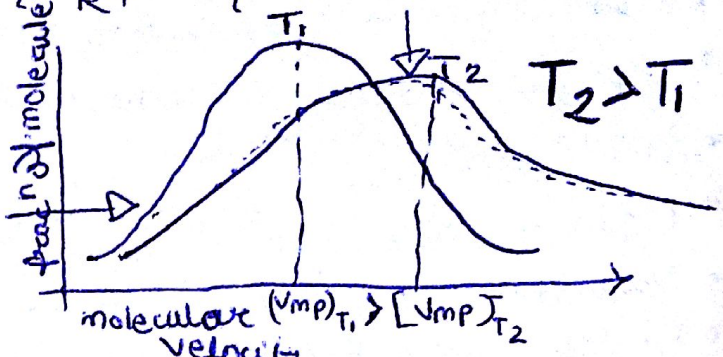
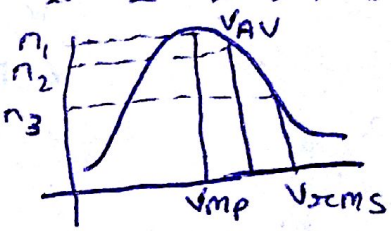
- Rate of diffusion / effusion $= \frac{\text{amount of gas migrated}}{\text{time taken}} = \frac{\text{no of moles effused}}{\text{time taken}}$.
- Avg. K.E of gas $= \frac{3}{2}RT$
- Vapour density $= \frac{\text{density of gas A}}{\text{density of } H_2} = \frac{\left\{\frac{P \times M_{gas}}{RT}\right\}}{\left\{\frac{P \times M_{H_2}}{RT}\right\}} = \frac{M_{gas}}{2}$ [Unitless]
- On ↑ temp, V_{mp} ↑
  At low temp. less molecule have velocity close to V_{mp}.
  At high temp. more molecule's velocity approaches toward V_{mp}.



- For a gas at certain temp. ⇒

$$V_{most\ probable} = \frac{V_{mp}}{\sqrt{\frac{2RT}{M}}} < V_{Average} = \frac{V_{AV}}{\sqrt{\frac{8RT}{\pi M}}} < V_{root\ mean\ square} = \frac{V_{rms}}{\sqrt{\frac{3RT}{M}}}$$

$$n_1 > n_2 > n_3$$

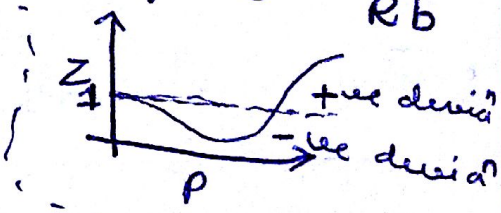


- Compressibility factor $(Z) = \frac{P_{obs} \times V_{obs}}{nRT}$
  - $Z > 1$ (+ve deviation) Less compressibility ⇒ $V > 22.4\,l$ [Repuls^n force dominates]
  - $Z < 1$ (−ve deviation) More compressibility ⇒ $V < 22.4\,l$ [attrac^n force dominates]
- $(P_{obs} + P_{correction})(V_{obs.} - V_{molecule}) = nRT$

$$\Rightarrow \left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

, a = Vander waal's const.
b = effective vol. occupied by 1 mole molecule $= 4 \times \left(\frac{4}{3}\pi r^3\right) \times N_A$

- H_2 & He → show +ve deviation [$a \to 0$]
- At very high or very low temp → real gas behaves ideally
- Boyle's temp → temp. above which real gas behaves ideally ⇒ $T_b = \frac{a}{Rb}$
- Critical state ⇒ $(\text{density})_{liq.state} = (\text{density})_{gaseous\ state}$



- Value of Z in critical state ⇒

$$Z = \frac{P_c \times V_c}{n \times R \times T_c} = \frac{\left(\frac{a}{27b^2}\right) \times 3b}{1 \times R \times \left(\frac{8a}{27Rb}\right)} = \frac{3}{8}$$

$$\boxed{Z = \frac{3}{8}}$$

$$(f')^x = \frac{n_1'/n_2'}{n_1/n_2} = f \quad \Big| \quad x = \frac{2\log f}{\log\left(\frac{m_2}{m_1}\right)}$$

- H-Bond > dipole–dipole > Vander waal's force
- Vapour density & degree of dissociation ⇒ $\alpha = \frac{(V \cdot D)_{th} - (V \cdot D)_{obs}}{(n-1)\,V \cdot D_{obs}}$

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(5)

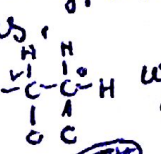
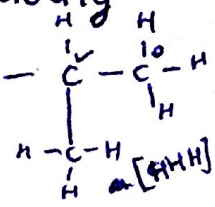
- **Chirality** : object whose mirror image are not superimposable.
- **Constitutional isomers** : same molecular formula but diff. connectivity.
- **Stereoisomers** : same connectivity but diff. orientation in space.
- **enantiomer** – non superposable mirror image; [chiral molecule] (no pos).
- **Diastereomers** – whose molecule are not mirror image of each other. [as $ trans → example]

meso – having plane of sym.

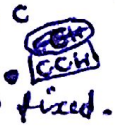
B B x not mirror image.

→ chiral molecule $ its mirror image → pair of enantiomer.

↳ a molecule with tetrahedral atom $ 4 diff. gps. attached to it.

- **Stereogenic carbon** – a carbon bearing gps of such nature that an interchange of any 2 gps will produce stereoisomers.
- All molecules with plane of symmetry are achiral.
- R $ S (Rectus $ sinister) → ① gps are given priority accor. to molecular wts of linked element
  ↓ CW ↓ ACW
  ② If same element are linked then, 2nd link is seen.
  ③ Rotate the formula so that gp with low priority is directed away from us.
  ④ $-CH=CH_2$ is treated as if it was  with high. priority. 

- **Identical comp.** :– ① fix one element $ rotate other.
  ② interchange nearby 2 element keeping other fixed.



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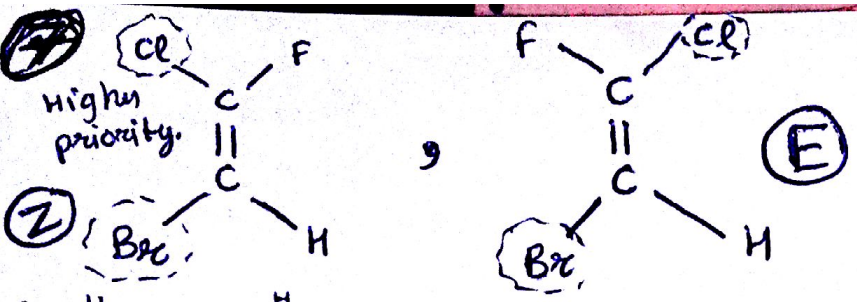
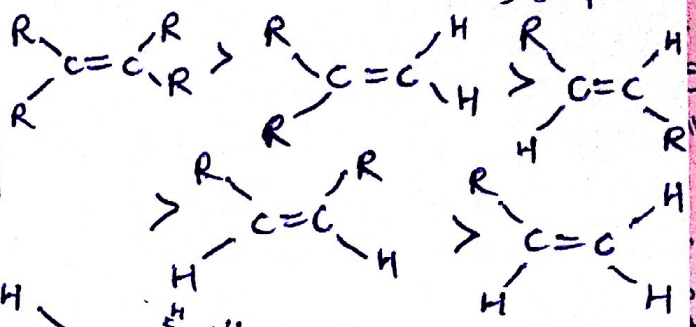
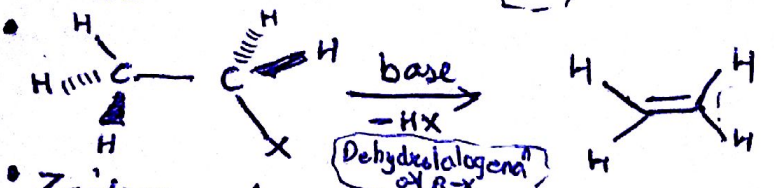
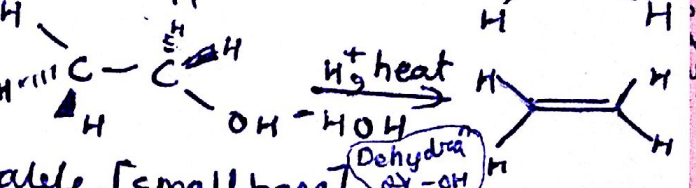


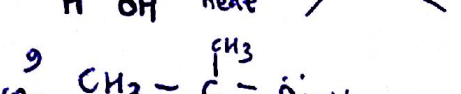
Relative stability of Alkene → olefins







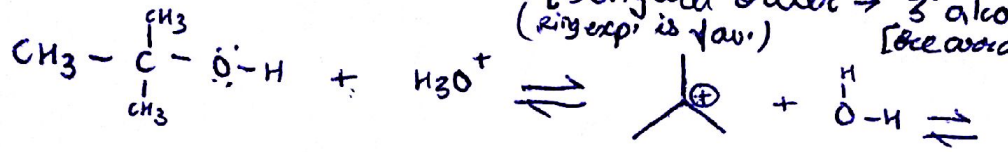
- Zaitsev rule: more sub. to = more stable. [small base]
- Hofmann rule: less sub. to = more stable [st. or bulky base]



Leaving gp is excellent as halide

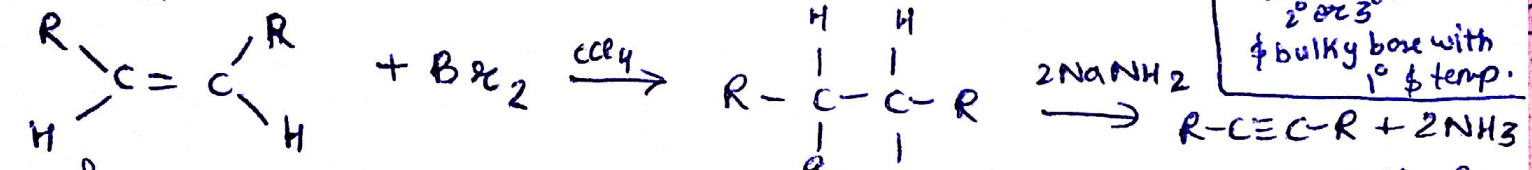
[Dehydration order → 3° alcohol > 2° alco. > 1° alcohol]
(Ring exp. is fav.) [Decarrang. of carbocation).

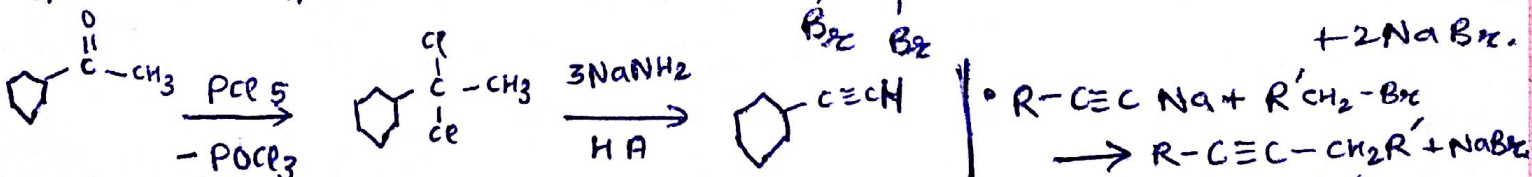
⇒ E_1 3° 2° 3° alco.



⇒ E_2 1° alco

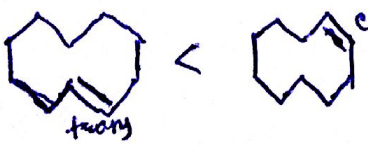
To favour E_2 :- 2° or 3° & bulky base with 1° & temp.





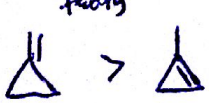
- syn addⁿ (hydrogenation): Lindlar's catalyst (H_2/Pd/$CaCO_3$, Ni_2B) (OsO_4) (quinoline) Lead acetate
- anti addⁿ: Li or Na & $C_2H_5NH_2$ −78°C & NH_4Cl, NH_3 or RNH_2.
- Degree of unsaturation (DU = x) or Index of H-deficiency → CHO $\xrightarrow{-O}$ (New form) $\xrightarrow{+x H_2}$ C_nH_{2n+2}
  CHN $\xrightarrow{-NH}$ new form, $\xrightarrow{+x H_2}$ C_nH_{2n+2}
  CHX $\xrightarrow[+H]{-X}$ new form $\xrightarrow{+x H_2}$ C_nH_{2n+2}

DU = 0, no multiple bond or ring
DU = 1, 1 = or 1 ring
DU = 2, 1 ≡ or 2 = or 1 = , 1 ring or 2 ring

- Effect's order: AEMHI
- 

cis > trans when ring is of less than 11 C

for D = 4, & C_6 or > C_6 try to introduce benzene ring.

-  ring strain

HC → stability of: alkene, carbocation, free radical or phenyl.
↳ & no of α-H.

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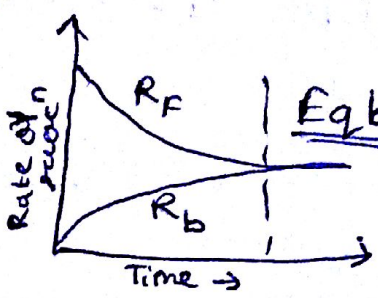
EQUILIBRIUM

- Condi^n at which, Rate of forward Rk^n = Rate of backward Rk^n.
- Law of mass action, $aA + bB \rightarrow$ product.
  Rate of rck^n $\propto [A]^a[B]^b \Rightarrow$ Rate $= K[A]^a[B]^b$

K → Rate constant [Depend only on temp] $\xrightarrow{\text{K}\uparrow \text{ on } \uparrow \text{Temp.}}$ $\xrightarrow{\text{K}\uparrow \text{ or } \downarrow \text{ on adding catalyst}}$ $\$$ catalyst



- $K_{eq} = \frac{K_f}{K_b} = \frac{[C]^c \times [D]^d}{[A]^a \times [B]^b}$ [for $aA + bB \rightleftharpoons cC + dD$]
- Eqb constant : Independent of → [] of RKt/Pdt → Pressure → Vol. of container → catalyst [↑ or ↓ by same rate]

PV = n RT
P = [] RT
P ∝ []
: Depend on Temp → forward exothermic $\$$ Bwd endo. ⇒ T↑ Keq↓ | T↓ Keq↑
→ fwd endo. $\$$ Bwd exother. ⇒ T↑ Keq↑ | T↓ Keq↓

- $K_c = \frac{[C]^c_{eq} \times [D]^d_{eq}}{[A]^a_{eq} \times [B]^b_{eq}} \Rightarrow K_p = \frac{(P_C)^c_{eq} \times (P_D)^d_{eq}}{(P_A)^a_{eq} \times (P_B)^b_{eq}}$ $\$$ $K_p = K_c (RT)^{\Delta ng}$

[Δng = n_{pdt} (ingaseous state) − n_{rct} (ingaseous state)]

- $A \rightleftharpoons B$ Keq. then $B \rightleftharpoons A$ $\left(\frac{1}{K_{eq}}\right)$
  $\frac{1}{2}A \rightleftharpoons \frac{1}{2}B$ $(K_{eq})^{1/2}$

$A \rightleftharpoons B \rightarrow a$
$B \rightleftharpoons C \rightarrow b$ (Keqa $\&$ b)
$A \rightleftharpoons C$ (a × b)

$A \rightleftharpoons B$ (a)
$A \rightleftharpoons C$ (b)
$B \rightleftharpoons C$ (b/a)

$R_b < R_f$ ← $R_f = R_b$ → $R_f < R_b$

Q = Reac^n quotient. $Q < K_{eq}$ (fwd rc^n) | $Q = K_{eq}$ At eqb. | $K_{eq} < Q$ (backward rc^n)

- [] of solid $\$$ Liquid is taken to be 1 (unity). No unit.
- degree of dissocia^n $\Rightarrow \frac{\text{Moles dissociated}}{\text{Initial moles}} \times 100 = \alpha$.
- $\alpha = \frac{(V.D)_{th} - (V.D)_{obs.}}{(n-1)(V.D)_{obs}}$ [$(V.D)_{th}$ = Vapour density theoritical; $(V.D)_{obs}$ = Vapour density observed.]
- DISTURBING EQb (Le Chatlier's principle) : If we disturb eqb then the reversible reac^n behave in such a manner, that it nullifies the disturbing factor or its effect.

① Effect of change of [Rkt]/[Pdt] :- [Rkt]↑ or [Pdt]↓ ($R_f > R_b$)
- [Rkt]↓ or [Pdt]↑ ($R_f < R_b$)

→ Addi^n or removal of solid $\$$ lqd has no effect.

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② Effect of heat ⇒ Addⁿ of heat (↑ in temp.) → Endothermic favoured
Removal of heat (↓ in temp.) → Exothermic favoured

③ Effect of press. ⇒ ↑ in press. → favour side producing less no of mole
↓ in press → favour to produce more no of mole.

④ Addⁿ of Inert gas ⇒ At const. vol. = no effect
at const press ⇒ addⁿ lead to side where more no of moles are produ

- $\Delta G_T^0 = -RT \ln K_{eq.} = -2.303 \cdot RT \log K_{eq.}$
- $\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$ | $\frac{d(\ln K)}{dt} = \frac{\Delta H^\circ}{RT^2}$
- $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$
- $\Delta H^\circ = -RT \ln K$

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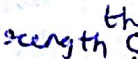
IONIC EQUILIBRIUM

- Water → $\mu \neq 0$ → Polar molecule → Liquid at room temp. → act as polar solvent
- Polar Solute [NaCl & NaOH [Ionic compound]
  HCl, CH_3COOH, CH_3COOH, NH_3, sugar, glucose [covalent compounds]
- ARRHENIUS CONCEPT [concept confined to aq. soln only.] → Acid – which produce H_3O^+ on adding to H_2O.
  → Base – which produce ^-OH upon added to H_2O.
- Bronsted Lowry concept → Acid – donate H^+ [All Arrhenius base are bronsted base but vice versa is not true.]
  Base – accepts H^+
  $NH_3 + HCl \rightarrow NH_4^+ + Cl^-$
- LEWIS CONCEPT → Acid – lone pair acceptor [e^- deficient] – electrophile
  → Base – lone pair donor [e^- surplus] – nucleophile.
- All bronsted acid are not lewis acid [H_2SO_4]
- Acid $\xrightarrow[+^-OH]{-H^+}$ conjugate Base similar Base. Base $\xrightarrow[+H^+]{-^-OH}$ conj. Acid.
- K_a (Acid) × K_b (conjugate base) = K_w [Strong acid $\xrightarrow{\text{conjugate}}$ weak base; SB → WA / WA → SB / WB → SA]
- if $K_a(A) > K_a(B)$ then $K_b(A) < K_b(B)$. [K_w → Ionic pdt. of H_2O]
- $K_w = [H^+][^-OH] \Rightarrow [H_3O^+] = [^-OH] = \sqrt{K_w}$ [T ↑ PH ↓ & T ↓ PH ↑]
- Strong acid → HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_3$, $HClO_4$. $\sqrt{K_w} = 10^{-7}$ at 25°C
- Strong base → All gp ① hydroxide & $Ca(OH)_2$ & $Sr(OH)_2$. & $Ba(OH)_2$
- Strong electrolyte → salt, St. acid & base. • Weak electrolyte → weak acid & base.
- Practically, Ionic comp's salt whose solubility is very very less [sparingly soluble salt]
- Ksp → solubility pdt.
- At given temp., degree of dissocia^n or ionisa^n of H_2O ⇒ $\alpha = \frac{\sqrt{K_w}}{55.5}$ [pure H_2O]
- $PH = -\log_{10}[H_3O^+]$
- At const. temp, upon dilu^n of aq. soln of weak acid, its degree of dissocia^n (α) increases. [Oswald's dilu^n law] $K_a = \frac{C\alpha^2}{1-\alpha} = \text{const.}$ [$1-\alpha \sim 1$]
  $K_a = C\alpha^2$ or $\alpha = \sqrt{\frac{K_a}{C}}$ ($\alpha \propto \sqrt{V}$)
- $RSO_3H > RCOOH > H_2CO_3 >$ $> H_2O > ROH >$  $> R-C\equiv C-H > H_2 > NH_3$
  acidic strength (then solve by quadratic eqn; if not $\alpha < 10^{-3}$)
- $RSO_3^- < RCOO^- < HCO_3^- <$ $< ^-OH < RO^-$ ... Conjugate basic strength.
- Left → Right Acidic strength increases. & Top to bottom
- The acid base neutralisa^n rek^n eqb proceed in the direc^n producing weaker acid/base
- When solvent reduces the acidic/basic strength of strong acid HA & HB to level the acidic/basic strength of H_3O^+ ⇒ Levelling effect of solvent.
- Hydrolysis of a salt ⟶ Salt + $H_2O \rightleftharpoons$ acid + base [h → degree of hydrolysis]
- $PH = \frac{PKa_1 + PKa_2}{2}$ [for small conc.]
- (WA & SB) → PH > 7 ⇒ $h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{K_w}{C \times K_a(WA)}}$ • Salt (WA & WB) ⇒ PH > 7, PH < 7, PH = 7
  $K_h = \frac{K_w}{K_a(WA) \times K_b(WB)}$
- Salt (SA & WB) (PH < 7) $K_h = \frac{K_w}{K_b(WB)}$ ⇒ $h = \sqrt{\frac{K_h}{C}}$ ⇒ $h = \sqrt{\frac{K_w}{C K_b(WB)}}$ ⇒ $PH = 7 + \frac{1}{2}[pK_a - pK_b]$

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• when both solute are weak acid/weak base, Ionisaⁿ of both weak acid will ↓ & $[H_3O^+] = \sqrt{(Ka_1 \times C_1) + (Ka_2 \times C_2)}$

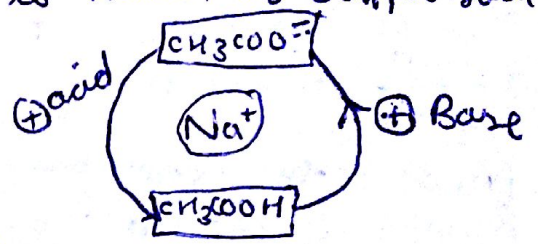
• for weak acid & its salt with st. base, $PH = PKa(\text{weak acid}) + \log \frac{[\text{salt}]}{[\text{acid}]_{\alpha \to 0}}$
(CH_3COOH) (CH_3COONa) CH_3COOH $[CH_3COO^-]$ salt $[CH_3COOH]_{\alpha \to 0}$
[100% dissociaⁿ of salt]

• for weak base & salt with st. acid → $POH = PKb + \log \frac{[\text{salt}]}{[\text{base}]_{\alpha \to 0}}$
(NH_3) (NH_4Cl) $[NH_4^+]$ salt. $[NH_3]_{\alpha \to 0}$

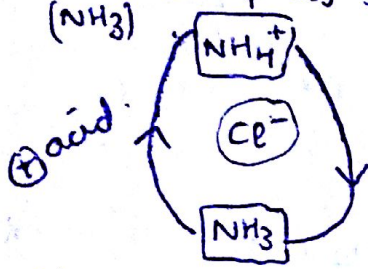
* COMMON ION EFFECT – The ionisaⁿ of weak electrolyte gets suppressed in presence of st. electrolyte with some common ion.
$\left[\frac{(\text{Common ion})_{\text{st. electrolyte}}}{(\text{weak electrolyte})_{\alpha \to 0}}\right]$

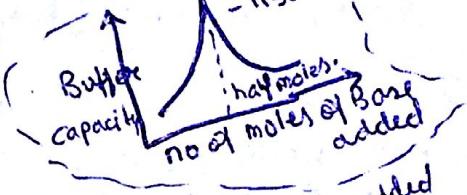
• Soluⁿ which resist change in PH or $[H^+]/[OH^-]$ is known as Buffer soluⁿ.

⇒ Weak acid & its salt with st. base → (CH_3COOH & CH_3COONa)



⇒ weak base & its salt with st. acid → (NH_3 & NH_4Cl)





• Buffer Capacity = No. of moles of H^+ & ^-OH added in 1ℓ Buffer soluⁿ to change its PH by 1.

= $\frac{\text{no. of moles of } H^+/^-OH \text{ in 1ℓ Buffer soluⁿ}}{\text{change in PH value}}$

equivalent pt. for

for maxᵐ buffer acⁿ, $PH = PKa(\underline{WA})^{CH_3COOH}$ & $PH = 14 - PKb(\underline{NH_3})$

• Sparingly soluble salt → Ionic compound's salt whose solubility is very less & makes ppt. easily in the soluⁿ,
• [AgCl, $PbCl_2$, $Al(OH)_3$, ZnS, CuS, $CaCO_3$, $SrCO_3$, $Fe(OH)_3$]
An eqb is established b/w ppt. salt & ion. +ⁿᵗ in soluⁿ.

• Binary salt – AB type.
• Ternary salt – AB_2 / A_2B type.
• Quaternary salt – AB_3 / A_3B type.

A_xB_y (ppt) ⇌ $x A^{y+} + y B^{x-}$

$K_{sp} = [A^{y+}]^x \times [B^{x-}]^y$.

$S = \left(\frac{K_{sp}}{x^x y^y}\right)^{\frac{1}{x+y}}$

⇒ Mass acⁿ ratio / reacⁿ quotient – Ionic pdt of sparingly soluble salt, K_{ip}.

• ① $K_{ip} < K_{sp}$ → Unsaturated
② $K_{ip} = K_{sp}$ → Saturated
③ $K_{ip} > K_{sp}$ → Super saturated [ppt occurs]

Solubility of AgCl ↑ with ↑ in NH_3.

SA + SB ⇒ 4–10 (all indicat - or can be used)
WA + SB ⇒ 7.5–10.5 Phenophthalen
WA + WB ⇒ 6.5–7.5
SA + WB ⇒ 2.5–5.5 methyl orange are used

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General Organic Chemistry

- Bond fission → Homolytic : equidistribu^n of e$^-$ [via free radical] → highly reactive & Paramagnetic
  → free radical does not participate in hybridisa^n. (is in pure 'p' orbital.)
- → Hetrolytic fission : opp. charges are formed:

electrophile	nucleophile
↓	↓
e$^-$ deficient ; e$^-$ loving.	e$^-$ rich ; nucleus loving
+ve charge	-ve charge.

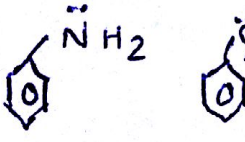
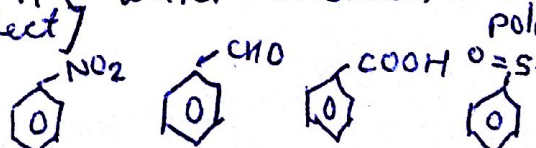
Note → electrophile have vacant orbitals [on making e$^-$ configura^n]
→ All e$^-$ phile are lewis acid.
→ Nucleophile [availability of lone pair
[paired e$^-$ in configura^n or energy level diag

→ CH_3^+ : sp^2 (6e$^-$) vacant 'p' orbital ⊥ to plane
CH_3^- : sp^3 (8e$^-$) no pure 'p' orbital.

- Inductive effect : the transfer of partial sigma e$^-$ which induces charge due to polarity [Permanent effect]
  [↓ with increase in distb/wthem]
  +I (all e$^-$ donating / pushing gps) [If 'p' orbital is +nt, H.C is also +nt, also side CH bond.]
  -I (all e$^-$ withdrawing gps) → $-NO_2$, $-CN=O$, $-C\equiv N$, $-F$, $-COOH$, $-Cl$ etc ; = ; ≡
- → stability of carbocation — (+charge) +I is favoured. ∴ -> => ≡ [s charac ↑ e$^-$-tivity ↑]
  stability order : > 3° > 2° > 1° (or by Hyper conjuga^n 's counting. HC ∝ stability.
- → stability of free radical (through hyper conjuga^n) → Resonance is more stable over HC.
  → alternate - $ = bond in resonance - more stable
  → free radical on = bond — unstable.
- → Stability of anion ($H^- > Phenyl^- > R^-$) → (H·C is not toward anion/-ve charge)
  $1° > 2° \geq 3°$ -I is favoured. ⊙ -I of NO_2 is more than -I of Cl or F
- * Acidic strength → -I favoured [toward func^{n}al gp]
  * Basic strength → +I favoured. [toward lone pairs]
- Resonance — alternate — $ = bonds
  - cationic
  - anionic
  - Lone pair
  - free radical

  more stable
  - all 'p' orbitals lie in same plane for conjuga^n.
  - Linear conj. — resonance from one side — effective reso.
  - Cross conj. — resonance from both side
  - When Lone pair participate in resonance it does not participate in hybridisa^n. [L.P. is ⊥ to plane]
  - Species without charge is more stable.

more resonance → more stable

- Mesomeric effect — complete transfer of π e$^-$ which induces charge or polarity.
  +M :  [Permanent effect]
  -M : 
  → e$^-$ deficient (acidic nature)
- Hyper conjuga^n — σ, π delocalisa^n [there must be CH bond at ∝ carbon]
  HC ∝ stability.
- Stability of cation → +M > H·C > +I [Halogens → -I dominating]

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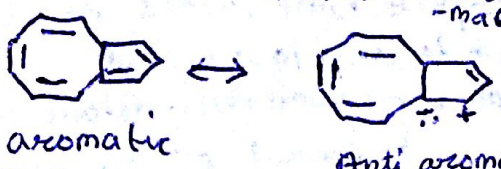
• Aromatic compounds (possess small) → condi^n of aromaticity – must be cyclic
– conj./resonance
– molecule must be planar
– sp^2 hybridised C.

HUCKLE RULE → $(4n+2)\pi e^-$ [n-whole no.]

→ antiaromatic → sp^2 → $(4n)\pi e^-$ while other non aromatic (sp^3 hybridi^n)

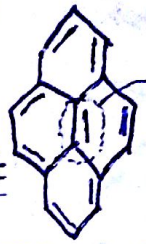
stability : Aromatic > non aromatic > antiaromatic

• Azulene ⇒



aromatic ⟷ Anti aromatic

• Pyrine



Aromatic

πe^- does not participate in conjugan.

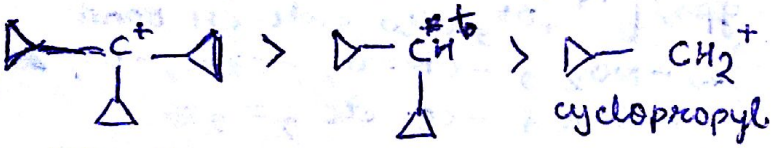
• 10-Azulene

non aromatic

due to H-H repuls^n orbital shift from its plane

Anti aromatic → tub shape Non aromatic

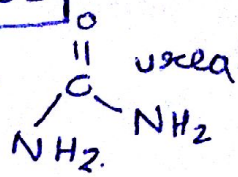
• Due to σ route conjugan –



$> CH_2^+$ cyclopropyl > Aromatic compounds >

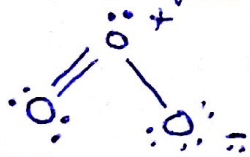
• Bond length –

Carbon Compounds & chemical bonds

- Friedrich Wohler (1828) — urea ⇒ $NH_4^+NCO^- \xrightarrow{\Delta}$  urea.
  ammonium cyanate.
- BP: ethyl alcohol > dimethyl ether.
  (MP & 1st H-bond; isomers)
- Ionic substance — high electrostatic force — more BP & MP.
- PCl_5, SF_6 → exceptional of octate rule.
  excess e^-
- BF_3 → Boron atom - only 6 e^-. [Incomp. octate]
- Formal charge = Free valency − no. of bond − unshared e^-
- In resonance, all = & - participating in resonance
  possess equal length.
  ↓ we are moving e^- for diff. st. not atom.
- Energy of actual comp. is lower than any structure.
- L.P can be moved same as BP in forming resi. st.
- Stability of resonatical structure —
  ① octate for 2nd row element (C,N,O,F)
  ② min formal charge & more bonds.
  ③ −ve charge on é-ve atom (O > N > C)
  ④ min charge sepn while keeping formal charge close together.

→ Polarised struc have higher energy [sep. of opp. charge req. energy]
→ More stable — possess less energy

Ozone

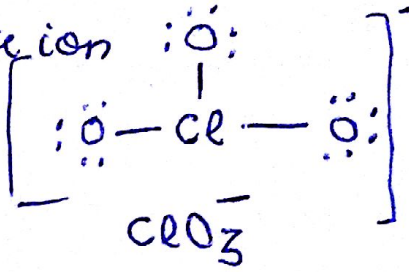


- Greater no. of node greater energy.
- Restricted rotaⁿ in C=C.
- Stereoisomers differ only in arrangement of their atom in space.
- (BL) C≡C < C=C < C−C

sp	sp^2	sp^3
50% s	33% s	25% s

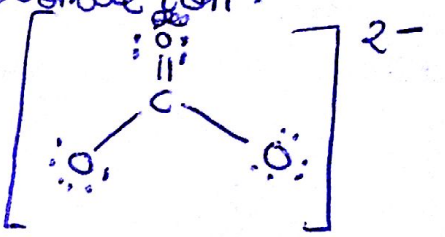
$\frac{1}{BO} \propto BL \propto \frac{1}{\text{s character}}$ ✓

- Chlorate ion

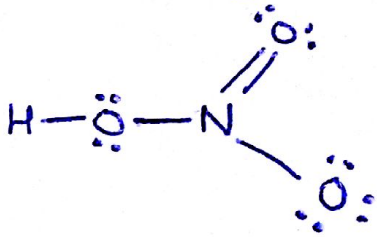


ClO_3^-

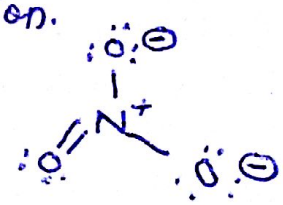
- Carbonate ion.



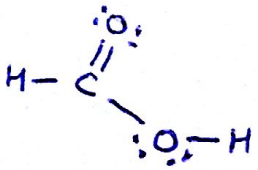
- HNO_3



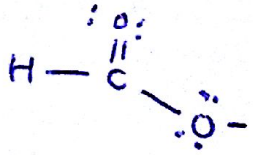
- Nitrate ion.



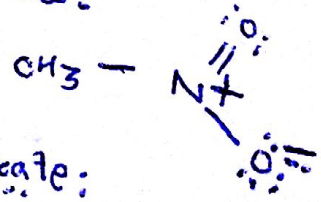
- Formic acid:



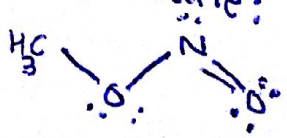
- Formate ion: anion



- Nitromethane:



- Methyl nitrate:



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(8)

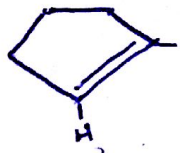
- Alkene – characteristic rekn : addin rekn
- Addin of H–X to Alkene → $>C=C< + H-X \rightarrow -\overset{H}{\underset{|}{C}}-\overset{X}{\underset{|}{C}}-$ accor. to markovnikov's rule. HI > HBr > HCl > HF adding silica gel or alumina ↑ rate
- Regioselective – when a rekn can potentially yield 2 or more isomers actually produces predominance of one. Ex markovnikov.
- Anti markovnikov → only with HBr in ↑ce of R–OO–R (peroxide)
- Addin of H_2SO_4 – (yellow fuming) $>C=C< + H-O-\overset{O}{\underset{O}{S}}-O-H$ (cold) $\rightarrow -\overset{HO_3SO}{\underset{|}{C}}-\overset{H}{\underset{|}{C}}- \xrightarrow[H_2O]{\Delta} -\overset{OH}{\underset{|}{C}}-\overset{H}{\underset{|}{C}}- + H_2SO_4$
- Hydrain – $>C=C< + HOH \xrightarrow{H_3O^+} -\overset{OH}{\underset{|}{C}}-\overset{H}{\underset{|}{C}}-$ ⇒ use dil acid so $[H_2O]$ is high
- Oxymercuration → high yield of alcohol. $>C=C< + H_2O + Hg(O-\overset{O}{\overset{||}{C}}-CH_3)_2 \xrightarrow{THF} -\overset{OH}{\underset{|}{C}}-\overset{Hg-O\overset{O}{\overset{||}{C}}CH_3}{\underset{|}{C}}- + CH_3COOH$
- Demercuration → $-\overset{OH}{\underset{|}{C}}-\overset{Hg-O\overset{O}{\overset{||}{C}}CH_3}{\underset{|}{C}}- + {}^-OH + NaBH_4 \rightarrow -\overset{HO}{\underset{|}{C}}-\overset{H}{\underset{|}{C}}- + Hg + CH_3COO^-$

[CH_3COO^- = acetate ion, –Ac]

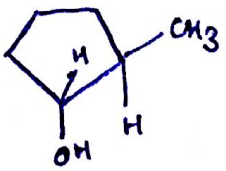
combined → $R-CH=CH-H \xrightarrow[②\ NaBH_4,\ OH^-]{①\ Hg(OAc)_2/THF-H_2O} R-\overset{H}{\underset{OH}{C}}-\overset{H}{\underset{H}{C}}-H$ markovnikov rule

- Anti M.N. syn hydran → hydroborin oxidisan

$3\ CH_3CH=CH_2 \xrightarrow[\text{hydroboran}]{BH_3:THF} (CH_3CH_2CH_2)_3B \xrightarrow{H_2O_2,OH^-} 3\ CH_3CH_2CH_2OH$



$\xrightarrow[H_2O_2,\ OH^-]{BH_3\ THF}$



$>C=C< + H-B$ (boron hydride) $\xrightarrow{\text{hydroboran}} -\overset{H}{\underset{|}{C}}-\overset{B}{\underset{|}{C}}-$

more subs. ↘ ↙ less subs.

$CH_3-CH=CH_2 \xrightarrow{1\ eq} CH_3-\overset{H}{\underset{|}{C}}-\overset{BH_2}{\underset{|}{CH_2}} \xrightarrow[CH_3CH=CH_2]{2^{nd}\ eq} CH_3-\overset{H}{\underset{|}{C}}-\overset{B(CH_2CH_2CH_3)_2}{\underset{|}{CH_2}} \xrightarrow{3^{rd}} CH_3CH_2CH_2-B(CH_2CH_2CH_3)_2$

anti M.N. ; $H + BH_2$

↳ due to steric factor → bulky boron cont. gp. can approach to less subs C atom.

B_2H_6 Diborane + $2:O:$ (THF) $\rightarrow 2\ H-\overset{H}{\underset{|}{B}}-\overset{+}{O}:$ ($BH_3:THF$)

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(9) **Radical RR^n's**

- $R-\ddot{O}:\ddot{O}$ (N) $\xrightarrow[\text{heat}]{h\nu}$ $2\,R-\ddot{O}\cdot$

Dialkyl peroxide — Alkoxyl radicals.

- Br is generally less reactive toward alkane than Cl. but more selective.
- Reac^n is promoted by → heat or light.
- halogena^n : $F^\cdot > Cl^\cdot > Br^\cdot > I^\cdot$

$$CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{H}{|}}{C}}-CH_3 \xrightarrow[\text{light, } 127^\circ C]{Br_2} CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{Br}{|}}{C}}-CH_3 + CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{H}{|}}{C}}-CH_2Br$$

Br (> 99%) — Cl (37%)

Br (trace) — Cl (63%)

$$:\ddot{X}\cdot + H:R \longrightarrow :\ddot{X}:H + R^\cdot$$

Bond forma^n is exothermic process.
→ bond break : ΔH +ve
→ bond form : ΔH −ve

stability of radical comp.: $3^\circ > 2^\circ > 1^\circ$

$R-H + X_2 \longrightarrow R-X + HX$

ⓐ Chain initia^n – crea^n of radical
ⓑ Chain propaga^n – one radical generate another.
ⓒ Chain termina^n – end of radicals.

⇒ Anti markovnikov Addi^n : [of HBr only]

* in +nce of R-O-O-R peroxide.
* non polar solvent are preferable.

① $R-\ddot{O}-\ddot{O}-R \xrightarrow{\text{heat}} 2\,R-\ddot{O}\cdot$

② $R-\ddot{O}\cdot + H:\ddot{Br}: \longrightarrow R-\ddot{O}H + \ddot{Br}\cdot$

③ $\ddot{Br}\cdot + H_2C=CH-CH_3 \rightarrow :\ddot{Br}:CH_2-\overset{\overset{CH_3}{|}}{CH}\cdot$ (2°)

④ $Br-CH_2-\overset{\overset{CH_3}{|}}{CH}\cdot - CH_3 + H:\ddot{Br}: \rightarrow Br-CH_2-CH_2-CH_3 + Br^\cdot$

- $2O_2^-\cdot + 2H^+ \xrightarrow[\text{dismutase}]{\text{superoxide}} H_2O_2 + O_2$.
- $2H_2O_2 \xrightarrow{\text{catalase}} 2H_2O + O_2$.

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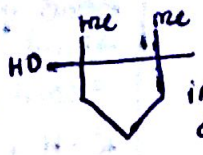
④ Rkⁿ involving free radical : Dimerisaⁿ

① Pinacol formaⁿ → $Me-\overset{O}{\overset{\|}{C}}-Me \xrightarrow[H_2O]{Al(Hg)/Mg(Hg)} Me-\overset{OH}{\overset{|}{C}}-\overset{OH}{\overset{|}{C}}-Me + Mg(OH)_2$ (Me, Me)

* $Me-\overset{O}{\overset{\|}{C}}-(CH_2)_3-\overset{O}{\overset{\|}{C}}-Me \xrightarrow[H_2O]{Mg(Hg)}$  intramolecular dimerisaⁿ

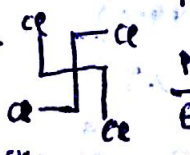
② Wurtz Rkⁿ → $R-X \xrightarrow[\text{dry ether}]{Na} R-R$ + (Alkane + Alkene, traces.) [symmetrical alkane → high yield]

$CH_3-CH_2-Cl \xrightarrow[\text{dry ether}]{Na} CH_3-CH_2-CH_2-CH_3 + CH_3-CH_3 + CH_2=CH_2$ (traces.)

$HO-CH_2-CH_2-Br \xrightarrow[\text{dry eth.}]{Na}$

$H-O-CH_2-CH_2-CH_2-CH_2-Br \xrightarrow[\text{dry eth}]{Na}$ THF

for 3° free radical → Disproportionaⁿ → $Me_3C-Cl \xrightarrow[\text{dry eth.}]{Na} Me_3CH + CH_2=CMe_2$

③ Intramolecular wurtz rkⁿ →  $\xrightarrow[Et_2O]{Na}$

④ + $CH_3-Cl \xrightarrow[\text{dry eth.}]{Na}$ + H_3C-CH_3 + Ph−Ph.

Wurtz-fittig. ether.

⑤ Fittig Rkⁿ → $Ph-Cl \xrightarrow[\text{dry ether}]{Na} Ph-Ph$ diphenyl

⑥ Ullmann rkⁿ → $Ph-I \xrightarrow[\Delta]{Cu} Ph-Ph$ [biaryls]

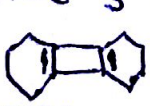
⑦ Frankland's rkⁿ → [alkane] $R-I \xrightarrow[\Delta]{Zn} R-R$.

⑧ Kolbe's electrolysis → $2Et-\overset{O}{\overset{\|}{C}}-OK \xrightarrow{\text{electrolysis}} Et-Et_{(maj)} + Eth + Ethene + Et-\overset{O}{\overset{\|}{C}}-OEt$

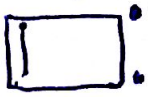
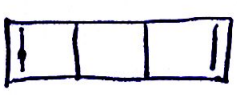
Anode : oxid CO_2
cathode : red. H_2
[as Rkⁿ go fwd PH ↑ formaⁿ of −OH]

• symm. alkane. [even no of C] → good yield.

⑨ Intramolecular Kolbe's electrolysis → $CH_3-CH-COOK$ / $CH_3-CH-COOK$ $\xrightarrow{\text{Kolbe's elysis.}}$ trans/major

Ex: (a) $\xrightarrow{\text{elysis}}$ 

Bp. C is possible for ring ≥ 8 Carbon.

(b) →  $\xrightarrow{\text{dimerise}}$ 

⑩ Acyloin condensation. →

$R-\overset{O}{\overset{\|}{C}}-OR' \xrightarrow[\text{or m-xylene}]{\text{• Na (in ether) or benzene}} R-\overset{O}{\overset{\|}{C}}-\overset{OH}{\overset{|}{C}}H-R$

α-hydroxy Ketone (Acyloin)

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THERMO

- system - anything under observation
- surrounging - everything except system.
- open system - transfer of mass & energy
- closed system - transfer of energy.
- Isolated system - No transfer.
- State func^n - independent of path / depend on final & initial posi^n.
- Δ state func^n in cyclic process = 0
- $\Delta E = \Delta U = q + w$ | $W = -P_{ext}\Delta V$
- $\Delta H = \Delta U + \Delta(PV) = \Delta U + V\Delta P + P\Delta V$
- $\Delta G = \Delta H - T\Delta S$
- Const. press. $= \Delta H = nC_p\Delta T = q_p$
  Const vol. $= \Delta U = nC_v\Delta T = q_v$
- $W = P\Delta V = q_p - q_v$.

Heat absorbed → +ve
heat released → -ve
work done by system → -ve
work done on system → +ve.

- C (Molar heat capacity) → heat req. of 1 mole by 1° to ↑ temp.
- C_p - molar heat capacity at const. press.
- C_v = molar heat capacity at const. volume.

- Degree of freedom for → monoatomic gas (3), diatomic (5) $C_v = \frac{f}{2}RT$
  & $C_p = C_v + R$.
  & W_{gas} = area (curve)
- ISOTHERMAL → $\Delta U = nC_v\Delta T = 0$ or $q = -w$
  ($\Delta T = 0$) (Rev) $w = -2.303\, nRT \log\frac{V_2}{V_1} = -2.303\, nRT \log\frac{P_1}{P_2}$
  (PV = const) (Irrev) $w = -P(V_2 - V_1)$
- ISOBARIC → $\Delta U = nC_v\Delta T$
  ($\Delta P = 0$, $\frac{V}{T}$ = const.) $w = -P_{ext}\Delta V$ where, $P_{ext} = P_{gas}$.
  $\Rightarrow w = -P_{gas}(V_2 - V_1) \times 101.3$ J [If P & V are in SI no need to multiply]
  (atm, lts)
  Expansion $\Rightarrow w = -ve$
  Compression $\Rightarrow w = +ve$.

NOTE → a gas is expanded or compressed against const. ext press → Irreversible proc. [$w = -P_{ext}\Delta V$]

- ISOCHORIC → $W = -P_{ext}\Delta V = 0$
  ($\Delta V = 0$, $P \propto T$) $\Delta U = nC_v\Delta T = q_v$.
- CYCLIC PROCESS → all state func^n are zero ($\Delta H, \Delta U, \ldots$) | $w = -ve$ (CW), $+ve$ (ACW)
  w = area (under PV graph) (toward V)

→ Poly tropic process : if PV^x = const. then $w = \frac{nR\Delta T}{x-1}$

on diff. → $\frac{dP}{dV} = -x\frac{P}{V}$

- ADIABETIC PROCESS → $w = \Delta U = nC_v\Delta T$, Also PV^γ = const.
  (q = 0)
  & $\gamma = \frac{C_p}{C_v}$ & $C_p - C_v = R \Rightarrow C_v = \frac{R}{\gamma - 1}$
  [$\gamma = 5/3$, monoatomic; $\gamma = 7/5$, diatomic; $\gamma = 4/3$ or $7/5$ (V) graph, triatomic ($\frac{7}{3}$ lines)]
  $\rightarrow w = nC_v\Delta T = \frac{nR\Delta T}{\gamma - 1} = \frac{\Delta(PV)}{\gamma - 1} \times 101.3$ J

- Calorimeter [Rx^n at const press.) & • Bomb. calorimeter [Rx^n at const. vol.]
  → If specific heat of calorimeter = x J/°C or x J/K then,
  Heat transfer = $(x \times \Delta T)$ J
- Δn_g = (no of moles of pdt in gaseous state) − (no of moles of rxt in gas state)

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- $\Delta(PV) = RT(\Delta n_g)$ & $\Delta H = \Delta U + \Delta(PV) \Rightarrow \Delta H = \Delta U + RT\Delta n_g$
- $(\text{Heat of rkt})_{\text{const press}} = (\text{Heat of rkt})_{\text{const vol.}} + RT(\Delta n_g)$
- Elements in natural state, $\Delta H_f = 0$.
- Heat of atomisaⁿ → change in enthalpy during formaⁿ of 1 mole of atom in gaseous state from natural state
- Hess's Law → ΔH – state funcⁿ & extensive prop. [addiⁿ & subt. of rkⁿ]
[Ex → $H^+ + {}^-OH \rightarrow H_2O$ $\Delta H = -13.7$ K cal. for 1 mole]
- $(\Delta H)_{\text{neutralisaⁿ}} = -13.7$ K cal $+ \Delta H_{\text{ion(acid)}} + \Delta H_{\text{ion(Base)}}$ [ΔH_{ion} for SA & SB = 0]
- $(\Delta H)_{\text{neut}}$ ↑ then Acidic st. ↓
- SECOND LAW → ΔS (entropy) $= \frac{q_{rev.}}{T}$ $\quad \Delta G_T = \Delta H - T\Delta S$
- $\Delta S = n C_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$
- $\Delta S = n C_p \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$ $\left[C_p - C_v = R, \ \frac{PV}{T} = \text{const.}\right]$
- Isothermal: $\Delta S_{sys.} = nR \ln \frac{P_1}{P_2} = nR \ln \frac{V_2}{V_1}$
- Isobaric: $\Delta S_{sys.} = n C_p \frac{T_2}{T_1} = nC_p \ln \frac{V_2}{V_1}$
- Isochoric: $\Delta S_{sys.} = n C_v \ln \frac{T_2}{T_1} = nC_v \ln \frac{P_2}{P_1}$
- Adiabetic: $\Delta q = 0 \Rightarrow \Delta S = \frac{q_v}{T} = 0$
- Cyclic: $\Delta S = 0$, state funcⁿ

$\Delta S_{total} = \Delta S_{sys} + \Delta S_{surr.}$
But for –
Rev. $\Rightarrow \Delta S_{sys} = -\Delta S_{surr}$
i.e. $\Delta S_{Total} = 0$
& for irrev. if sign is opp → value can be found accurately

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- Allylic halides → = के पास वाले C पर X.
- Benzylic halide → ring के पास वाले C पर X
- Vinylic halides → = पर X.
- Acyl halide ⇒ benzene पर X.

gem-dihalide: $CH_3-CH(Cl)_2$

vic-dihalide: $H_2C(Cl)-CH_2(Cl)$

⇒ Prepration

$R-OH + HX \xrightarrow{ZnCl_2} R-X + H_2O$

$R-OH + KBr \xrightarrow{H_2SO_4} RBr + KHSO_4 + H_2O$

$3R-OH + PX_3 \longrightarrow 3-R-X + H_3PO_3$ [X → Cl, Br]

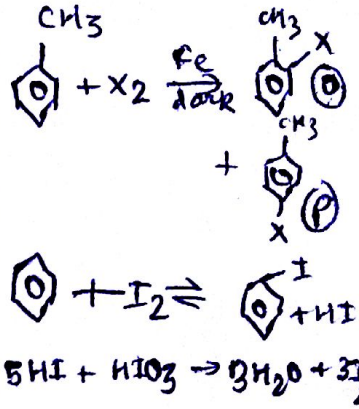
$R-OH + PCl_5 \longrightarrow R-Cl + POCl_3 + HCl$

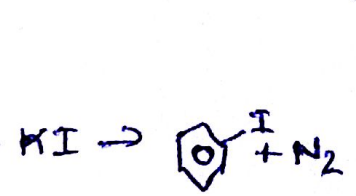
$R-OH \xrightarrow[X_2 \to Br_2, I_2]{Red\ P/X_2} R-X$

$R-OH + SOCl_2 \longrightarrow R-Cl + SO_2 + HCl$

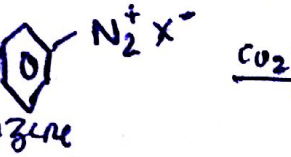
thionyl chloride

1° > 2° > 3° alc. reactivity

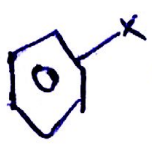


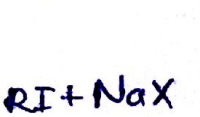


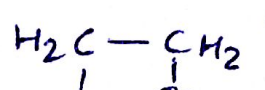
Sandmeyer's rxn.



benzene diazonium halide





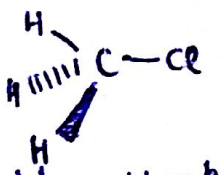
⇒ $H_2C=CH_2 + Br_2 \xrightarrow{CCl_4}$ 

$\Rightarrow CH_3Br + AgF \rightarrow CH_3-F + AgBr$.

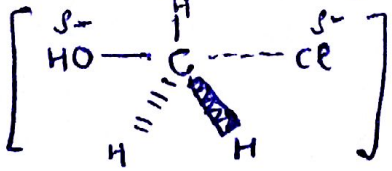
$R-X + NaI \rightarrow RI + NaX$

order in nucleophilic sub. rxn

$I > Br > Cl > F$

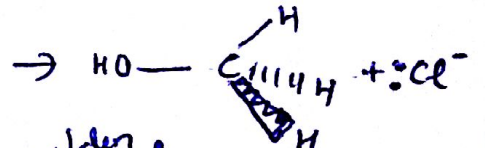
S_N2: $^{-}OH +$ 

1° > 2° > 3°



back side attack.

transition state



walden inversion

[DMF, DMSO] S_N2 ↑ ←

CH_3-X $R-CH_2-X$ $R-CH(R')-X$ $R-C(R')(R'')-X$

1° 2° 3°

S_N1 ↑ ← in which acid. α-H is + ve

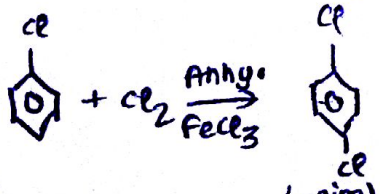
$2CHCl_3 + O_2 \xrightarrow{h\nu} 2COCl_2 + 2HCl$

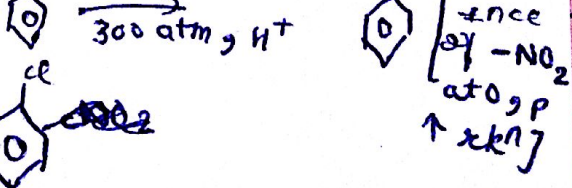
Phosgene

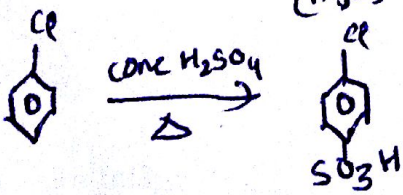
• $CH_3-CH_2-Br + Mg \xrightarrow[ether]{dry} CH_3-CH_2-Mg-Br$.

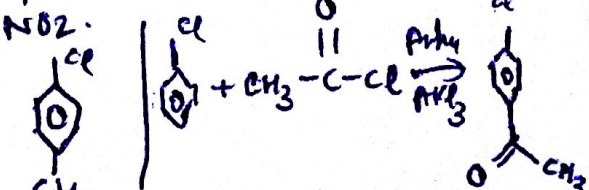
grignard's reagent.

$RMgX + H_2O \rightarrow RH + Mg(OH)X$









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LIQUID SOLUTION

- solvent → lqd state ; solute → solid / lqd / gas [Like dissolves like] → PP / NP-NP
- Alcohol CH_3-CH_2-OH → Polar part / non polar part } ⇒ soluble in both Polar & N.P.
- Vol based units [T. dep.] T ∝ vol. → Molarity, Normality
- mass based units (T. indep.) → Molality, Mole fraction } All are intensive prop.

- Molarity (M) = $\frac{n_{moles\ of\ solute}}{Vol.\ of\ soln\ (in\ l)}$ [for lqd-lqd soln $V_{soln} = V_{solute} + V_{solvent}$]
- Molality (m) = $\frac{n_{solute}}{M_{solvent}\ (kg)}$ [assume variables in ques^n, e.g. $2 = \frac{x/60}{\frac{280-x}{1000}}$]

ppt
- AgCl
- $BaSO_4$

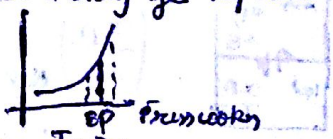
[2m & 3m are mixed ⇒ Resultant b/w 2 & 3 m]

$$d = M\left(\frac{1}{m} + \frac{M_{solute}}{1000}\right)\ gm/ml\ ,\ kg/l$$

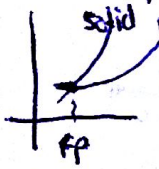
- x% soln (m/v) 100 ml soln → x gm solute
- x% soln (m/m) 100 gm soln → x gm solute

$$x_{solute} = \frac{m \rightarrow molality}{m + \frac{1000}{M_{solvent}}}$$ (M solvent → molar mass)

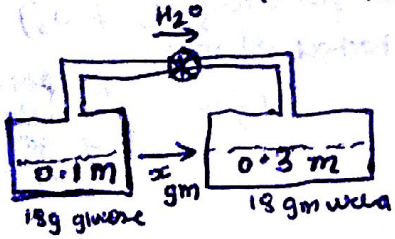
- $E_{H_3PO_3} = \frac{M_{H_3PO_3}}{2}$, $E_{H_3BO_3} = \frac{M_{H_3BO_3}}{1}$
- $E_{H_3PO_2} = \frac{M_{H_3PO_2}}{1}$, $E_{H_3PO_4} = \frac{M_{H_3PO_4}}{3}$
- Normality = molarity × basicity / acidity.
- $K_p = \frac{(P_{vap})_{eq}}{(P_{lqd})_{eq}}$ = V.P. of lqd. — independent of: Amount, Volume, Shape, S.A — depends on: Temp. [T ↑ VP ↑]
- At certain temp, (V.P)$_{lqd.}$ = Atm press ⇒ Boiling pt. [Before BP molecules ↑nt at lqd surf. could only get vapourised]
  - on ↑ A.P → BP ↑
  - VP = 200 torr
  - VP = 400 torr → more volatile → more vapourising tendency [Intermolecular forces ↓] → less BP
  - $VP = A' e^{-\frac{\Delta H_{vap}}{RT}}$



- Freezing pt: temp at which lqd state remain in eqb with solid state
  - T > FP → lqd amount ↑ solid amount ↓
  - (VP)$_{lqd}$ > (VP)$_{solid}$, lqd ↓ & solid ↑
  - $|\Delta H|_{sublimation} > |\Delta H|_{vap.}$
  - (slope)$_{solid\ VP}$ > (slope)$_{lqd\ VP}$

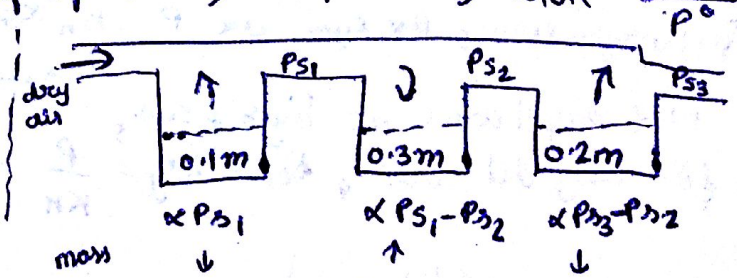


- H_2O + sugar → शर्बत ⇒ At eqb no of moles of vap is less, P_{vap} is less, $x_{solute} = \frac{P^\circ - P_s}{P^\circ}$



$$\frac{18/180}{\frac{1000-x}{1000}} = \frac{18/60}{\frac{1000+x}{1000}}$$

$x = 500$



⇒ Solvent + non volatile solute = soln

BP = T_1 , BP = T_2 [$T_2 > T_1$]

$\Delta T_b = K_b m$ [$K_b (H_2O) = 0.52\ K\text{-}kg/mole$]

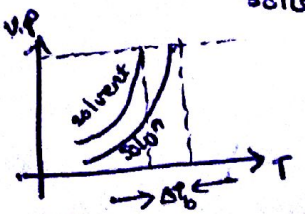
$$K_b = \frac{RT_b^2}{1000\ L_v}$$

$$L_v = \frac{(\Delta H_{vap})_{solvent}}{\text{Molar mass of solvent}}$$

T_b = $BP_{solvent}$

L_v = Latent heat of vap of solvent (/gm)

K_b = molal elevation constant.
- depends only on nature of solvent not solute



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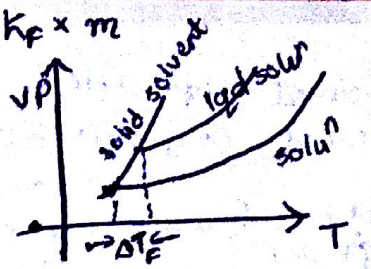
• Depression in freezing pt ⇒ $\Delta T_f = (FP)_{solvent} - (FP)_{soln} = K_f \times m$

↳ depend only on nature of solvent not solute.

$K_f = \frac{R \times T_f^2}{1000 \times L_f}$; $L_f = \frac{\Delta H_{fusion}}{M_{solvent}}$



⇒ Osmosis:

अर्धपार	पानी
(solⁿ)	← H_2O → (solvent)
high []	Low []

$\pi = C \delta T \rightarrow (K)$

↓ osmotic press. ; ↓ universal solⁿ constant [= R value]

• solute may be diff. but same solvent

$\pi = |C_1 - C_2| \delta T$

π ↳ π is ext press app on solⁿ side to ~~prevent~~ stop osmosis $\pi = P_A - P_B$

⇒ $A \rightarrow n\,B$

$\frac{1}{1-\alpha}$ $\frac{n\alpha}{}$

[for complete dissociation $\alpha = 1$]

⇒ $nA \rightarrow B$

$\frac{1}{1-\alpha}$ $\frac{0}{\frac{\alpha}{n}}$

$(\Sigma n_{solute})_{obs} = [1 + \alpha(n-1)]$

$i = 1 + (n-1)\alpha$

dissociaⁿ

$i = 1 - \left(1 - \frac{1}{n}\right)\alpha$

Associaⁿ

Ex of associaⁿ = RCOOH in benzene

Phenol in CCl_4

$(\alpha = 3) \leftarrow$ trimer formaⁿ in solⁿ

$(n_{solute})_{obs} = i \cdot (n_{solute})_{th.}$

$(M_{th}) = i \cdot m_{obs}$

for dil solⁿ, $\left(\frac{P^\circ - P_s}{P_o}\right)_{obs} = i \left(\frac{P^\circ - P_s}{P_o}\right)_{th.}$

• 0.1m $MgCl_2$ ($\alpha = 0.8$) $i = 2.6$ — m: 0.26 — VP ⇒ I > III > II

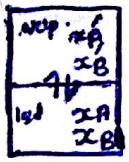
• 0.2m $AlCl_3$ ($\alpha = 0.4$) $i = 2.2$ — 0.44 — BP ⇒ II > III > I

• 0.3m glucose ($\alpha = 1$) $i = 1$ — 0.30 — F.P ⇒ I > III > II

↳ press. due to V.P of A only.

• Raoult's law: $P_A = x_A (liq) \times P_A^\circ$ (mix of liq A & B)

$\Rightarrow P_T = x_A P_A^\circ + x_B P_B^\circ$



$\left(\frac{x_A}{x_B}\right)_{vap} = \left(\frac{x_A}{x_B}\right)_{liq} \times \frac{P_A^\circ}{P_B^\circ}$

$F_{AA} = F_{A-B} = F_{B-B}$	$F_{AB} > F_{AA}$	$F_{AB} < F_{AA}$
	$F_{AB} > F_{BB}$	$F_{AB} < F_{BB}$
↓	↓	↓
Ideal	-ve deviaⁿ	+ve deviaⁿ
$\Delta H = 0$	$\Delta H < 0$	$\Delta H > 0$
$\Delta V = 0$	$\Delta V < 0$	$\Delta V > 0$

• Azeotropic mixture – molar compositⁿ of liq & vap. state is same.

• +ve deviaⁿ azeotr. mix → min B.P.

• -ve deviaⁿ azeo. mix → maxᵐ B.P

⊛ Solubility of gas in water ⊕ Factor aff. → Nature of gas: acidic / basic – more; other – less soluble

(very dil solⁿ)

gas(free) ⇌ gas(diss.)

⊕ Temp → ↑ temp solu. ↓

⊕ Press → P↑ [] free gas ↑ solubility ↑

⇒ Henry's law: $P \propto x_{gas}$ or $P = K_H x_{gas}$ [P = partial press for only N_2 & O_2]

↳ depends on temp and nature of gas.

Not applicable at high press.

• for very dil solⁿ, solubility = $\frac{P}{K_H} \times 55.5\ mol\,l^{-1}$

$x_{gas} = \frac{P}{K_H}$ for more soluble gas K_H is less

• $\ln \frac{c_2}{c_1} = \frac{\Delta H}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

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ALCOHOLS

- CH_3-OH ⇒ carbenol

GMP of monohydric alcohol → from alkene

hydration (Markovnikov addition, Reaction) → H_2O/H^+, KOH

hydroboration oxidation → B_2H_6(THF), H_2O_2

oxide mercuration (Oxymercuration) → $Hg(OCOCH_3)_2$ / $NaBH_4$ (reducing agent)

(a) Hydration → $R-CH=CH_2 + H^+ \longrightarrow R-\overset{\oplus}{C}H-CH_3$ (2° carbocation)

$\equiv \xrightarrow{NaNH_2} =$

$R-\overset{\oplus}{C}H-CH_3 \xrightarrow{H_2\ddot{O}:} R-CH(\overset{\oplus}{O}H_2)-CH_3 \xrightarrow{-H^+} R-CH(OH)-CH_3$

(b) Hydroboration → $R-CH=CH_2 + BH_3 \longrightarrow (R-CH_2-CH_2)_3B$ (trialkyl borane)

Isotopic effect

$(R-CH_2-CH_2)_3B \xrightarrow[H_2O_2]{\text{anti Markovnikov}} R-CH_2-CH_2-OH$

$(R-CH_2-CH_2)_3B \xrightarrow{H^+} R-CH_2-CH_3$

B_2H_6 can be used for Anti Markovnikov addition

$R-\overset{\delta+}{C}H-CH_2-\overset{\delta-}{B}H_2$ (Intermediate)

(c) Oxide mercuration – generally syn addition (Markovnikov reaction)

$R-CH=CH_2 + Hg(OCOCH_3)_2$ in THF, H_2O

oxymercuration → $R-CH(\overset{\oplus}{O}H_2)-CH_2-Hg-OCOCH_3$

$\xrightarrow{-H^+}$ deprotonation → $R-CH(OH)-CH_2-Hg-OCOCH_3$

$\xrightarrow{NaBH_4}$ demercuration → $R-CH(OH)-CH_3 + Hg + CH_3COOH$

Ex → $R-CH=CH_2 \xrightarrow[NaBH_4]{D_2O/THF/Hg(CH_3COO)_2} R-CH(OD)-CH_2D$

(2) From ester –

$R-COOR' + H_2O \xrightarrow[H^+]{Aq\ H_2SO_4} R-COOH + R'-OH$

(acid, alcohol, excess)

molecularity = 2, order = 1, Pseudo order reaction

mech: $R-C(=O)-O-R' + H^+ \longrightarrow R-C(OH)=O-R' \longrightarrow R-C(OH)(\overset{\oplus}{O}H)-OR' \xrightarrow{H_2O:} R-C(OH)_2-OR' \xrightarrow{-H^+} R-C(OH)_2-OR'$

(engaged in conj.) → unstable

$R-C(OH)_2-OR' \longrightarrow R-COOH + R'OH$

(3) From ether – hydrolysis in acidic medium

$ROR' + H_2O \xrightarrow{H_2SO_4} R-OH + R'-OH$

$R-\ddot{O}-R' + H^+ \longrightarrow R-\overset{\oplus}{O}(H)-R' \longrightarrow ROH + R'^+$

$R'^+ + H_2O \longrightarrow R'-OH_2 \xrightarrow{-H^+} R'-OH$

Except: $CH_3-C(CH_3)_2-O-CH_3 \xrightarrow[H^+]{H_2O} CH_3-C(CH_3)_2-\overset{+}{O}(H)-CH_3$

$\xrightarrow{-CH_3OH} CH_3-\overset{\oplus}{C}(CH_3)_2 \longrightarrow CH_3-\overset{\oplus}{C}(CH_3)-CH_2-H \longrightarrow CH_3-C(CH_3)=CH_2$

- $R-MgX + HCHO$ (formaldehyde) $\longrightarrow R-CH_2-OMgX \xrightarrow{H_2O/H^+} R-CH_2-OH$ (1° alco.)
- $R-MgX + R'CHO \longrightarrow R-CH(R')-OMgX \xrightarrow{H_2O/H^+} R-CH(R')-OH$ (2° alco.)
- $RMgX + R'COR'' \longrightarrow R-C(R')(R'')-OMgX \xrightarrow{H_2O/H^+} R-C(R')(R'')-OH$ (3° alco.)
- $RMgX + \frac{1}{2}O_2 \rightarrow R-OMgX \xrightarrow[H^+]{H_2O} R-OH + Mg(X)(OH)$

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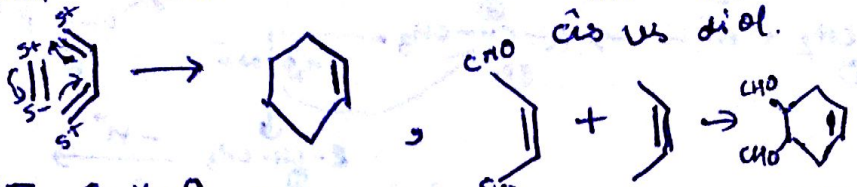
• $RMgX + CH_2-CH_2$ (oxirane) $\longrightarrow R-CH_2-CH_2-OMgX \xrightarrow[H^+]{H_2O} R-CH_2-CH_2-OH$

• $>C=C<$ $\xrightarrow[\text{MCPBA}]{\text{Peroxide acid}}$ epoxide $\xrightarrow{\text{corrosion}}$ cis vic diol.

• Diels alder reⁿ – 

• from fermentation – $C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{invertase}} C_6H_{12}O_6 + C_6H_{12}O_6$

$C_6H_{12}O_6 \xrightarrow[\text{yeast}]{\text{Zymase}} C_2H_5OH + CO_2$

$C_6H_{12}O_6$: $R-C(=O)-R$ (keto sugar.) ; $R-C(=O)-H$ (aldo sugar.)

* $R-NH_2 + NaNO_2 \xrightarrow[HCl]{dil} R-OH + N_2$.

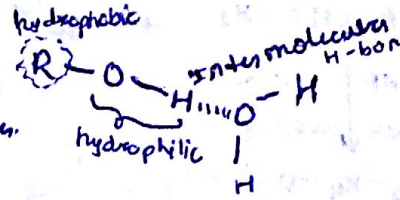
$R \neq CH_3$; [for $R=CH_3$; dimethyl ether is pdt]

• Phy prop : C_1-C_4 gas. ; C_4-C_8 lqd ; C_8 ... solid

B.P ⇉ Alcohol > ether

B.P ∝ mw in alco. [due to ↑ C-chain dominant over H-bond]

Solubility ⇒ alkane < alcohol > ether (H-bond)



⇒ MW ↑ ; Hydrophobic por^n ↑ ; H-bond ↓ ; solubility ↓

• Acidic st^r of alcohol – $R-O\overset{(H)}{\underset{\text{acidic H}}{}} + Na \longrightarrow R-O-Na + \frac{1}{2}H_2\uparrow$ [$-I = NO_2 > CN > F > Cl$]

ord^n → 0° > 1° > 2° > 3°

$\propto -I \propto \frac{1}{+I}$

Acidic H st^r order : Int·reso > reso· > sp > sp^2 > sp^3

* chem prop – (1) Reac^n containing $\overset{|}{\underset{|}{-O-H}}$ bond fission.

(a) $R-OH + H-O-C(=O)-R' \longrightarrow R-O-C(=O)-R'$ ($1^\circ > 2^\circ > 3^\circ$ acid H)

(b) $R-OH + Na \rightarrow RONa + H_2\uparrow$

(c) $R-OH + R'MgX \rightarrow R'H$

(2) Reac^n involving $C \div O$ bond fission : (a) $R-OH + HX \rightarrow R-X + H_2O$ [$3^\circ > 2^\circ > 1^\circ$ Reactivity : HI > HBr > HCl > HF]

Mechanism ⇒ $R-O-H + H^+ \rightleftharpoons R-\overset{\oplus}{O}(H)-H \rightarrow R^+ \xrightarrow{X^-} RX$.

• Dehydrogena^n : $R-CH_2-OH \xrightarrow[\Delta]{Cu} R-C(=O)-H$; $R-CH(R)-OH \xrightarrow[\Delta]{Cu} R-C(=O)-R$ (Rk^n takes place in acidic med only)

$R-C(R)(CH_3)-OH \xrightarrow[450^\circ C]{Cu-Al_2O_3}$ Alkene [$R-C(R)=CH_2 + H_2O$]

• $R-CH_2-OH \xrightarrow[HI\,;\,\Delta]{Red\ P} R-CH_3$

• $R-OH + NH_3 \xrightarrow[-H_2O]{} R-NH_2$

• Oxidisa^n → $1^\circ \xrightarrow[\text{mild oxid. agent}]{KMnO_4}$ aldehyde $\xrightarrow{[O]}$ carboxylic acid (comp. oxi) [C atom unchanged]

$2^\circ \xrightarrow[\text{st^r ox. Ag}\ K_2Cr_2O_7]{[O]} R-C(=O)-R' \xrightarrow[K_2Cr_2O_7]{[O]}$ carboxylic acid (1-carbon less)

unchanged C-chain

$3^\circ \xrightarrow[K_2Cr_2O_7]{[O]} R-C(=O)-R \xrightarrow[\text{1C less}]{[O]}$ carboxylic acid

Lucas test :–

Anhy $ZnCl_2$ + dil HCl → Lucas reagent

3° alco $\xrightarrow{L.R}$ RCl [spont.]

2° alco $\xrightarrow{L.R}$ within 5 min

1° alco $\xrightarrow[+\Delta]{L.R}$ no reⁿ (takes long time)

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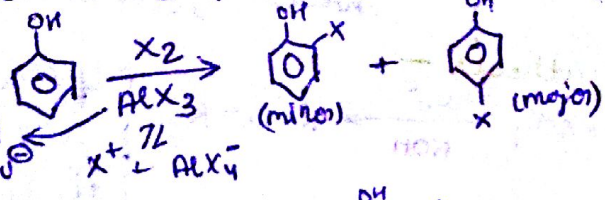
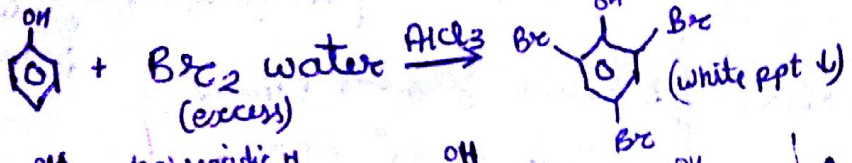
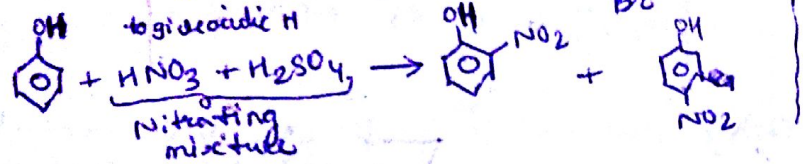
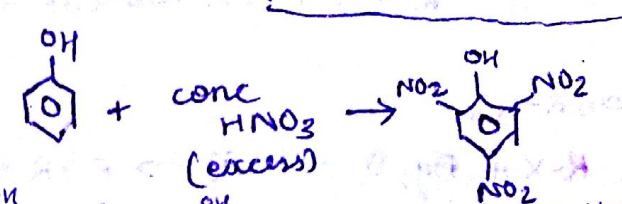
PHENOL

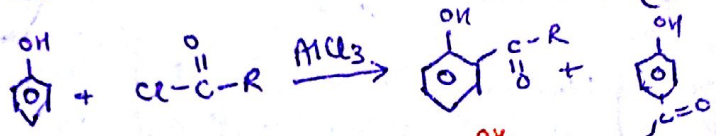
- Acidity : RCOOH > H_2O > Ph-OH > R-OH
- o, p directive (Activating for $E^{\oplus}$ subs. rekn)
- Halogenation :

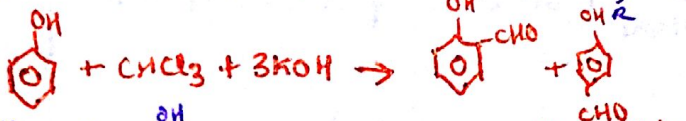


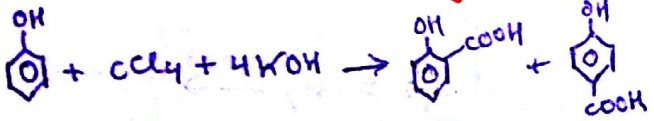
- 

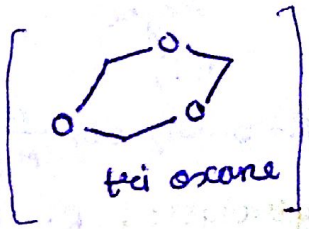
- 

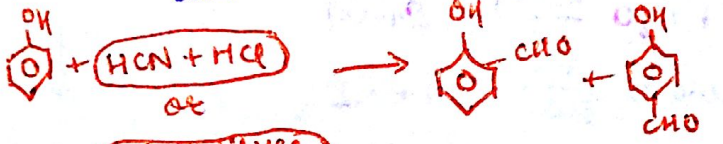
- 

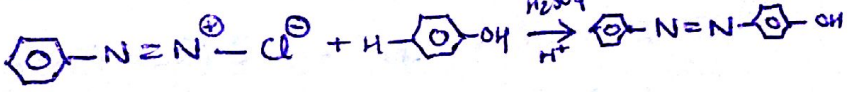
- Friedal craft Rkn : 

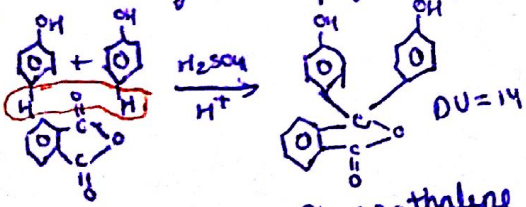
- Riemer Tiemann Rkn : 

- 

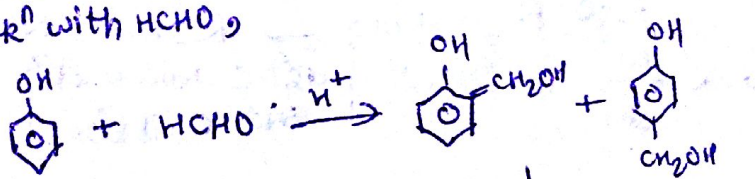


- Gattermann Rkn : 

- Rkn with benzene diazonium chloride (coupling rekn of phenol) : 

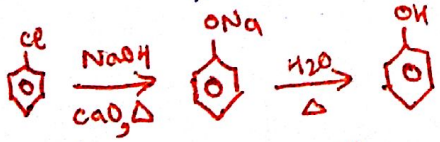
- 

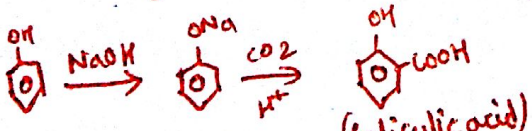
Phenopthalene

- Rkn with HCHO : 

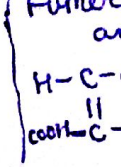
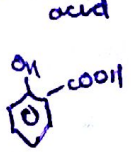
(Bakelite) ← C_6H_5OH (excess)

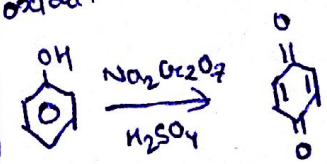
-

- Dow's process : 

- Kolbe's rekn : 

(salicylic acid)

Fumeric acid	Salicylic acid
	

- oxidation : 

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ETHER

- General formula: $C_nH_{2n+2}O$, IUPAC: Alkoxy (small C chain) alkane (large C chain). [metamerism is possible]

- **Williamson continuous ether synthesis –**

[Intermediate carbocation] $R-OH + H_2SO_4 \xrightarrow{100^\circ-110^\circ} R-HSO_4 \xrightarrow[ROH]{140^\circ} R-O-R$.

[if temp. raised to 180°C
i) $R \equiv R$
ii) $+H_2SO_4 \rightarrow$ (no ether, o,p derivatives formed.)]

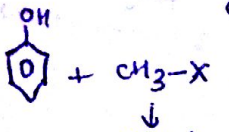
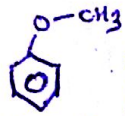
- **Williamson ether synthesis –**

$R-X + NaOR' \longrightarrow ROR' + NaX$.

[3° halide favours Elimination
max^m yield → 1° → S_N2 fav.]

- from R–X,

$R-X + Ag_2O + X-R \rightarrow ROR + 2AgX$

-  + CH_3-X (may be 1°, 2°, 3°) →  Anisole [Lewis acid must be Absent otherwise friedal craft Rxn]

- $R-O-CH_2-CH_3 \xrightarrow[Al_2O_3]{\Delta} R-CH=CH_2 + alcohol$.

- **Hydrolysis:** $R-OR + H_2O \xrightarrow{H_2SO_4} 2ROH$

- with HX,

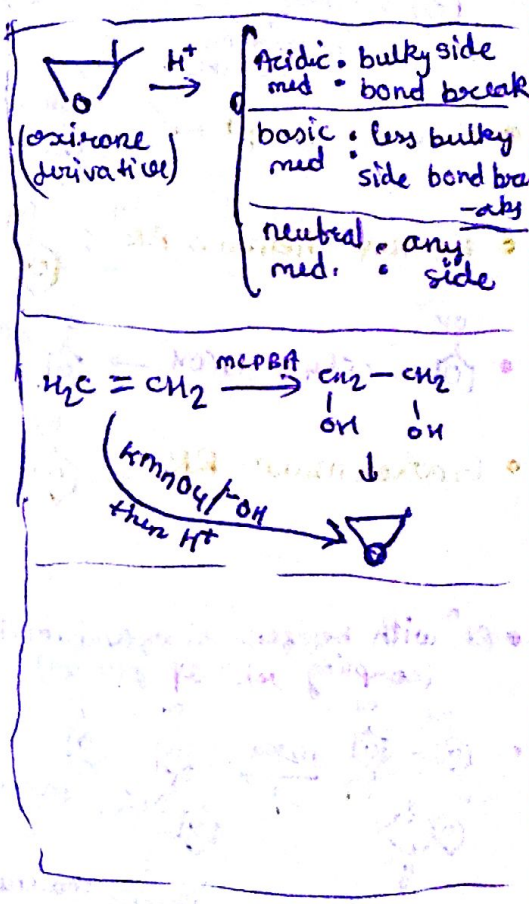
$R-O-R + cold\ HI \xrightarrow[fav.]{S_N2} ROH + RI$

$R-O-R + hot\ HI \longrightarrow RI$

$R-O-R + PCl_5 \longrightarrow 2RCl + POCl_3$

$R-O-R + NH_3 \longrightarrow ROH + RNH_2$.

Ⓚ $R-O-R' \xrightarrow[200\ atm]{CO} R-\overset{O}{\overset{\|}{C}}-OR'$ [बड़ी C chain ओ पहले छोटी से Ⓡ]



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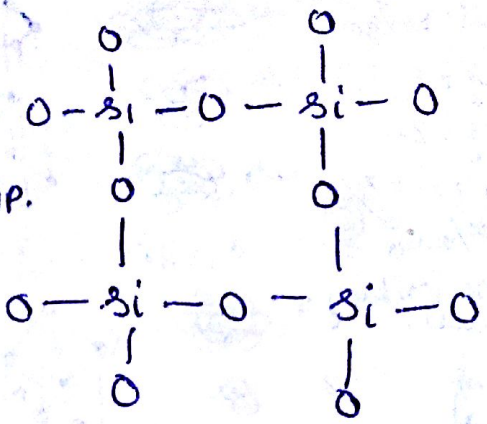
SOLID STATES

- fixed shape and volume due to strong interparticle force.
- Ionic solid – NaCl – electrostatic force b/w anion & cation.
- Metallic solid – attrac^n b/w metal ion & the stream of electron cloud
- H-bond – Ice – H_2O molecule.
- Dipole–dipole → Dry ice – solid CO_2.
- Van der waal's force – solid argon.

} molecular solid

- Network solid → SiO_2 (covalent bond) →

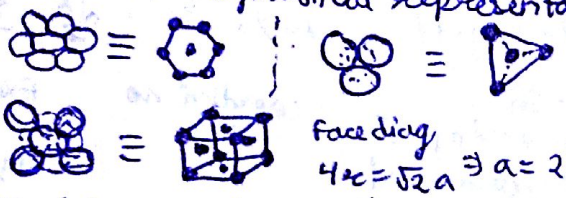


⇒ Crystalline → Regular arrangement with fix MP.
→ Aniso tropic in nature
↳ diff. direc^n diff. prop.

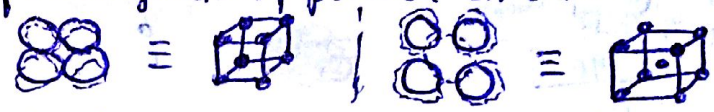
(1 amu = $\frac{1}{N_a}$ gm)

⇒ Amorphous → not regular
→ Range of MP.
→ Isotropic in nature.

- Space lattice – symbolical represenata^n of arrangement of particle in 3-D.



Face diag
$4r = \sqrt{2}a \Rightarrow a = 2\sqrt{2}r.$

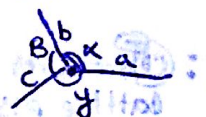


(diagonal)
$\sqrt{3}a = 4r$
$a = \frac{4r}{\sqrt{3}}$

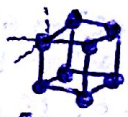
Unit cell → smallest 3-D geom. repeating itself in space lattice

Crystal system	Sides	axial angles	Bravais lattices	
cubic	$a=b=c$	$\alpha=\beta=\gamma=90°$	simple, BCC, FCC	3
rhombic	$a=b=c$	$\alpha=\beta=\gamma\neq90°$	simple	1
tetragonal	$a=b\neq c$	$\alpha=\beta=\gamma=90°$	simple, BCC	2
hexagonal	$a=b\neq c$	$\alpha=\beta=90°, \gamma=120°$	simple	1
orthorhombic	$a\neq b\neq c$	$\alpha=\beta=\gamma=90°$	simple, BCC, FCC, ECC	4
monoclinic	$a\neq b\neq c$	$\alpha=\gamma=90°, \beta\neq90°$	simple, ECC	2
triclinic	$a\neq b\neq c$	$\alpha\neq\beta\neq\gamma\neq90°$	simple	1
				14

Element of symm. in cubic crystal : POS = 9, Axis OS = 13, COS = 1, Total (23)



Simple cubic system –
eff particle → 1
Pack. eff. = 0.52
coordina^n no → (6)



BCC

eff par. = 2
Pack. eff = 0.68
coordina^n no → (8)

$4r = \sqrt{3}a.$

Types of voids →

triangular voids
$r = 0.155R$

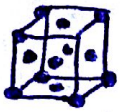
tetrahedral voids
$r = 0.225R$

octahedral voids
$r = 0.414R.$
↳ Best packing efficiency (74%)

cubic void
$r = 0.732$

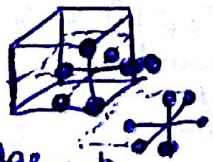
FCC

eff par. = 4
Pack eff = 0.74
coordina^n no = 12



In FCC,

Body centre



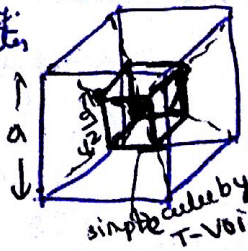
edge centre.

O-voids

no. of o-voids / unit cell = 4
no. of sphere / unit cell = 4
no. of O-void / sphere = 1

T-voids

each corner particle + related 3 face centre
Loca^n of T-void = mid pt of corner & body centre
no. of T void / unit cell = 8
no of T void / sphere = 2.
min dist b/w two T-voids = $\frac{a}{2}$.
min dist b/w T-void & O-void = $\frac{\sqrt{3}a}{4}$



simple cube by T-voids

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• close packing of sphere → (A) Spheres are put over T-void formed b/w preceding 2 layers. each 1st & 3rd layer are of similar pattern ABAB AB type.

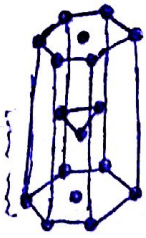
(B) spheres are put over O-voids formed b/w preceding 2 layers. each 1st & 4th layer are same ABC ABC ABC type.

→ Hexagonal closed packing (hcp) fig (3) (2) (1)
AB type [unit cell - hexagon]

- face centre → $2 \times \frac{1}{2} = 1$
- corner → $12 \times \frac{1}{6} = 2$.

(*) 3 particle inside → $3 \times 1 = 3$

no of particle/unit cell = 6



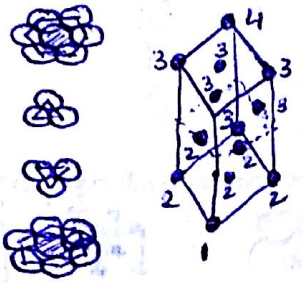
no of T-voids/unit cell = 12
no of O-voids/unit cell = 6
coordination no = 12
Pack eff. = $\frac{\text{Vol occ}}{\text{Vol as.}} = \frac{6 \times \frac{4}{3}\pi r^3}{\left[6 \times \frac{\sqrt{3}}{4}(2r)^2\right]\left[4\sqrt{\frac{2}{3}}r\right]}$

0.74 or 74%

d_{min} b/w layers $= 2\sqrt{\frac{2}{3}}r$

(b) FeO(s) ⇒ Fe_xO (x<1)
Fe^{2+} ⇒ Fe^{3+}
for main neut. → + 2 Fe^{3+}
- 2 Fe^{2+}
≡ Metal deficiency

→ (ccp) Cubic close packing
unit cell = cube.



fcc, no of particle/unit cell = 4
no of O.voids/unit cell = 4
no of T-voids/unit cell = 8
coordination no = 12
Packing eff. = 74%.

(b) NaCl $\xrightarrow[\text{vap.}]{Na}$ Na_xCl (x<1)
(metal excess)
- Paramag.
- good cond.
- light yell.

Colour: e^- (F-centre) absorbs visible light

c^+ in	coordination no	Example.
Cubic void of A^-	3	X
tetrahedral void	4	Na_2O, ZnS
O-void	6	NaCl
cubic void	8	CsCl

• Radius ratio — $\frac{r_{c^+}}{r_{A^-}} = x$

$0.155 \le x < 0.225$	
$0.225 \le x < 0.414$	
$0.414 \le x < 0.732$	
$0.732 \le x < 1$	

(I) NaCl: (Rock salt str.)
Cl^- occupies all ccp/fcc lattice pts.
Na^+ occupies all O-voids
unit cell → cube.
- Cl^- → 6 face centre + 8 corners
- Na^+ → body centre + 12 edge centre

no of Cl^-/unit cell = 3 + 1 = 4
no of Na^+/unit cell = 1 + 3 = 4
no of formula unit/unit cell = 4

density ratio or formula unit ratio

COORDINATION NO →
1st neigh [Na^+ → 6 Cl^- ion; Cl^- → 6 Na^+ ion.
2nd neigh [Na^+ → 12 Na^+ ion; Cl^- → 12 Cl^- ion

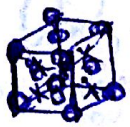
d.D = 1, 2, 3, 4

$(d_{Na^+})_{min} = (r_{Na^+} + r_{Cl^-})\sqrt{2}$.

$3d' = \sqrt{3}a = \sqrt{3}(x+y)\sqrt{2} \times 2$

Interchange Na^+ & Cl^- → no effect on charge.

(II) Na_2O (Anti-fluorite structure)
O^{2-} → ccp/fcc lattice
Na^+ → occupies T-voids
no of O^{2-}/unit cell = 4
x- . Na^+ - - - = 8.
Na_2O
4 formula unit
coordination no →
1st near neigh → Na^+ → 4 O^{2-}
O^{2-} → 8 Na^+
2nd → Na^+ → 6 Na^+
O^{2-} → 12 O^{2-}



(III) CaF_2 str. (Fluorite str.)
F^- in T-void of Ca^{2+} ion
Ca^{2+} → ccp/fcc & F^- → T-void
n formula unit = 4.
coordination no → 1st: Ca^{2+} → 8 F^-
F^- → 4 Ca^{2+}
2nd → Ca^{2+} → 12 Ca^{2+}
F^- → 6 F^-

Defects: (a) stoichiometric
(b) non stoichiometric

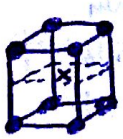
(a) Schottky: missing of c^+ & A^- pair to maintain neutrality. c^+ → cubic or O-void. density ↓ / r_{c^+} & r_{A^-} comparable

(a) Frenkel defect (Edis is Ca defect): c^+ → T-void. r_{c^+} will be very less.

(b) ZnO ⇒ Zn_xO (x>1) (metal excess)
ZnO(s) ⇒ $Zn^{2+} + 2e^- + \frac{1}{2}O_2(\uparrow)$
(crystal void) (interstitial void (e))

(IV) ZnS str. (1:1)
- wurtzite str. → Zinc blende str.
- S^{2-} → ccp lattice pt
- Zn^{2+} → half T-void.
- S^{2-}/unit cell = 4
- Zn^{2+}/unit cell = 4
- F.U = ZnS
- no of F.U = 4
- coordination no. Zn^{2+} → 4 S^{2-} & S^{2-} → 4 Zn^{2+}
- formula unit ZnS
- no of F.U = 4

(V) CsCl → cubic
Cl^-/unit cell = 1
Cs^+/unit cell = 1
no of formula unit = 1
coordi. geom → 1st Cs^+ → 8 Cl
Cl^- → 8 Cs^+
$r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}a}{2}$



2nd Cs^+ → 6 Cs^+
Cl^- → 6 Cl

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ALDEHYDE & KETONE

- $C_nH_{2n}O$ → cyclic ether, enol, cyclic alcohol, ketone, aldehyde

- Isomerism: chain, Posin, functional, metamerism (only in ketone), tautomerism } known as metamers in ketone.
  for same m.w of Isomers are aldehyde & ketone.

$C_nH_{2n}O$

- $H-C(=O)-R$ — prefix: formyl; suffix: al; IUPAC: Alkanal
- $R-C(=O)-R'$ — prefix: oxo/keto; suffix: one; IUPAC: Alkanone

- $RMgX + H-C\equiv N \longrightarrow R-C(H)=NMgX \xrightarrow[\text{hydrolysis}]{2\text{ time}} R-C(=O)-H + Mg(X)NH_2$ [HCN or $R'-CN$]

- $H-C(=O)-OH + H-C(=O)-OH \xrightarrow[MnO]{MnO_2} HCHO + CO_2 + H_2O$
- $CH_3-C(=O)-OH + CH_3-C(=O)-OH \xrightarrow{MnO_2} CH_3-C(=O)-CH_3 + CO_2 + H_2O$

[Fn of MnO_2 is to provide active surface for rekn & neutralise H_2CO_3; half [1] of carboxylic act as Proton donor & half as proton acceptor. In diff, st. acid – donor, weak acid – acceptor.]

- $(CH_3-C(=O)-O)_2Ca + Ca \xrightarrow{\Delta,\ \text{distillation}} CH_3-C(=O)-CH_3 + 2CaCO_3$

[chloral: CCl_3-CHO]

- from Alkene, $\xrightarrow[-OH,\ \Delta]{KMnO_4}$ $\xrightarrow[OH]{KMnO_4}$

- **Dehydrogenation** —
  $CH_3-CH_2-OH \xrightarrow[300^\circ C]{Cu} CH_3CHO + H_2\uparrow$
  $(CH_3)_2CH-OH \xrightarrow[300^\circ C]{Cu} (CH_3)_2C=O + H_2\uparrow$
  [in case of 3° carbocation, dehydrogn takes place]

- **Rosenmund's rekn** — $R-C(=O)-Cl \xrightarrow[BaSO_4]{H_2/Pt/Pd} R-C(=O)-H$
  [$BaSO_4$ act as poisonous catalyst preventing further reduc^n of aldehyde ~~& ketone~~]

- **Stephen's Rkn** —
  $R-C\equiv N \xrightarrow[HCl + SnCl_2 + H_2]{HCl + Sn} R-CH=NH \xrightarrow{H_2O/H^+} R-CHO + NH_3$
  (aldemine)

- $H_2 + CO \xrightarrow[Cu]{375-450K} CH_3OH \xrightarrow[Cu]{\Delta} HCHO$

- $R-C(=O)-Cl + Cd(R)R \xrightarrow{-CdCl_2} 2R-C(=O)-R$
  $R-C(=O)-Cl$
  dialkyl cadmium.

- with HCN, $R(R')C=O + HCN \longrightarrow R(R')C(OH)(CN)$ — Ketone cyano-hydrin; for $R'=H$, Aldehyde cyanohydrin

$(-I)_2C=O + N.u^-$ (Reactivity ↑)
Resimisan takes place.

Rktivity of $>C=O$ gp toward $Nu^\ominus$
$\propto -I \propto \frac{1}{+I}$

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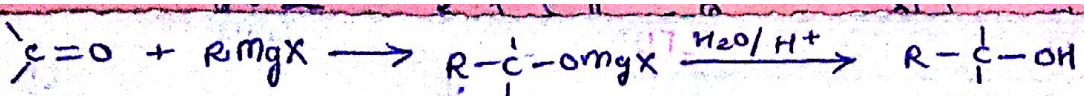
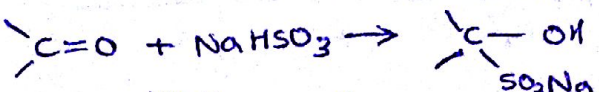
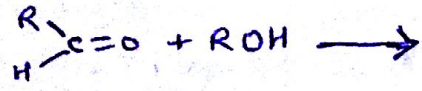
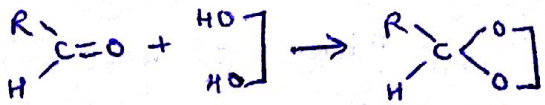
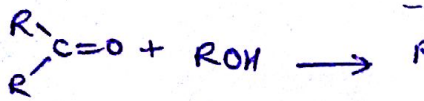
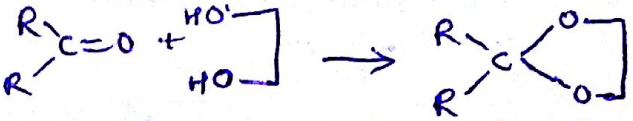
- 

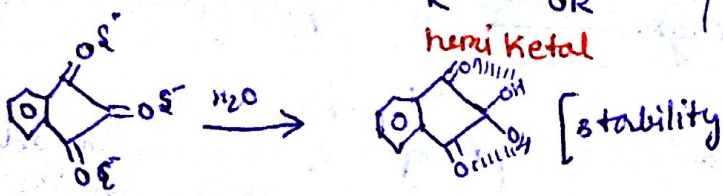
- Addⁿ of sodium bi sulphate ($NaHSO_3$) : 

- 
  hemiacetal

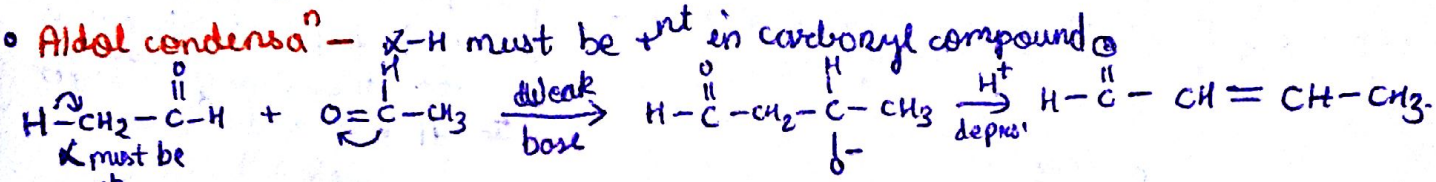
- 
  Acetal

- 
  hemi Ketal

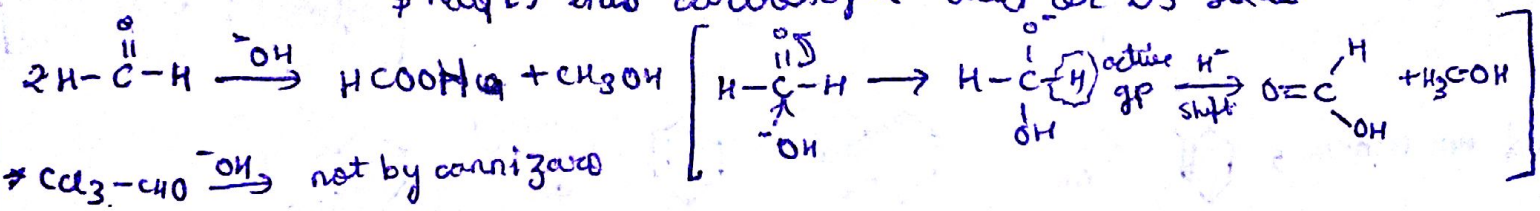
- 
  Ketal

- 
  [stability of hydrate ∝ −I]

- **Aldol condensaⁿ** – α-H must be +ʳᵗ in carbonyl compound



- **Cannizaro reckⁿ** – In this reckⁿ, half [½] of carbonyl gp convert into alcohol & half [½] into carboxylic acid or its salt



≠ $CCl_3-CHO \xrightarrow{OH^-}$ not by cannizaro

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CHEMICAL KINETICS

• $2A(g) \rightarrow 3B(g)$ $\quad -\frac{\Delta[A]}{2\Delta t} = +\frac{1}{3}\frac{\Delta[B]}{\Delta t} =$ Rate of reac^n

• $\frac{\Delta P}{\Delta t}$ (atm/time) $= RT\,\frac{\Delta[\,]}{\Delta t} \rightarrow mole^{-1}t^{-1}$

for $2A \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} 3B$ $\quad \left\{\frac{-1}{2}\frac{\Delta[A]}{\Delta t}\right\}_{overall} = \left\{\frac{1}{3}\frac{\Delta[B]}{\Delta t}\right\}_{overall} = (R_f - R_b)$

• for $aA + bB \rightarrow cC + dD$, a = molecularity of rek^n w.r.t A

Rate of rek$^n \Rightarrow K[A]^x[B]^y$ $\quad x$ = order w.r.t A

• If $x=a$ & $y=b$ for $\forall$ rekt $\rightarrow$ elementary / simple rek^n (single step rek^n)

If not, complex reac$^n \rightarrow$ multistep rek^n [complex rek^n] Rkt $\rightarrow$ Intermediate $\rightarrow$ pdt

• $n'A \xrightarrow{K}$ pdt [order of rek$^n = n$]

t=0	a	
t=t	a−x	x

$\frac{1}{n'}\frac{dx}{dt} = K(a-x)^n \quad \int\frac{dx}{(a-x)^n} = \int n'K\,dt$

for $n \neq 1$, $\frac{1}{[Rkt]_t^{n-1}} - \frac{1}{[Rkt]_0^{n-1}} = \underbrace{n'(n-1)Kt}_{m}$ $\quad \downarrow x$

$(y = mx + c$ st. line$)$ $\quad$ slope $\rightarrow n'(n-1)K$ (+ve) $\quad n>1$ (slope +ve), $n<1$

Intercept $\rightarrow \frac{1}{[Rkt]_0^{n-1}} = [Rkt]_0^{1-n}$

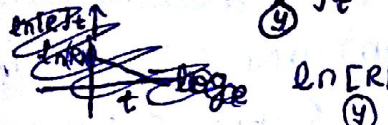
★ $[Rkt]^{1-n}$ vs time graph $\rightarrow$ st. line [order of rek$^n \rightarrow n$]

$t_{1/2}$ = time period in which Rkt become half, $\quad t_{1/2} = \frac{2^{n-1}-1}{n'(n-1)K[Rkt]_0^{n-1}}$ $\quad \& \quad t_{1/2} \propto [Rkt]_0^{1-n}$

Ex ⇒ 0 order : 16M $\xrightarrow{20min}$ 8M $\xrightarrow{10min}$ 4M $\xrightarrow{5min}$ 2M

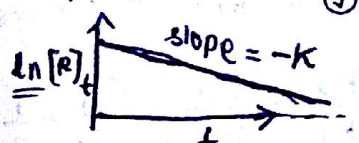
⇒ 2nd order : 16M $\xrightarrow{20min}$ 8M $\xrightarrow{40min}$ 4M $\xrightarrow{80min}$ 2M

for $n=1$, $[Rkt]_t = [Rkt]_0\,e^{-Kt}$ ② $\quad$ time taken in comp. of 1st order rek$^n = \infty$



④ $\ln[Rkt]_t = \ln[Rkt]_0 - Kt$ (c, m, ②) $\Rightarrow$ slope $= -K$ $\quad \& \quad \log_{10}[Rkt]_t = \log_{10}[Rkt]_0 - \frac{Kt}{2.303}$

slope $= \frac{-K}{2.303}$

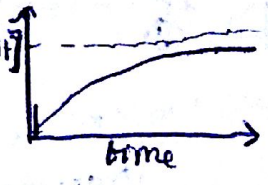


Also, $(\text{Rate of rek}^n)_t = (\text{Rate of rek}^n)_0\,e^{-Kt}$

• Pdt ↔ time $\quad A \rightarrow B$

t=0	a	0
t=t	a−x	x

$a - x = ae^{-Kt} \Rightarrow x = a[1 - e^{-Kt}]$



• Unit of K : $(mole^{-1})^{1-n}\,t^{-1}$, $\quad K = \frac{2.303}{t}\log_{10}\frac{[Rkt]_0}{[Rkt]_t}$ — ①

• In the first order rek^n, time taken in comp. of a certain % of rek^n is independent of initial [ ] of Rkt^n.

Ex → 27M $\xrightarrow{1hr}$ 9M $\xrightarrow{1hr}$ 3M $\xrightarrow{1hr}$ 1M

→ $t_{1/2} = \frac{2.303\log 2}{K} = \frac{\ln 2}{K} = \frac{0.693}{K}$ $\quad \{t = x\,t_{1/2} \quad [Rkt]_t = \frac{[Rkt]_0}{2^x}$

⇒ In ① any factor proportional to [ ] can be used. i.e → Partial press. of gas.

Ex: $K = \frac{2.303}{t}\log_{10}\frac{P_{t=0}}{P_{t=t}}$

• Prolong time : $t \rightarrow \infty$ [100% dissociation, $\alpha = 1$]

• Volume and optical rotation can be used

$A \xrightarrow{Z_1} B \quad (Z_2)$

t=0	a	0	$Q_0 = Z_1 a$
t=t	a−x	x	$Q_t = Z_1(a-x) + Z_2 x$
t=∞	0	a	$Q_\infty = Z_2 a$

$Q_0 - Q_\infty = a(Z_1 + Z_2)$

$Q_0 - Q_t = x(Z_1 + Z_2)$

$K = \frac{2.303}{t}\log\frac{Q_0 - Q_t}{Q_t - Q_\infty}$

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• $\left[\frac{-d[A]}{dt}\right]_{path\,I} = \frac{K_1[A]}{(K_1+K_2)[A]} = \frac{K_1}{K_1+K_2}$

$\left[\frac{-d[A]}{dt}\right]_{total}$

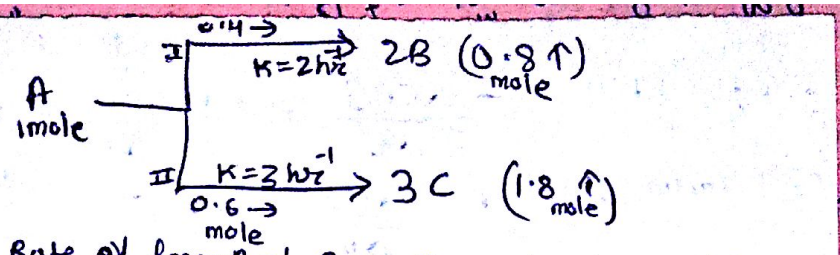
• $A \rightarrow bB + cC$.

$\frac{[B]_t}{[C]_t} = \frac{bK_1}{cK_2}$.

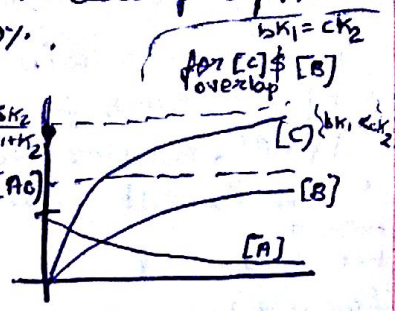
Pseudo 1st order Rkⁿ →
• Acidic hydrolysis of ester.
Rate of rkⁿ : $K\left[R-\overset{O}{\overset{\|}{C}}-OR'\right][H_2O]^0[H^+]$ → used as catalyst
→ used as solvent (const)

$K_{exp} = \frac{2.303}{t}\log\left\{\frac{V_\infty - V_0}{V_\infty - V_t}\right\}$

$V \propto \frac{1}{C}$

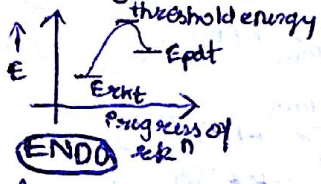


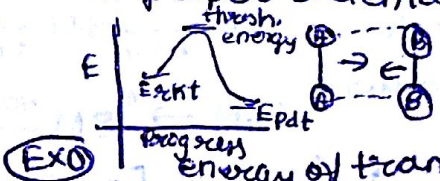
Rate of formaⁿ of B = 80% decomp. of A
% A decomposed via I = 40%.
At any time $\frac{[B]}{[C]} = \frac{0.8}{1.8}$



• Inversion of cane sugar : sugar + H_2O ⇒ fructose + glucose
• Rate constant : $[mol\,l^{-1}]^{1-n}\,t^{-1}$ → T↑ Rate constant↑ & use of catalyst → K↑
• Arhenius hypothesis of collision → Energy of colliding Rkt molecule ≥ min value
→ they must collide in proper orientation

Rkt → Transiⁿ state → Pdt
(Less stable than Rkt/Pdt.)





⇒ Min. amount of energy req. to be supplied to rkt molecule for crossing the threshold energy barrier. (energy of transiⁿ state = threshold energy)

ACTIVATION ENERGY

Ea ⇒ Erkt. → Etrans.

• no. of moles of collision per unit time → Total Collision freq. (Z_i)
• $K = \{ZP_iE$ P → orientaⁿ factor
$= Ae^{-E_a/RT}$ E → energy factor.
A → arhenius factor.

$f = e^{-E_a/RT}$

$\log\left(\frac{temp.}{coeff.}\right) = \frac{E_a}{2.303R}\left[\frac{1}{T} - \frac{1}{T+10}\right]$

T & f = b/w 2 & 3.

$\ln K = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right)$ (y) (x)

K↑ → T↑ Ea↓

$\log\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

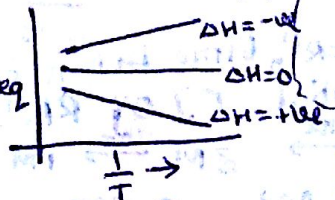
Units of R.

⇒ $\log\left(\frac{K_{efficiency\ of\ catalyst}(F)}{K_{ab.\ of\ catalyst}(F)}\right) = \left[\frac{E_{a(F)_{nc}} - E_{a(F)_{pc}}}{2.303\,RT}\right]$

$\frac{K_{PC(F)}}{K_{PC(B)}} = \frac{K_{AC(F)}}{K_{AC=(b)}}$

$(K_{eq})_{PC} = (K_{eq})_{AC}$.

$E_{pdt} - E_{rkt} = E_{a(F)} - E_{a(b)}$
$= (\Delta H)_R$



⇒ In enzyme mech. only lock key type system of catalyst

• Complex Rkⁿ → ① we assume formaⁿ of Intermediate
② seperate into no of steps = no. of Intermediate + 1
③ Slowest step → (RDS)order = (Rkⁿ)order.
× ④ order is always expressed wrt rckt/pdt not Intermediate.

⇒ use of K_{eq} for eliminaⁿ of Intermediate.

Homogeneous catalytic syst
Rkt & catalyst are in same phase.

Heterogeneous catalytic sys
Rkt & catalyst → diff. phase
Ex: Ni → surface area. for adsorpⁿ
highly exothermic
E_a ↑ which reduces

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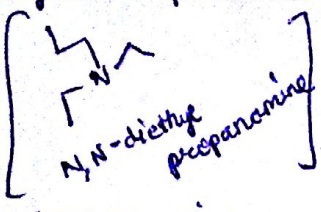
AMINES - ammino derivatives of alkane.

- General formula : 1^0 RNH_2 | 2^0 R_2NH | 3^0 R_3N



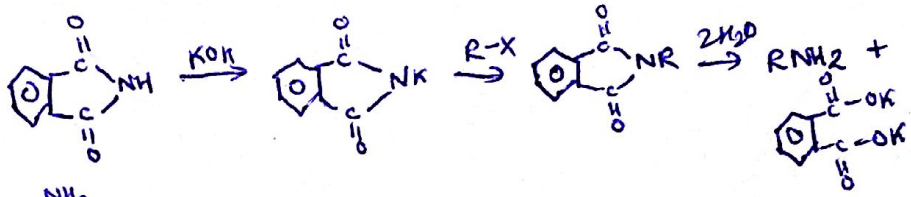
1° amine
- chain
- posⁿ
- optical
- Fn
- geom.

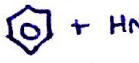
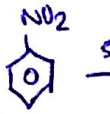
2° amine
- Fn
- metamerism
  - chain
  - posⁿ

3° amine
- Fn
- metamerism
  - chain
  - posⁿ

⇒ GMP of amine -

- $R-CN \xrightarrow[Sn+HCl]{Reducⁿ} R-CH_2-NH_2$ (also $\xrightarrow[C_2H_5OH]{Na}$)
- $R-NO_2 \xrightarrow{LiAlH_4} R-NH_2 + H_2O$
- $R-OH + NH_3 \rightarrow R-NH_2 + H_2O$
- $R-X + NH_3 \rightarrow R-NH_2 + HX \xrightarrow{\downarrow AX} R-NH-R$.
- $R-MgX + Cl-NH_2 \rightarrow R-NH_2$. (chloroamine)
- Gabriel phthalimide synthesis -



-  $+ HNO_3 + H_2SO_4 \rightarrow$  $\xrightarrow{Sn+HCl}$

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NUCLEAR CHEMISTRY

- Mass of nucleons > mass of nucleus [mass defect, Δm = mass of nucleons − mass of nucleus]
- Binding energy = $(\Delta m)c^2$
- for getting stability,

I will try to ↓ $\left(\frac{n}{p}\right)$

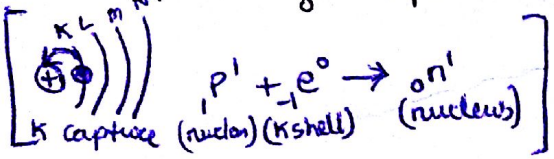
$n \rightarrow p$ (β emission)

${}_0n^1 \rightarrow {}_1p^1 + {}_{-1}e^0$

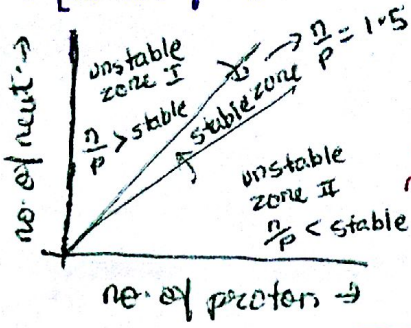
II will try to ↑ $\left(\frac{n}{p}\right)$

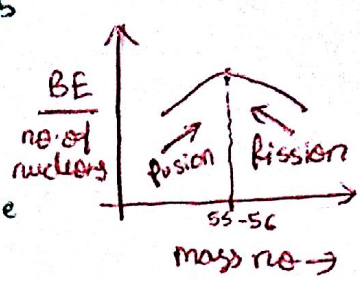
$p \rightarrow n$ (e^+ → positron emission or K capture)

${}_1p^1 \rightarrow {}_0n^1 + {}_1e^0$



${}_1p^1 + {}_{-1}e^0 \rightarrow {}_0n^1$ (nucleus) (K shell) (nucleus)





(−1) β particle : At. no ↑ by 1 & mass no unchanged
(−1) e^+ : At no ↓ by 1 & mass no unchanged
(−1) α particle : At no ↓ by 2 ; mass no ↓ by 4
(−1) γ emmision : No change

- Rate of radioactive disintegration (in disintegration per sec or dps) −

SI : 1 Becquerel (Bq) = 1 dps.

other : 1 Rutherford = 10^6 dps

1 curie (Ci) = 3.7×10^{10} dps.

$\frac{-dN}{dt} \propto N_t$ [λ = decay const.]

- $N_t = N_0 e^{-\lambda t}$
- $N_t = \frac{N_0}{2^{t/t_{1/2}}}$, $t_{1/2} = \frac{\ln 2}{\lambda}$, $t_{avg} = \frac{1}{\lambda}$
- $\lambda = \frac{2.303}{t} \log \frac{N_0}{N_t}$

	A^{200}	$\rightarrow$	B^{150}
N_0	x mol		0
	−K		+K
N_t	x−K		K

$\frac{n_A}{n_B} = \frac{5}{3}$.

$\frac{x-K}{K} = \frac{5}{3}$, $x = \frac{8K}{3} \& N_0$

- Fossil : Age determination. C^{12} (Non radioactive), C^{14} (radioactive)

$\frac{C^{14}}{C^{12}}$ or $\frac{C^{14}O_2}{C^{12}O_2}$ = fixed ratio

$\left(\frac{C^{14}}{C^{12}}\right)_{fossil} = N_t$, $\left(\frac{C^{14}}{C^{12}}\right)_{nature} = N_0$

$t_{1/2}$ of C^{14} = 5730 yrs.

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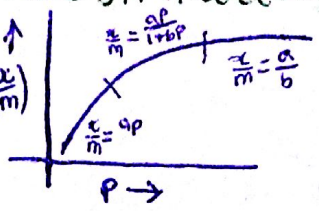
SURFACE CHEMISTRY

- Adsorp^n : exothermic in nature.
- Adsorbent – solid/lqd surface on which adsorp^n takes place
- Adsorbate – gas molecule whose adsorp^n takes place.

Gas (free) → gas (adsorbed)
$\Delta S = -ve$
$\Delta H = \Delta G + T\Delta S = -ve$
(↓ -ve, ↓ -ve)

- In liquid, adsorp^n is due to inward pull experienced by molecule at lqd surface.
- In solid, adsorp^n is due to unsatisfied valency over the surface of solid [more corners, more adsorp^n]

→ Factors on which adsorp^n depend : Nature of gas (adsorbate); more Intermolecular force of attrac^n, more adsorp^n
: Surface area of adsorbent; more SA more adsorp^n (for same S.A. more corners more adsorp^n)
: Temp ↑ adsorp^n ↓
: Pressure; Langmuir



$(1+bP \to bP)$ high press^n $\Rightarrow \frac{x}{m} = \frac{a}{b} = const$
$(1+bP \to 1)$ low press^n $\Rightarrow \frac{x}{m} = aP$

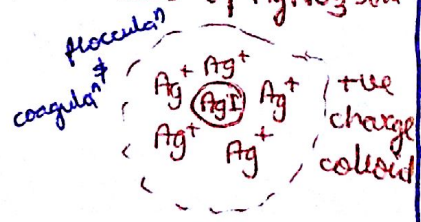
$\frac{x}{m} = \frac{aP}{1+bP}$ a & b depend upon nature of adsorbate & adsorbent

- Colloid ┬ dispersed phase
  └ dispersing medium

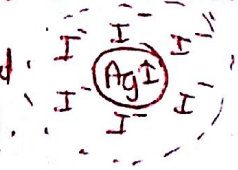
sol^n : size of particle < 1nm
suspens^n : size of particle > 1000 nm
colloid : 1nm < size of particle < 1000 nm.

- lyophilic colloid : when their would be attrac^n force b/w DP & DM.
- lyophobic colloid : when their would be repuls^n force b/w DP molecule. → charged in nature
- Preferential adsorp^n theory : colloid particle will preferably adsorb its cation or anion on its surface preferably.

→ small amount of KI is added in excess of $AgNO_3$ soln^n
we added a -ve charge ion [I^-]



→ small amount of $AgNO_3$ is added in excess of KI soln^n
we added a +ve charge ion (Ag^+)



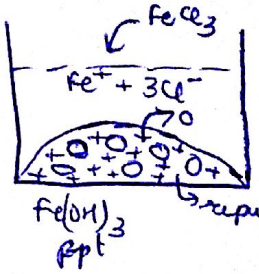
-ve charged colloid.

LYOPHILIC COLLOID	LYOPHOBIC COLLOID
• more stable	• less stable
• Reversible	• irreversible
colloid $\xrightarrow{-DM}$ DP $\xrightarrow{+DM}$ colloid	colloid $\xrightarrow{-DM}$ DP $\xrightarrow{+DM}$ no colloid
• surf. tens^n of colloid < ST of DM	• almost equal
• Viscosity of colloid > viscosity of DM	• Almost equal

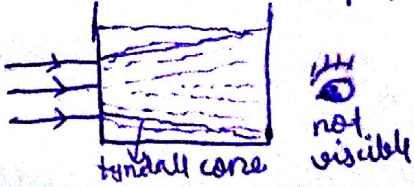
- for better flocculn^n or coaguln^n –
  ① Charge on ion should be more : $Al^{3+} > Ca^{2+} > Na^+$
  $PO_4^{3-} > SO_4^{2-} > Cl^-$
  ② Size of ion should be small : $Li^+ > Na^+ > K^+$
  $Mg^{2+} > Ca^{2+} > Sr^{2+}$
  $F^- > Cl^- > Br^-$

Also known as HARDY SCHULZE LAW

- freshly ppt $Fe(OH)_3$
  Peptizing agent ($FeCl_3$)
  Process : Peptisn^n



⇒ Tyndall effect : Path of light gets illuminated in colloid if observed ⊥ to direc^n of light
Condi^n : ① n of DM ≠ n of DP
② λ of light ∼ size of DP



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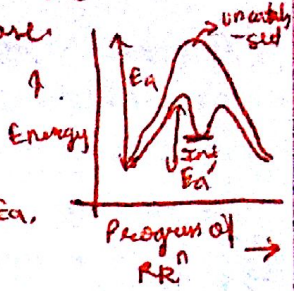
• Catalysis : ↑ Rate of RR^n ⇕ Rate const. ↓ activa^n energy [$K = A \cdot e^{-E_a/RT}$

Ⓐ Homogeneous catalytic system – RRt & catalyst are in same phase.
[intermediate is formed.



Ⓑ Heterogeneous catalytic system – Rkt & catalyst are in diff. phase.
↳ provide surface for adsorp^n

Rkt got adsorbed → heat is released (exothermic), the released heat ↓ value of E_a.

Catalyst – ① specific [diff. for diff. process]
② specific condi^n [~~amount does not chan~~ Alkaline, acidic etc]
③ Amount does not change before & after rek^n.

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USEFUL REAGENTS ①

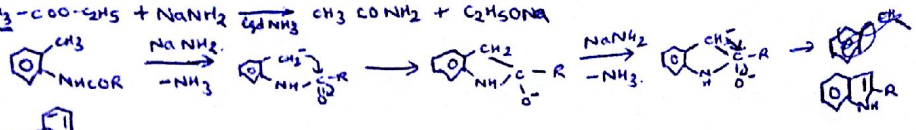
① **Sodium** → To detect the presence of active hydrogen → In Wurtz rᵏⁿ
→ As reducing agent with ethanol or water → for detecⁿ of element. org. comp's + Na → NaCN; 2Na + S → Na_2S; 2Na + X_2 → 2NaX.

② **Zn-Cu couple** : used as red. agent with alcohol [Zn-Cu + C_2H_5OH] , $CH_3I \xrightarrow[C_2H_5OH]{Zn-Cu} CH_4 + HI$

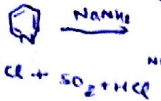
③ **Soda lime** (NaOH + CaO) → both in solid form : It rᵏt with sodium salt of fatty acid to produce & undergoes decarboxylaⁿ to give hydrocarbon with one carbon less. → $C_6H_5-Z + Na_2CO_3$ [Z may be $-NO_2$, $-OH$, $-CH_3$ etc]

④ **Sodamide** [$NaNH_2$] → $NaNH_2 \rightleftharpoons Na^+ + NH_2^-$

- $-CHBr-CHBr- + 2NaNH_2 \xrightarrow{Liq\,NH_3} -C\equiv C- + 2NaBr + 2NH_3$.
- $CH_3-C\equiv C-CH_3 \xrightarrow[Liq\,NH_3]{NaNH_2} CH_3-CH_2-C\equiv CH$
- Rᵏⁿ of ether with sodamide to form alkene : $NH_2^- + H-CH_2-CH_2-O-C_2H_5 + Na^+ \longrightarrow NH_3 + CH_2=CH_2 + C_2H_5ONa$
- with ester having α-H : $CH_3-COO-C_2H_5 + NaNH_2 \xrightarrow{Liq\,NH_3} CH_3CONH_2 + C_2H_5ONa$
- cyclizaⁿ of anilide comp's:



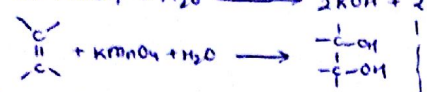
- with pyridine :



⑤ **Thionyl chloride** : [$SOCl_2$] $ROH + SOCl_2 \rightarrow RCl + SO_2 + HCl$; $RCOOH + SOCl_2 \rightarrow R-COCl + HCl + SO_2$; $CH_2(OH)-COOH \xrightarrow{SOCl_2} CH_2Cl-COCl$

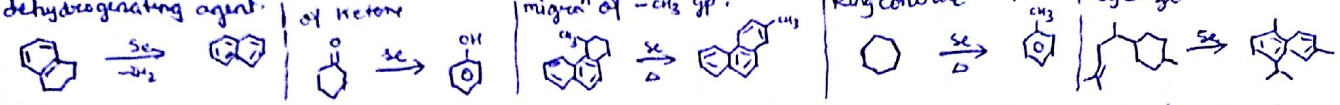
⑥ **Potassium permanganate** : ($KMnO_4$) $2KMnO_4 + H_2O \xrightarrow{Na_2CO_3} 2KOH + 2MnO_2 + 3[O]$

Known as Bayer's reagent in alkaline medium.



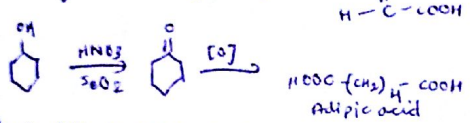
used to detect unsaturaⁿ

⑦ **Selenium** : dehydrogenating agent. (Se)



⑧ **Selenium dioxide** : oxidaⁿ of reactive methyl & methylene gp ♦ (SeO_2)

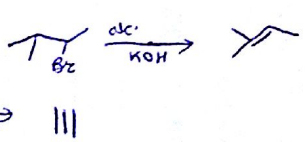
attacks $-CH_2-C(=O)-$ to give $-C(=O)-C(=O)-$

- $CH_3-CO-CH_2-CH_2-CO-CH_3 \xrightarrow{SeO_2} CH_3-CO-CH_2-CH=CH-CO-CH_3$
- catalytic trans hydroxylaⁿ : $H-C(=O)-COOH \xrightarrow[\text{tert-butyl alcohol}]{H_2O_2,\ SeO_2}$ meso-tartaric acid
- 

⑨ **Aq caustic potash** : (Aq KOH) $RX + KOH \rightarrow R-OH + KX$

$HC(Cl)_3 \xrightarrow{KOH} H-C(OH)_3 \rightarrow HCOOH + 3KCl$

⑩ **Alcoholic caustic potash** : (alc. KOH)



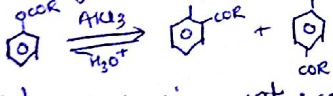
⑪ **Hydriodic acid** : (HI) Good reducing agent which convert -OH to -H

$C_2H_5OH + 2HI \longrightarrow C_2H_6 + H_2O + I_2$

$CH(OH)-COOH \xrightarrow{4HI} CH_2-COOH + 2H_2O + 2I_2$.

⑫ **Anhydrous $AlCl_3$** : friedal ocraft Rᵏⁿ (help in producing electrophile from given reagent)

→ fries rearrangement.



→ Gattermann Koch aldehyde synthesis : $C_6H_6 + CO + HCl \xrightarrow{AlCl_3} C_6H_5CHO + HCl$.

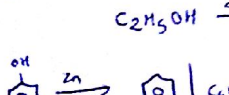
→ Gattermann aldehyde synthesis : $C_6H_6 + HCl + HCN \xrightarrow[H_2O]{AlCl_3} C_6H_5CHO$

→ Isomerisaⁿ : $CH_3-CH_2-CH_2-Cl \xrightarrow{AlCl_3} CH_3-CH(Cl)-CH_3$ [also with alkane]

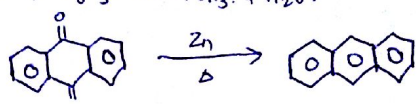
⑬ **Anhydrous Zinc chloride** ($ZnCl_2$) : dehydrating agent : can be used in place of $AlCl_3$ in friedel ocraft rᵏⁿ's

$C_2H_5OH \xrightarrow{anhy\ ZnCl_2} CH_2=CH_2 + H_2O$; $C_6H_5NH_2 + HOOCCH_3 \xrightarrow{ZnCl_2} C_6H_5-NH-COCH_3 + H_2O$.

⑭ **Zinc dust** :



$C_6H_5CH_2OH \xrightarrow{Zn\ dust} C_6H_5CH_3 + ZnO$ [Zn & alc KOH — Reducing agent]



Zn with CH_3COOH, HCl & KOH is used as reduc. agent.

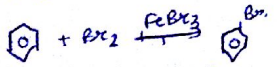
⑮ **Bromine** :

if brown colour of bromine is discharged by shaking with comp. this comp is unsaturated

$CH_2=CH_2 + Br_2 \xrightarrow{CS_2} CH_2Br-CH_2Br$.

no effect of this addiⁿ on aromatic ring.

$R-OH + P-Br_2 \longrightarrow RBr$.



$RCOOAg + Br_2 \rightarrow RBr + AgBr + CO_2$

⑯ **Bromine & caustic potash** : (Br_2 + KOH) convert $-CONH_2$ to $-NH_2$.

Known as Hofmann Bromamide rᵏⁿ

$BrO^\ominus / OH^\ominus$

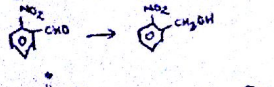
$CH_3CONH_2 + Br_2 + 4KOH \rightarrow CH_3NH_2 + 2KBr + K_2CO_3 + 2H_2O$

[similar for aromatic amides]

⑰ **Aluminium isopropoxide** $[(CH_3)_2CHO]_3Al$:

→ specific red. agent for carbonyl gp only.

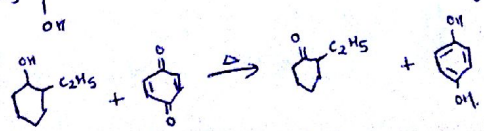
$CH_3-CH=CH-CHO \longmapsto CH_3CH=CH-CH_2OH$



⑱ **Aluminium tri-ter-butoxide** $[(CH_3)_3CO]_3Al$: as oxidising agent.

no effect on double bond

$CH_3-CH(OH)-CH_2-CH_3 + CH_3-CO-CH_3 \xrightarrow{\Delta} CH_3-CO-CH_2CH_3 + CH_3-CH(OH)-CH_3$ [opposite of oxidaⁿ]



$CH_3-CH(OH)-CH=CH \cdot CH-CH=CH_2 \longrightarrow CH_3-CO-CH=CH-CH(CH_3)-CH=CH_2$

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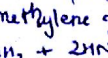
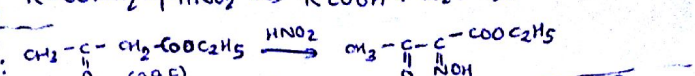
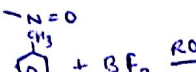
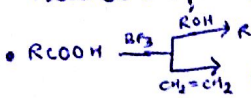
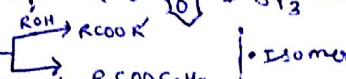
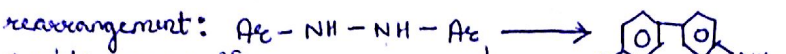
USEFUL REAGENTS (2)

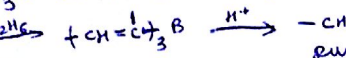
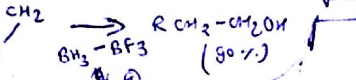
(19) Nitrous acid (HNO_2): ice cold $NaNO_2$ + cold HCl → NaCl + HNO_2.
beyond 5°c, it undergoes autoxidaⁿ to give HNO_3.

- 1° amine aliphatic $R-NH_2 \xrightarrow{HNO_2} R-OH + N_2 + H_2O$
- active methylene gp forms an oxime with HNO_2.:
- $NH_2CONH_2 + 2HNO_2$ (excess) → $2N_2 + CO_2 + 3H_2O$
- + HONO → 

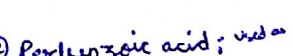
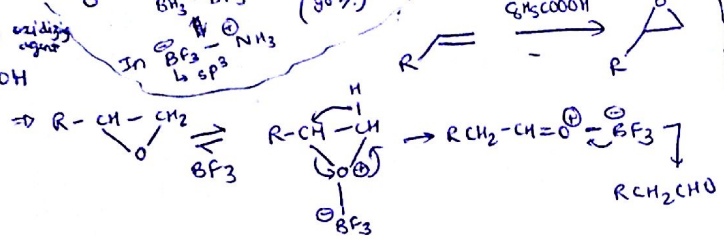
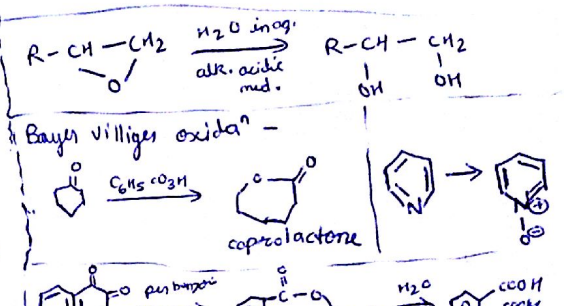
- aliphatic & aromatic amides:
  $R-CONH_2 + HNO_2 \rightarrow RCOOH + N_2 + H_2O$
- 

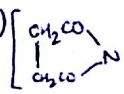
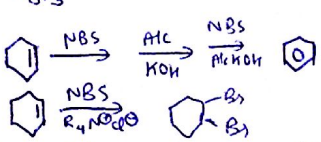
(20) Boron trifluoride: Friedel craft Rkⁿ: 

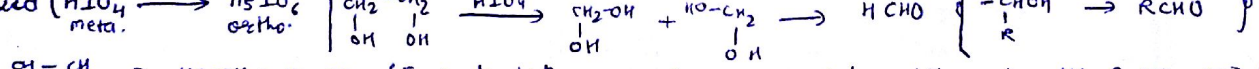
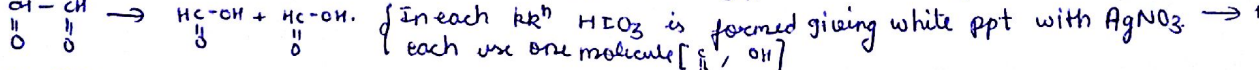
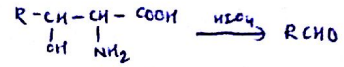
- 
- Isomerisaⁿ: 
- Benzidine rearrangement: 
- Methylaⁿ of acid: $RCOOH \xrightarrow[CH_3OH]{BF_3} RCOOCH_3$

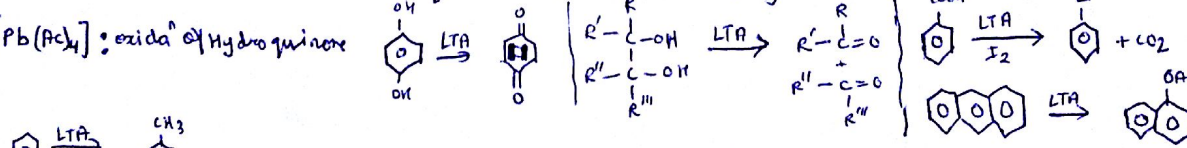
(21) Diborane (B_2H_6): Hydroboraⁿ
: $(R-CH_2-CH_2)_3B \xrightarrow{H^+} 3RCH_2CH_3$
: 
- 

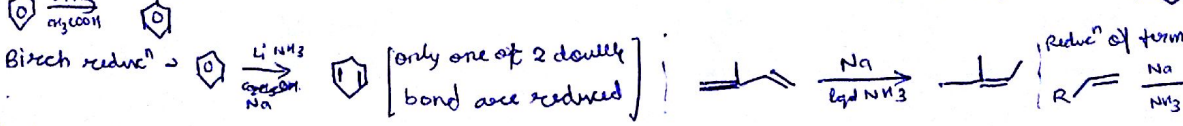
B_2H_6 in THF is reducing agent. [Ex: $-COOH \rightarrow -CH_2OH$]
does not reduces $-COCl$, RNO_2, $ArNO_2$, RX,

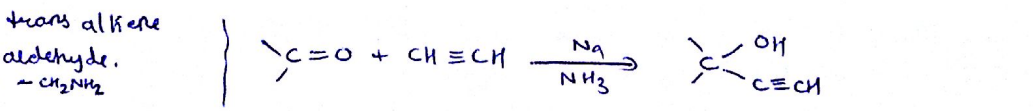
(22) Perbenzoic acid: used as oxidising agent




(23) N-Bromosuccinimide (NBS) 
⇒ as oxi. agent with some reagent.

⇒ NBS in aq. solⁿ decarboxylate α-ammino acid of protein & peptide.

(24) Periodic acid (HIO_4 meta, H_5IO_6 ortho)


- 

(25) Lead tetra acetate [$Pb(Ac)_4$]: oxidaⁿ of Hydroquinone (LTA)


(26) Liquid Ammonia: Birch reducⁿ → 
- Red of ≡ to trans alkene
  amide → aldehyde.
  $-C\equiv N$ to $-CH_2NH_2$



(27) CH_2N_2 (diazo methane)

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	Crystal Systems	Sides	axial angles	Bravais lattices	
C	Cubic	$a=b=c$	$\alpha=\beta=\gamma=90^\circ$	Simple, BCC, FCC	3
R	Rhombic	$a=b=c$	$\alpha=\beta=\gamma\neq90^\circ$	Simple	1
T	tetragonal	$a=b\neq c$	$\alpha=\beta=\gamma=90^\circ$	Simple, BCC	2
H	hexagonal	$a=b\neq c$	$\alpha=\beta=90^\circ, \gamma=120^\circ$	Simple.	1
O	orthorhombic	$a\neq b\neq c$	$\alpha=\beta=\gamma=90^\circ$	Simple, BCC, FCC ECC	4
M	monoclinic	$a\neq b\neq c$	$\alpha=\gamma=90^\circ, \beta\neq90^\circ$	Simple, ECC.	2
T	triclinic	$a\neq b\neq c$	$\alpha\neq\beta\neq\gamma\neq90^\circ$	Simple.	1
					14

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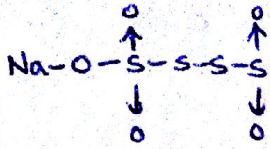
ELECTROCHEMISTRY



$23 \Rightarrow +5$
$23 \Rightarrow \pm 0$

- charge on 1 mole of $e^- = 96500\ C = 1\ F$
- mass of OA reduced by 1F = Eq mass of OA

- Potential of all other half shell are measured wrt SHE | Electrochemical series

Li —— (RP = (-ve), max metal) —— H —— (RP = +ve, max non metal) —— F_2

$E^\circ_{Li^+/Li}$ = max^m ⊖ve (Best reducing agent)

$E^\circ_{H^+/H_2} = 0$

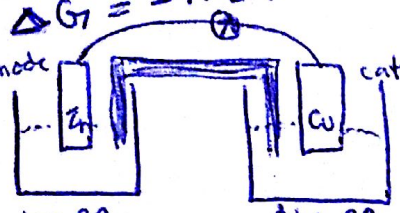
Oxidising power increases.

$E^\circ_{F_2/F^-}$ = ⊕ max^m (Best oxidising agent)

Cu, Ag, Hg, Au, Pt $\not\Downarrow +H^+$ not liberate H_2.

- RP ↑ → will be Reduced → better oxidising agent [act as cathode]
- RP ↓ → will be oxidised → better red^n agent [act as anode]

$\Delta G = -nEF$



$E^\circ_{cell} = (RP)^\circ_{cathode} - (RP)^\circ_{anode}$

feasibility : $E_{cell} = +ve$

$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log Q$

$E_{cell} = 0.0591\ [PH(A) - PH(C)]$

- Eqb of galvanic cell : $E_{cell} = 0$, $E^\circ_{cell} = \frac{0.0591}{n} \log K_{eq}$.
- $K_{sp}(AgCl) = [Ag^+] \times [Cl^-]$ [think at $E_{cell} = 0$]
- Faradays law of electrolysis ① mass deposited ∝ charge passed

$w = zIt$

$w = z \cdot Q$ → electrochemical unit ⇒ gm/C equ.

$Ca^{2+} + 2e \rightarrow Ca$ (2×96500 : 40 gm)

$Z_{Ca} = \frac{40}{2 \times 96500}$ gm/C

$2Cl^- \rightarrow Cl_2 + 2e$

$Z_{Cl_2} = \frac{71}{2 \times 96500}$

- 1C charge ⇒ mass deposited ⇓ Electrochemical equivalence of substance (z)
- 1F charge ⇒ mass deposited ⇒ Eqv mass of substance (E)

$E = 96500 \times z$

- Electrolysis [electric energy → Redox rk^n]
  - → of molten NaCl : Cl_2 at anode, Na^+ deposit at cathode
  - → of aq. soln of NaCl : Cl_2 at anode, H_2 at cathode.
  - → of acidic / alk H_2O : n_{H_2} liberated = $2n_{O_2}$ liberated
  - → of aq. soln of $CuCl_2$: Cl_2 at anode, Cu is deposited

② If the charge passed is same the mass of diff substances deposited in resp electrodes is ratio of their resp equivalent mass. $w_1 = z_1 Q$, $w_2 = z_2 Q$

$\frac{w_1}{w_2} = \frac{z_1}{z_2} = \frac{E_1}{E_2}$

Case I) molten $AlCl_3$

$\frac{w_{Al}}{w_{Cl_2}} = \frac{E_{Al}}{E_{Cl_2}}$

Case II) three diff elect. cell

① molten NaCl, $CaCl_2$, $AlCl_3$

$w_{Na} : w_{Ca} : w_{Al} = E_{Na} : E_{Ca} : E_{Al}$

$E_{Na} = \frac{23}{1}$, $E_{Ca} = \frac{40}{2}$, $E_{Al} = \frac{27}{3}$

- Conductance in electrolytic soln is due to migra^n of ions in soln.

→ more free migra^n more conductance.

- Electrolyte is that solute which gets ionised in aq. soln.
- Factors affecting migra^n of ion — Ion ion soln^n [movement of ion are in hydrated form]
  - Ion solv^n interac^n
  - solvent solvent interac^n

conductance [$Na^+ > Li^+$]

- Among all cation, conductance due to H^+ is max^m in aq soln
- among all anion, conductance due to ^-OH is highest.
- Diff in conductance in
  - electrolytic soln ≡ migra^n of A^+, B^- — T↑ C↑
  - Metallic wire — migra^n of e^- — T↑ C↓

- $\frac{1}{\rho} = \frac{1}{R} \times \frac{\ell}{A}$

$K = c \times$ cell const

specific conductance ↓ conductance

- Charge is made fixed.

① Molar conductance (Λ_m) — conductance shown by that vol. of soln containing 1 mole of electrolyte.

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- unit of Λ_m : SI = $ohm^{-1} m^2 mole^{-1}$ & $\Lambda_m = K\left(\frac{1}{C} \times 10^{-3}\right)$ $ohm^{-1} m^2 mole^{-1}$
- Equ conductance : conductance shown by that vol of soln which contains 1F of cationic & (Λ) Ex) 1 mole $Al_2(SO_4)_3 \rightarrow$ 2 mole Al^{3+} of 1F of anionic charge
  3 mole SO_4^{2-} of & 1 eqv of $Al_2(SO_4)_3 = \frac{1}{C}\Lambda_m$

$$\Lambda = K \times \frac{1}{N} \times 10^{-3}\ ohm^{-1} m^2 eqv^{-1}$$

- Kohlrausch's law : $\Lambda_\infty = \Lambda_{C^+} + \Lambda_{A^-}$ & $\alpha = \frac{\Lambda}{\Lambda_\infty}$

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Salt analysis

• **Acid Radical :**

dil gp	conc. gp	Specific gp
SO_3^{2-}, S^{2-}, CO_3^{2-}, NO_2^-, CH_3COO^-,	Cl^-, Br^-, I^-, BO_3^{2-}, $C_2O_4^{2-}$,	F^-, SO_4^{2-}, PO_4^{3-}, CrO_4^{2-}, BO_3^{2-}.

• CO_3^{2-} : $CO_3^{2-} \xrightarrow{dil\ HCl}$ CO_2 gas (colourless odourless) $\xrightarrow{lime\ H_2O}$ milkiness $\xrightarrow{excess\ CO_2}$ milkiness disappear.

• $SO_3^{2-} \xrightarrow{dil\ HCl} SO_2\uparrow$ (smell of burning sulphur) $\xrightarrow{lime\ H_2O}$ milkiness $\xrightarrow{excess\ SO_2}$ Milkiness disappear.

$Cr_2O_7^{2-} + CO_2 \rightarrow$ no rxn

$Cr_2O_7^{2-} + SO_2 \rightarrow SO_4^{2-} + Cr^{3+}$ (orange) (green)

• $S^{2-} \xrightarrow{dil\ HCl}$ H_2S gas (smell of rotten egg) $\xrightarrow{Paper\ dipped\ in\ Pb(Ac)_2}$ Paper is blackened. ($PbS\downarrow$)

* with ↑ in PH oxid. prop. of dichromate ion ↓

• $S^{2-} \xrightarrow{CdCl_2} \overset{+2}{Cd}S\downarrow$ (yellow) $\xrightarrow{KCN} [\overset{+2}{Cd}(CN)_4]^{2-}$ (colourless) Less stable $\xrightarrow{H_2S} CdS\downarrow$ (yell. ppt)

• $S^{2-} \xrightarrow{CuCl_2} \overset{+2}{Cu}S\downarrow$ (black) $\xrightarrow{KCN} [\overset{+1}{Cu}(CN)_4]^{3-}$ (colourless) more stable $\xrightarrow{H_2S}$ No ppt obs.

Na... test : $S^{2-} + Na_2[Fe(\overset{-1}{CN})_5(\overset{+1}{NO})] \rightarrow Na_4[Fe(CN)_5(NOS)]$ (purple)

(Grp 2 & 4 sulphide would not give this test)

N,N-Dimethyl-p-phenylene-diamine test

$+ S^{2-} \xrightarrow[H^+]{Fe^{3+}}$ methylene blue [H_2S also give this test]

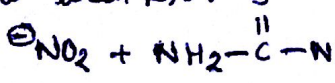
• $NO_2^- \xrightarrow{dil\ H^+}$ Pale blue colour HNO_2 $\xrightarrow{HNO_2 + HNO_2}$ after some time evolution of Reddish brown gas. ($HNO_3 + NO$)

$NO + O_2 \rightarrow NO_2$

KI starch test: $NO_2^- + KI \rightarrow NO + I_2$

I_2 + starch = blue colour adsorption complex.

Thio-urea test:



$SCN^- + Fe^{3+}$ soln $\rightarrow Fe(SCN)_3\downarrow$ (blood red ppt)

$NO_2^- \xrightarrow{dil\ H^+}$ Pale blue HNO_2 $\xrightarrow{FeSO_4}$ Brown ring $[\overset{+1}{Fe}(H_2O)_5(\overset{+2}{NO})]^{2+}$

transfer of e^- from NO to Fe^{2+}

$^{+1}N_2O$, ^{+2}NO — Neut

$^{+3}N_2O_3 - HNO_2$

$^{+4}NO_2 - HNO_2 + HNO_3$

$^{+5}N_2O_5 - HNO_3$

• CH_3COO^- ion $\xrightarrow{dil\ H^+}$ CH_3COOH vinegar type smell $\xrightarrow{alcohol}$ fruity smell (estn)

$\xrightarrow{FeCl_3}$ $Fe(CH_3COO)_3$ (brick red ppt) $\xrightarrow{boil}$ $Fe[(OH)_2(CH_3COO)]\downarrow$ (deep red ppt)

• $Cl^- \xrightarrow[H_2SO_4]{conc}$ HCl gas $\xrightarrow{glass\ rod\ in\ NH_4OH}$ NH_4Cl (white fumes)

• $Cl^- \xrightarrow[conc\ H^+]{K_2Cr_2O_7}$ Red colour gas CrO_2Cl_2 (chromyl chloride) $\xrightarrow{NaOH}$ yellow colour soln (Na_2CrO_4) $\xrightarrow{Pb(Ac)_2}$ $PbCrO_4\downarrow$ (yell ppt)

[$[Cl^-]$ must be significant — not by Ag^+, Pb^{2+}, Hg_2^{2+}]

$PbCrO_4 + H^+ \rightarrow Pb^{2+} + Cr_2O_7^{2-}$ (yellow) (orange)

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$Cl^- + MnO_2 + H^+ \xrightarrow{\Delta}$ yellow colour gas $\xrightarrow{KI}$ I_2 $\xrightarrow{starch}$ blue colour adsorbed complex.
(Cl_2)

$Cl^- + AgNO_3 \longrightarrow$ white ppt ($AgCl$) $\xrightarrow{NH_4OH}$ $Ag[(NH_3)_2]^+ Cl^-$ (ppt gets dissolved) $\xrightarrow{HNO_3}$ again white ppt $AgCl\downarrow + 2NH_4$

- $Br^- \xrightarrow[H_2SO_4]{conc} Br_2\uparrow$ (Brown vapour)

$Br^- + MnO_2 + H^+ \nearrow$

$Br^- \xrightarrow{AgNO_3} AgBr$ (light yell) Partially soluble in NH_4OH.

$Cl_2 H_2O$ test : {aq. soln of Cl_2 + small amount of org solvent}

$Br^- \xrightarrow{Cl_2} Br_2 \longrightarrow$ will get dissolve in org layer to provide brown colour.

$AgCl$, $AgBr$, AgI — NH_4OH → more soluble / spar. soluble / not soluble

- $I^- \xrightarrow[H_2SO_4]{conc} I_2$ (violet vap.)

$I^- + MnO_2 + H^+ \longrightarrow I_2\uparrow \longleftarrow I^- + Cr_2O_7^{2-} + H^+$

$I^- + AgNO_3 \longrightarrow AgI$ (yell. ppt) $\Rightarrow$ Insoluble in NH_4OH.

$I^- \xrightarrow{Cl_2} I_2 \longrightarrow$ will get dissolve in org. layer to provide pink colour.

I_2 + starch $\longrightarrow$ blue colour adsorp^n complex.

starch iodide test: $Br_2 \xrightarrow{KI} I_2 \xrightarrow{starch}$ blue colour complex

- Br_2 + starch $\longrightarrow$ orange red adsorp^n complex.

* Cl_2 water is not taken in excess [$Br_2 + Cl_2 \rightarrow 2Br^+Cl^-$; $I_2 + Cl_2 \rightarrow IO_3^-$]

- $NO_3^- \xrightarrow[H_2SO_4]{conc}$ Reddish brown gas (NO_2) $\xrightarrow{Paper\ ball}$ Cu turning $\rightarrow$ colour of gas intensifies.

$FeSO_4$ soln + $NO_3^- \rightarrow Fe^{3+} + NO$.

$\downarrow H_2O$

$[Fe^{II}(OH_2)_5(\overset{II}{N}O)]^{2+}$ brown. colour.

- $C_2O_4^{2-}$ + conc $H_2SO_4 \longrightarrow H_2C_2O_4 \xrightarrow{\Delta} H_2O + CO_2 + CO$

(neut, liquid at room temp) (acidic, lime water) (neut, burns with blue flame)

- $SO_4^{2-} \xrightarrow{BaCl_2} BaSO_4\downarrow$ (white) $\xrightarrow{O.A}$ not soluble

$SO_3^{2-} \xrightarrow{BaCl_2} BaSO_3\downarrow$ (white) $\xrightarrow{O.A}$ soluble,

- PO_4^{3-} + $(NH_4)_2MoO_4$ (ammonium molybedate) $\longrightarrow (NH_4)_3PO_4 \cdot 12MoO_3\downarrow$ ammonium phospho molybedate. (canary yell. ppt)

$PO_4^{3-} + Mg^{2+} + NH_4OH \rightarrow Mg(NH_4)PO_4\downarrow$ white.

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POLYMERS

- Classificⁿ on basis of source – ① Natural polymer : protein, cellulose, starch, resin, rubber
② Semi-Synthetic polymers : cellulose derivative as rayon, cellulose nitrate
③ Synthetic polymers : polythene, synthetic fibre & syn. rubber (Nylon-6,6) (Buna S)

- on basis of structure – ① linear polymer – long st. chain – high density polythene, PVC
② Branched chain po– low density polythene,
③ cross linked or network – formed from bi or tri funcⁿal gp – bakelite, melamine

- based on mode of polymerisaⁿ – ① Addiⁿ polymers. – double or triple bond. [homopolymer, copolymer.
② Condensaⁿ polymers – eliminiⁿ of H_2O

Ex – Copolymer : $n\ CH_2=CH-CH=CH_2 + n\ C_6H_5CH=CH_2 \longrightarrow \{CH_2-CH=CH-CH_2-CH_2-CH(C_6H_5)\}_n$
1,3-Butadiene, styrene (Butadiene – styrene copolymer) Buna-S

– Condensaⁿ polymer : $n\ H_2N(CH_2)_6NH_2 + n\ HOOC(CH_2)_4COOH \longrightarrow \{NH-(CH_2)_6-NH-C(=O)-(CH_2)_4-C(=O)\}_n + nH_2O$
hexamethylene diamine, adipic acid, Nylon-6,6

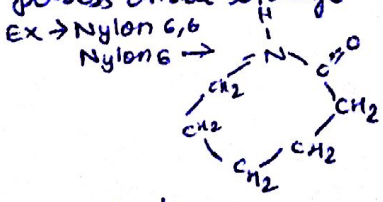
- based on molecular forces :-
① Elastomer – weakest intermolecular force [buna-S, buna-N, neo prene) $\{CH_2-C(Cl)=CH-CH_2\}_n$ neo prene.
② Fibres – high tensile strength (H-bond) – crystalline nature – Polyamide, polyester Nylon-6,6
③ Thermoplastic polymer – repeatative softening on heating & hardening on cooling – intermolecular forces intermediate b/w elastomer & fibre – polythene, polystyrene, polyvinyl.
④ Thermosetting polymers – on heating undergo extensive cross linking in moulds and again become infusible, cannot be reused, bakelite / urea formaldehyde resin.

Some imp • oddiⁿal polymer : Polythene [LDP – high press (1000 to 2000 atm) – 350 to 370 K; Inert, tough, flexible, poor conductor
HDP – use of Ziegler Natta [triethyl aluminium & titanium tetra chloride]; P = 6 to 7 atm; Inert, more tough, harder]

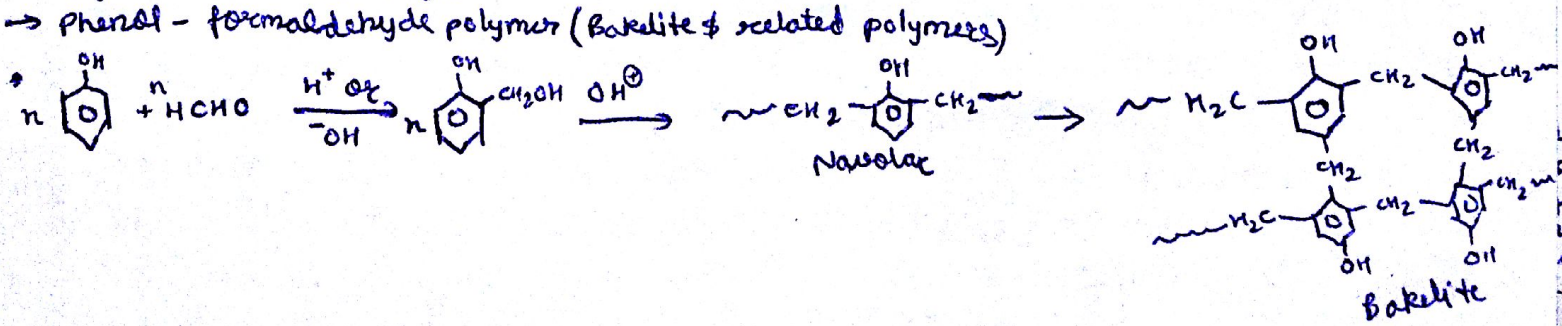
: Teflon $\{CF_2-CF_2\}_n$

: Polyacrylonitrile $\{CH_2-CH(CN)\}_n$

- Condensaⁿ polymer : $n\ HOH_2C-CH_2OH + n\ HOOC-C_6H_4-COOH \longrightarrow \{OCH_2-CH_2-O-C(=O)-C_6H_4-C(=O)-O\}_n$
ethylene glycol, Terephthalic acid, Terylene or dacron.

→ Polyamide – possess amide linkage
Ex → Nylon 6,6
Nylon 6 → 
$\xrightarrow[H_2O]{\Delta} \{C(=O)-(CH_2)_5-NH\}_n$ Nylon 6.

→ Polyesty. – Dacron or terylene.
→ Phenol – formaldehyde polymer (Bakelite & related polymers)



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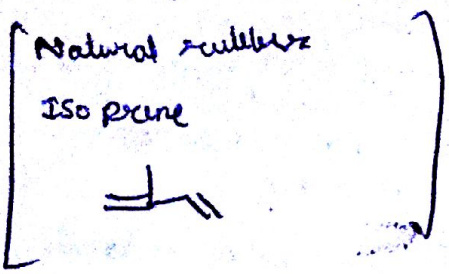
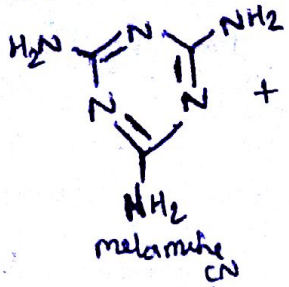
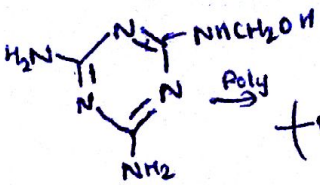
- melamine formaldehyde polymer :

Natural rubber
Isoprene







- Buna N : $n CH_2=CH-CH=CH_2$ (1,3-butadiene) + $n CH_2=CH(CN)$ $\xrightarrow{\text{copolymerisation}}$ $+CH_2-CH=CH-CH_2-CH(CN)+_n$

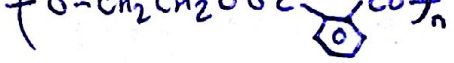
- PHBV – Poly-Bhydroxybutyrate – co-β-hydroxyvalerate ⇒ Biodegradable polymer

$CH_3-CH(OH)-CH_2-COOH$ (3-hydroxy butanoic acid) + $CH_3-CH_2-CH(OH)-CH_2-COOH$ (3-hydroxypentanoic acid) $\longrightarrow$ $+O-CH(CH_3)-CH_2-C(=O)-O-CH(CH_2CH_3)-C(=O)+_n$

- Nylon-2-nylon-6

H_2N-CH_2-COOH (glycine) + $H_2N(CH_2)_5COOH$ (amino caproic acid) ⇒ biodegradable

- glyptal : (a) Ethylene glycol (b) Phthalic acid



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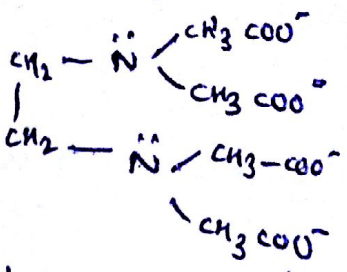
COMPLEX COMPOUND

- neutral ligand : $H_2\ddot{O}$, $\ddot{N}H_3$, $C_2H_5\ddot{O}H$, $CH_3-\ddot{O}-CH_3$
- +ve charge : $:\overset{\oplus}{N}=O$, $:NH_2-\overset{\oplus}{N}H_3$, -ve charged : Cl^-, Br^-, CN^-, $C_2O_4^{2-}$.
- Denticity — monodentate Cl^- Br^- OH^- $H_2\ddot{O}$
  - bidentate — Ethylene diamine (en) : $:NH_2-CH_2-CH_2-:NH_2$
  - hexadentate — ethylene diamine tetra acetate (EDTA)



- for any metal , effective atomic number ,

EAN : (At.no) − (O.N) + 2×(CN)

⇒ Werner's theory -

- Primary valency (ON) = ionizable, non direc^{n}al satisfied by anion (attached with ionic bond)
- Secondary valency (C.N) = non ionizable, direc^{n}al, satisfied by Ligands. (attached with coordina^n bond)

* If Ligand is neutral it satisfy only primary valency.
* If ligand is charged it satisfy both primary and second. valency.

- Being direc^{n}al in nature the secondary valency is oriented in symmetric fashion in 3D space.

→ could^{nt} explain : magnetic property : No theory
: colour property : coord. No 4 (tetrahedral / sq. planar), 5.

⇒ Valence bond theory : sp^3d^2 → outer octahedral complex (high spin complex)
: d^2sp^3 → inner octahedral complex (low spin)

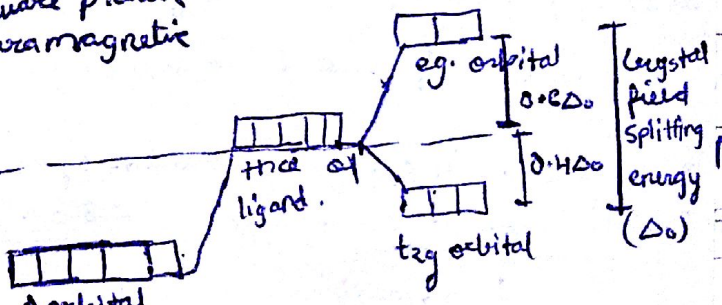
* Ni^{2+} always outer octahedral complex.

Exceptional : $Cu^{2+} \xrightarrow{NH_3} [Cu(NH_3)_4]^{2+}$
(blue) (intense, deep blue colour)

⇒ sp^2d square planar paramagnetic

- Crystal field theory ⇓

Violation of degeneracy of d orbital of ligand. ⇓ crystal field effect



in octahedral complex : ① $\Delta_o < P$ [weak ligand]
② $\Delta_o > P$ [St. ligand]

weak ligand ← → St ligand.

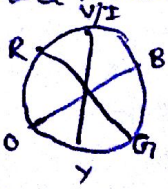
→ Spectrochemical series : $I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < {}^-OH < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

Stability obs : − $0.4(t_{2g}) \rightarrow 0.6(Eg)$

- Whenever d^0 & d^{10} seems coloured , charge transfer from anionic part to metal ion.

d^0	$Cr_2O_7^{2-}$ orange	MnO_4^- Purple	
d^{10}	CrO_4^{2-} yellow	Ag_2S Black	CdS yellow.

d^1-d^9 − coloured
d^0 & d^{10} − colourless



- in square planar complex : $\Delta_t = \frac{4}{9}\Delta_o$ (in tetrahedral)

energy $d_{xz} = d_{yz} < d_{z^2} < d_{xy} < d_{x^2-y^2}$

* In octahedral complex -
  - t_{2g} orbital will experience lesser repuls^n force (uneven distribu^n of e^- will not distort geom)
  - eg orbital will exp. more repuls^n force, (unsym. distribu^n will distort geom) Jahn teller effect.

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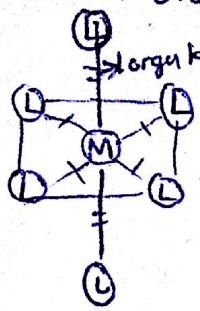
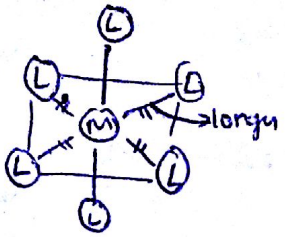
• unsymmetrical distribⁿ – d^4 (WL) & d^7 (SL) | d^9 (WL) & (SL)

one of the eg orbital is containing one more e⁻ than other.

(I) d_{z^2} contain one more e⁻ (II) $d_{x^2-y^2}$ cont. 1 more e⁻



tetragonal elongaⁿ (general case)



tetragonal compression (Rare case)

(III) d^5 (WL) $t_{2g}^3 e_g^2$

M^{2+} ion (Mn^{2+}) } d^5 confi

M^{3+} ion (Fe^{3+}) }

→ LE in metal halide –

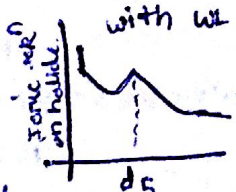
$MnCl_2 < CoCl_2$ | $FeCl_3 < CoCl_3$

$MnCl_2 < FeCl_2$ | $FeCl_3 < MnCl_3$

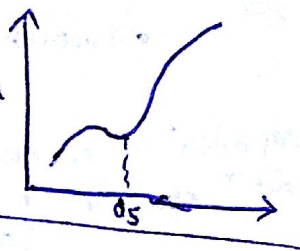
→ Ionic size of metal ion in halide –

$Mn^{2+} > Co^{2+}$ | $Fe^{3+} > Co^{3+}$

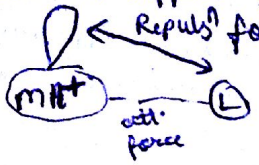
$Mn^{2+} > Fe^{2+}$ | $Fe^{3+} > Mn^{3+}$



Hydraⁿ energy – opp. of previous data



(IV) Factor affecting value of Δ :



• more repuln force more Δ

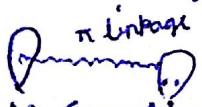
• more charge on metal ion, less size ⇒ Δ more

belonging to – 3d, 4d, 5d, 6d ↓ Δ will ↑

• geometry of complex, Δ(oct. complex) > Δ(tetra/sq planar complex)

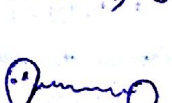
• ligand → st. ligand more Δ, normally st. base Δ more, but [CO – neut – st ligand] [$^{\ominus}OH$ – st base – weak ligand than H_2O]

case① π linkage



Δ ↓ π donor ligand ($^{\ominus}OH$)

case② CO



Δ ↑ π acceptor ligand

(V) more value of Δ, more stability of complex

• $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, $\Delta G^\circ = -2.303 RT \log K_{eq}$

• Δ = EDTA < NH_3 < en accor. to electrospectral curin

but → stability : $[Pt(NH_3)_6]^{3+} < [Pt(en)_3]^{3+} < [Pt(EDTA)]^{\ominus}$

chelaⁿ effect; macro cyclic effect.

denticity of ligand ↑ stability ↑

Isomerism.

• Ionizaⁿ isomers $[Pt(NH_3)_5Cl]Br_2$ & $[Pt(NH_3)_5Br]Cl_2$

• Hydraⁿ isomers $[Pt(H_2O)_4Cl_2]Cl_2 \cdot 2H_2O$, $[Pt(H_2O)_6]Cl_4$

• Ambidentate ligand $[Co(H_2O)_5(CN)]^{3+}$, $[Co(H_2O)_5(NC)]^{3+}$

salt/linkage isomerism

• Polymeriza^n polymer $[Pt(en)_2][PtCl_4] = [Pt(en)Cl_2]_2$

• Coordinate isomerism $[Co(H_2O)_6][CrCl_6]$, $[Cr(H_2O)_6][CoCl_6]$

• Bridge ligand coordina^n/posi^n isomerism

$\left[(NH_3)_4Co\begin{smallmatrix}OH\\OH\end{smallmatrix}Cr(H_2O)_4\right]^{4+}$, $\left[(NH_3)_2(H_2O)_2Co\begin{smallmatrix}OH\\OH\end{smallmatrix}Cr(H_2O)_2(NH_3)_2\right]$

[bridge isomerism]

• geometrical isomerism : coordination no. = 4,

Tetrahedral geometry : no GI.

Sq. planar – GI may exist.

• Coordina^n no 6, MA_2C_4 NoGI M_{ABCDEF} : 15 GI

$MAsB_4$

(VI) C≡O, BL of C–O = x

$[Ni(CO)_4]$, Ni ← CO, BL of CO = y.

soln) y > x

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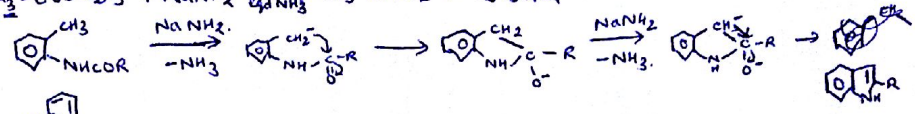
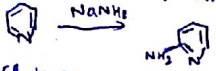
USEFUL REAGENTS (1)

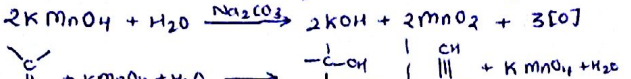
① **Sodium** → To detect the presence of active hydrogen → In wurtz rekn
→ As reducing agent with ethanol or water → for detecn of element.
[Na + C + N → NaCN
2Na + S → Na_2S
$2Na + X_2$ → 2NaX.]

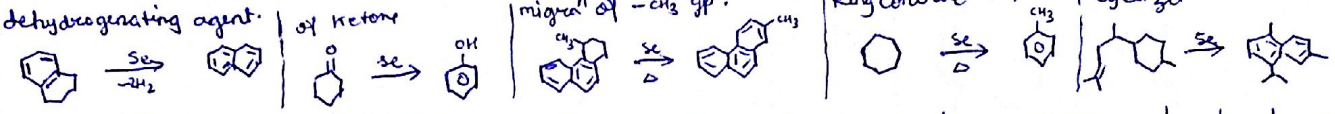
② **Zn–Cu couple** : used as red. agent with alcohol [Zn–Cu + C_2H_5OH] , $CH_3I \xrightarrow[C_2H_5OH]{Zn-Cu} CH_4 + HI$

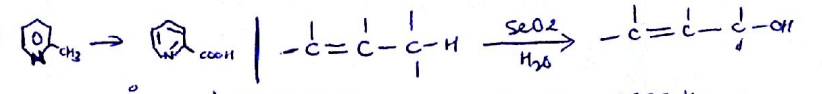
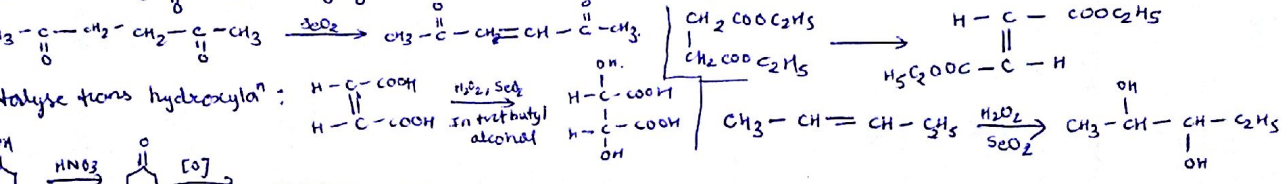
③ **Soda lime** ($NaOH + CaO$) → both in solid form : It rekt with sodium salt of fatty acid to produce & undergoes decarboxylan to give hydrocarbon with one carbon less. → $C_6H_4(Z)(COONa) \xrightarrow[\Delta]{NaOH+CaO} C_6H_5-Z + Na_2CO_3$ [Z may be $-NO_2$, $-OH$, $-CH_3$ etc]

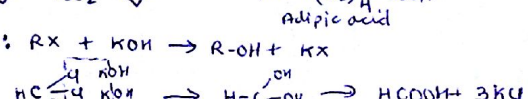
④ **Sodamide** [$NaNH_2$] → $-CH(Br)-CH(Br)- + 2NaNH_2 \xrightarrow{liq\ NH_3} -C\equiv C- + 2NaBr + 2NH_3$.
$NaNH_2 \rightleftharpoons Na^+ + NH_2^-$
$CH_3-C\equiv C-CH_3 \xrightarrow[liq\ NH_3]{NaNH_2} CH_3-CH_2-C\equiv CH$
• Rkn of ether with sodamide to form alkene : $NH_2^- + H-CH_2-CH_2-O-C_2H_5 + Na^+ \longrightarrow NH_3 + CH_2=CH_2 + C_2H_5ONa$
• with ester having α-H : $CH_3-COO-C_2H_5 + NaNH_2 \xrightarrow{liq\ NH_3} CH_3CONH_2 + C_2H_5ONa$
• cyclizan of anilide comp's:

• with pyridine :


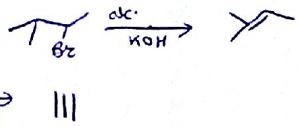
⑤ **Thionyl chloride** : [$SOCl_2$] $ROH + SOCl_2 \rightarrow RCl + SO_2 + HCl$, $RCOOH + SOCl_2 \rightarrow R-COCl + HCl + SO_2$, $CH_2(OH)-COOH \xrightarrow{SOCl_2} CH_2(Cl)-COCl$

⑥ **Potassium permanganate** : ($KMnO_4$) $2KMnO_4 + H_2O \xrightarrow{Na_2CO_3} 2KOH + 2MnO_2 + 3[O]$
Known as Bayer's reagent in alkaline medium.

used to detect unsaturaen

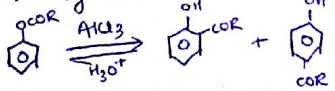
⑦ **Selenium** : dehydrogenating agent. (Se)


⑧ **Selenium dioxide** : oxidan of reactive methyl & methylene gp & (SeO_2)

attacks $-CH_2-C(=O)-$ to give $-C(=O)-C(=O)-$


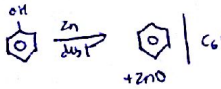
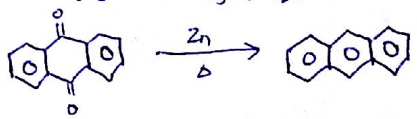
⑨ **Aq caustic potash** : (Aq KOH) $RX + KOH \rightarrow R-OH + KX$


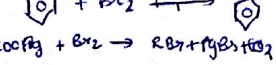
⑩ **Alcoholic caustic potash** : (alc. KOH)


⑪ **Hydriodic acid** ; good reducing agent which convert –OH to –H (HI)
$C_2H_5OH + 2HI \longrightarrow C_2H_6 + H_2O + I_2$
$CH(OH)COOH \xrightarrow{4HI} CH_2-COOH + 2H_2O + 2I_2$.
$CH(OH)COOH \qquad CH_2-COOH$

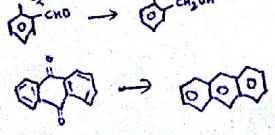
⑫ **Anhydrous $AlCl_3$** : friedal acraft Rkn (help in producing electrophile from given reagent)
→ fries rearrangement.

→ Gattermann Koch aldehyde synthesis : $C_6H_6 + CO + HCl \xrightarrow{AlCl_3} C_6H_5CHO + HCl$.
→ Gattermann aldehyde synthesis : $C_6H_6 + HCl + HCN \xrightarrow[H_2O]{AlCl_3} C_6H_5CHO + NH_4Cl$
→ Isomerisan : $CH_3-CH_2-CH_2-Cl \xrightarrow{AlCl_3} CH_3-CH(Cl)-CH_3$ [also with alkane]

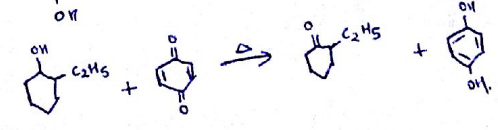
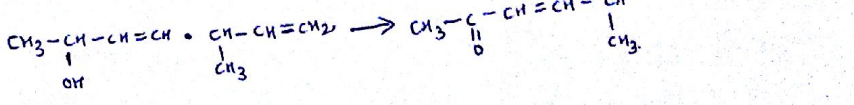
⑬ **Anhydrous Zinc chloride** ($ZnCl_2$) : dehydrating agent : can be used in place of $AlCl_3$ in friedal acraft rek's
$C_2H_5OH \xrightarrow[anhy\ ZnCl_2]{} CH_2=CH_2 + H_2O$
$C_6H_5NH_2 + HOOCCH_3 \xrightarrow{ZnCl_2} C_6H_5-NH-COCH_3 + H_2O$.

⑭ **Zinc dust** :

$C_6H_4(OH)CH_3 \xrightarrow{Zn\ dust} C_6H_5CH_3 + ZnO$ [Zn & alc KOH — Reducing agent]

Zn with CH_3COOH, HCl & KOH is used as redu. agent.

⑮ **Bromine** :
If brown colour of bromine is discharged by shaking with comp. then comp is unsaturated
$CH_2=CH_2 + Br_2 \xrightarrow{CS_2} CH_2(Br)-CH_2(Br)$.
no effect of this addin on aromatic ring.
$R-OH + P-Br_2 \longrightarrow RBr$.

$RCOOAg + Br_2 \rightarrow RBr + AgBr + CO_2$

⑯ **Bromine & caustic potash** : ($Br_2 + KOH$) convert $-CONH_2$ to $-NH_2$.
Known as Hofmann Bromamide rekn
BrO^-/OH^-
$CH_3CONH_2 + Br_2 + 4KOH \rightarrow CH_3NH_2 + 2KBr + K_2CO_3 + 2H_2O$
[similar for aromatic amide.

⑰ **Aluminium isopropoxide** [$(CH_3)_2CHO]_3Al$] :
→ specific acid reagent for carbonyl gp only.
$CH_3-CH=CH-CHO \rightarrow CH_3CH=CH-CH_2OH$


⑱ **Aluminium tri-ter-butoxide** [$(CH_3)_3CO]_3Al$] : as oxidising agent.
no effect on double bond
$CH_3-CH(OH)-CH_2-CH_3 + CH_3-CO-CH_3 \xrightarrow{\Delta} CH_3-C(=O)-CH_2CH_3 + CH_3-CH(OH)-CH_3$ [oppenauer oxidan]



USEFUL REAGENTS

(19) Nitrous acid (HNO_2) : ie cold $NaNO_2$ + cold HCl → NaCl + HNO_2.
beyond 5°c, it undergoes autoxidaⁿ to give HNO_3.

- 1° amine aliphatic $R-NH_2 \xrightarrow{HNO_2} R-OH + N_2 + H_2O$
- active methylene gp forms an oxime with HNO_2.
- $NH_2CONH_2 + 2HNO_2$ (excess) → $2N_2 + CO_2 + 3H_2O$
- Ph-OH + HONO → $C_6H_4(OH)-N=O$

(20) Boron trifluoride : Friedel craft Rkⁿ : Ph-CH₃ + BF_3 → (ROH) Ph-CH₃ (R)

- $RCOOH \xrightarrow{BF_3} RCOOR$ (ROH) ; $CH_2=CH_2 \xrightarrow{BF_3} RCOOC_2H_5$
- Isomerisaⁿ : $Ar_2C=NOH \xrightarrow{BF_3} ArCONHAr$
- Benzidine rearrangement : $Ar-NH-NH-Ar \longrightarrow H_2N-C_6H_4-C_6H_4-NH_2$
- Methylaⁿ of acid : $RCOOH \xrightarrow[CH_3OH]{BF_3} RCOOCH_3$

(21) Diborane (B_2H_6) : Hydroboraⁿ
: $(RCH_2CH_2)_3B \xrightarrow{H^+} 3RCH_2CH_3$
: $-C\equiv C- \xrightarrow{B_2H_6} +CH=CH_3B \xrightarrow{H^+} -CH=CH-$ pure cis
- $R-CH-CH_2$ (epoxide) → RCH_2-CH_2OH (90%) ; In BH_3-BF_3 → BF_3-NH_3 ($\oplus$ $\ominus$, sp^3)

B_2H_6 in THF is reducing agent. [Ex: $-COOH \rightarrow -CH_2OH$] does not reduces $-COCl$, RNO_2, $ArNO_2$, RX, ...

(22) Perbenzoic acid ; used as oxidising agent
$C_6H_5-C(=O)-O-O-H$, C_6H_5COOH

- $R-CH=CH_2 \xrightarrow{C_6H_5COOH}$ epoxide
- $R-CH-CH_2$ (epoxide) $\xrightarrow{BF_3}$ $R-CH(-O^{\oplus}-BF_3^{\ominus})-CH_2 \rightarrow RCH_2-CH=O^{\oplus}-BF_3^{\ominus} \rightarrow RCH_2CHO$
- $R-CH-CH_2$ (epoxide) $\xrightarrow[\text{acid. acidic med.}]{H_2O \text{ impg.}}$ $R-CH(OH)-CH_2OH$

Bayer villiger oxidaⁿ –
- Ph-CHO $\xrightarrow{C_6H_5CO_3H}$ coprolactone
- Ph-CO-Ph $\xrightarrow{\text{perbenzoic acid}}$ $Ph-C(=O)-O-Ph \xrightarrow{H_2O}$ COOH + OH

(23) N-Bromosuccinimide (NBS) [CH_2CO / CH_2CO N-Br]
⇒ as oxi. agent with some reagent.
- Ph-CH₃ $\xrightarrow{NBS}$ Ph-CH₂Br $\xrightarrow[KOH]{AlC}$ $\xrightarrow[MeOH]{NBS}$ Ph-CH(OMe)-CH₃
- Ph-CH₃ $\xrightarrow[R_4NO_2]{NBS}$ Ph-CH(Br)-CH₃
⇒ NBS in aq. solnⁿ decarboxylate α-amino acid, protein & peptide.

(24) Periodic acid (HIO_4 meta. $\xrightarrow{2H_2O}$ H_5IO_6 ortho.)
- $CH_2(OH)-CH_2(OH) \xrightarrow{HIO_4} CH_2(OH) + HO-CH_2 \rightarrow HCHO$
- $-CH_2OH \rightarrow HCHO$; $-CHOH \rightarrow RCHO$
- $O=CH-CH=O \rightarrow HC(=O)-OH + HC(=O)-OH$
- In each Rkⁿ HIO_3 is formed giving white ppt with $AgNO_3$ → $AgIO_3$
- each use one molecule [$\overset{\diagup}{C}$, OH]
- $R-CH(OH)-CH(NH_2)-COOH \xrightarrow{HIO_4} RCHO + CHO-COOH + NH_3$ [cleave epoxide to aldehyde]

(25) lead tetra acetate [$Pb(Ac)_4$] : oxidaⁿ of Hydroquinone (LTA)
- Ph(OH)(OH) $\xrightarrow{LTA}$ Ph(=O)(=O)
- $R'-CH(OH)-R \xrightarrow{LTA} R'-C(=O)-R$; $R''-CH(OH)-R''' \xrightarrow{LTA} R''-C(=O)-R'''$
- Ph(COOH)(I) $\xrightarrow[I_2]{LTA}$ Ph(I)(I) + CO_2
- Ph-Ph-Ph $\xrightarrow{LTA}$ Ph-Ph-Ph(OAc)
- Ph-H $\xrightarrow[CH_3COOH]{LTA}$ Ph-CH₃

(26) Liquid Ammonia : Birch reducⁿ (as solvent)
- Ph-Ph $\xrightarrow[\text{crystal Na}]{Li/NH_3}$ [only one of 2 double bond are reduced]
- $R-C\equiv C-R \xrightarrow[\text{liq } NH_3]{Na}$ trans alkene
- Reducⁿ of terminal ≡ bond : $R-C\equiv CH \xrightarrow[NH_3]{Na} R-CH=CH_2$
- Red of ≡ to trans alkene
- amide → aldehyde.
- $-C\equiv N$ to $-CH_2NH_2$
- $>C=O + CH\equiv CH \xrightarrow[NH_3]{Na} >C(OH)-C\equiv CH$

(27) CH_2N_2 (diazo methane)

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CHEMICAL BONDING

- Bond length → $- > = > \equiv$
- Hybridisa^n → less than or eq. to 8, divide by 2.
  → More than 8, divide by 8 and remainder by 2.
- After 2nd period, all elements have vacant 'd' orbital.
- Acidity → Alkyne > alkene > alkane or $sp > sp^2 > sp^3$ [Acidity ∝ scharacter]
- Excluding d.p →
  - → sp^3 – Tetrahedral – B.A – 109° – CH_4
  - → sp^2 – Trigonal planar – B.A – 120° – BF_3
  - → sp – linear – BA – 180° – $BeCl_2$.
  - → sp^3d – Trigonal planar bipyramidal – PCl_5 → B.A – Axial (90°), equitorial (120°)
  - → sp^3d^2 – Octahedral – B.A – 90° (12) – SF_6
  - → sp^3d^3 – Pentagonal planar tripyramidal – B.A – 90° (10) & 72° (5)
- Repuls^n → L.P–L.P > L.P–B.P. > B.P–B.P
- least repuls^n → more stable → least energy.
- Geometry is always seen against bond pair or bonds.
- Same atom → on same plane → more stable.
- π bond does not participate in hybridisation.
- Dipole movement (μ) = $q \times d$ = charge × dist. of seppa^n. also, $\mu \propto \frac{1}{a}$

[more electronegative atom ↓ on axial – more stable]

- All cis are polar in nature but trans may be polar or non polar in nature.
- $\mu = 0$, Non polar & $\mu \neq 0$, Polar.
- Alene – continious pair of bond
- Odd Alene → $\mu \neq 0$/polar / even alene → $\mu = 0$, non polar.
- Along gp → size of atom ↑ – B.A ↑
- Stability → $p_\pi p_\pi > p_\pi d_\pi > d_\pi d_\pi$ [vacant 'p' orbital i.e. 2p–2p] (back bonding resonance)
- Bonding orbital have min^m energy as they have high stability.
- Para magnetic nature (B_2, O_2) [O_2^-] unpaired e$^-$ – as
  Di magnetic nature – Paired e$^-$ – sum
- Bond order → $\frac{\text{Bonding } e^- - \text{Antibonding } e^-}{2}$ [0 or more than 3 not possible]
- Bond order ∝ Bond energy ∝ $\frac{1}{\text{Bond length}}$ ∝ $\frac{1}{p.E.}$ ∝ stability ∝ dissocia^n energy
- B.O → $\frac{\text{total bond}}{\text{side resonance}}$
- 1 – 13 e$^-$ → BABA(B)B & for 14 – 20 e$^-$ → BABAB(A)A
  B → Bonding e$^-$
  A → Anti bonding e$^-$
- H-bond → Polar in nature → (FON)H → covalent compound.
- H-bond length > Covalent bond length / H-bond strength < covalent bond strength
- H-bond ∝ boiling pt ∝ solubility ∝ Molecular mass. ∝ hydra^n energy or solva^n energy.
- Size of cation is small ∝ force of attrac^n ∝ lattice energy ∝ stability ∝ $\frac{1}{\text{solubility}}$
- Fazan's rule → size of cation (small) +ve, size of anion (large) –ve

- Lewis acid – vacant 'p' orbital.

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- Formal charge = [total no. of valence e^- in free state] − [lone pair e^-] $-\frac{1}{2}$[shared e^-]

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